

Daily Briefing - November 05, 2026

Your Daily Tech & Programming Digest

Thursday, November 05, 2026

1000	123983	1173	38
ARTICLES	WORDS	MIN READ	SOURCES

📰 Today's Top Stories

There are 13 ways to analyse a replication study, but only one of them is coherent.

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-05-03 📖 6 min 📝 1361 words

TWENTY PERCENT STATISTICIAN

> **Summary:** <p>Paul Simon’s says there are 50 ways to leave your lover, and a similar abundance characterizes the contemporary literature on how to analyse a replication study. The recent SCORE replication project in Nature makes this explicit by reporting “13 replication success metrics along with the number o...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/there-are-13-ways-to-analyse.html

Evaluating Dr. Cuddy’s Claim that the Debunking of Power Posing is a Myth

noreply@blogger.com (Daniel Lakens)
2026-05-25 30 min 6008 words

TWENTY PERCENT STATISTICIAN

Summary: In this blog post I will analyse the arguments that Dr. Amy Cuddy provided in a LinkedIn post “The "Power Posing Was Debunked" Myth: What the Research Actually Shows — and Why Scientific Discourse Matters” on February 26. You can find the LinkedIn post here: <span lang="...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/05/evaluating-dr-cuddys-claim-that.html

Which Data Repository Should You Use?

noreply@blogger.com (Daniel Lakens)
2026-08-15 17 min 3434 words

TWENTY PERCENT STATISTICIAN

Summary: Update 28-10-2026: ResearchBox now has an API. This new information has been incorporated in the text below. The Center for Open Science has announced that from November 16, 2026, no new projects can be created on the Open Sci...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/08/which-data-repository-should-you-use.html

Episode 29: Q&A with Dr. Anahita Kyani, Director of Neuroscience, Data Science and Digital Health at Johnson & Johnson Innovative Medicine

👤 Adriel Carridice 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/episode-29-qa-with-dr-anahita-kyani-director-of-neuroscience-data-science-and-digital-health-at-johnson-johnson-innovative-medicine/

Q&A with Felix Schwock, PhD student at University of Washington and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-06-22 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-felix-schwock-phd-student-at-university-of-washington-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Neuroethical Perspectives on Neurotechnology in Latin America

Adriel Carridice 2026-09-23 1 min 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/publications/research-and-white-papers/neuroethical-perspectives-on-neurotechnology-in-latin-america/

Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor in Chief Search

Deidre Artis 2026-07-17 1 min 34 words EMBS

Summary: Transactions on Neural Systems and Rehabilitation Engineering (TNSRE) Editor-in-Chief Search Submit your Application The post <a href="https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/">Transactions on Neural Syst...

🔗 **Read full article:**
https://www.embs.org/featured-news/transactions-on-neural-systems-and-rehabilitation-engineering-tnsre-editor-in-chief-search/

Understanding the Perception of Transliterated Offensive Social Media Posts Among Banglish and Hinglish Users

👤 Dhiman Goswami, Zhicong Lu, Sanchari Das
📅 2026-11-05 📖 1 min 📝 222 words ARXIV CS HC

Summary: arXiv:2611.06226v1 Announce Type: new Abstract: Effective content moderation depends on understanding not just what harmful content looks like, but how users perceive and respond to it. When offensive language is written in transliterated forms such as Banglish or Hinglish (Bangla or Hindi in Latin...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06226

ProGram: A Grammar for Progressive Computation Pipelines

👤 Seunghoon Jung, Jaemin Jo
📅 2026-11-05 📖 1 min 📝 146 words ARXIV CS HC

Summary: arXiv:2611.06188v1 Announce Type: new Abstract: Progressive Visual Analytics (PVA) systems have demonstrated the effectiveness of time-bounded, incremental computation for exploring large-scale data. However, our survey reveals that existing PVA implementations are often difficult to reuse and exte...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06188

Growing One's AI Tools: A Paradigm of Practice-Grown Tools

Zeming Zhao 2026-11-05 1 min 158 words ARXIV CS HC

Summary: arXiv:2611.06132v1 Announce Type: new Abstract: Professionals have long used software built for them by others. This article reports the growth of a professional computer-assisted interpreting (CAI) tool built by its own user, a conference interpreter without programming training. Drawing on an eig...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06132

Fluidic Body: Swarm Robots as a Fluid Extension of the Hand

Michikuni Eguchi, Shigeo Yoshida, Mai Nishimura, Takefumi Hiraki 2026-11-05 1 min 274 words ARXIV CS HC

Summary: arXiv:2611.06068v1 Announce Type: new Abstract: Swarm robots have attracted growing interest as a physical interface for human--computer interaction, since a swarm can reshape its formation to fit the objects and surroundings it works with. For a human operator, however, adapting the swarm's behavi...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06068

Robots in the City: Persona-Based Perceptions of Urban Robots

Keonhee Jang, Kanghoon Lee, Seokwoo Yun, Namwoo Kim, Yoonjin Yoon

2026-11-05 1 min

219 words

ARXIV CS HC

Summary: arXiv:2611.05954v1 Announce Type: new Abstract: Pedestrians increasingly encounter robots in urban public spaces without interacting with them, but how robots change the perceived safety of a street remains unclear. We examine how robot presence and type affect perceived safety across urban context...

🔗 **Read full article:**
https://arxiv.org/abs/2611.05954

Compiling, Validating, and Using Human Reference Atlas Cell Type Annotation Crosswalks

Nicole Vasilevsky, Aleix Puig-Barbe, Andreas Bueckle, Ellen M. Quardokus, Yongxin Kong, Divya Prasanth Paraman, Jie Zheng, Yongqun He, Yashvardhan Jain, Bruce W. Herr II, Katy B\"orner

2026-11-05 1 min

176 words

ARXIV CS HC

Summary: arXiv:2611.05888v1 Announce Type: new Abstract: The Human Reference Atlas (HRA) v2.5 includes 4,807 anatomical structures, 1,347 cell types, and 2,215 biomarkers (genes, proteins, lipids); 81 organs with 1,392 anatomical structures modeled in three dimensions (3D). Mapping new data into the HRA req...

🔗 **Read full article:**
https://arxiv.org/abs/2611.05888

Sure Enough to Share? Overconfidence in GenAI Video Discernment is Associated with Sharing Intentions

Haoning Xue, Sijia Qian, Christopher Calabrese, Cuihua Shen, Laramie Taylor, Yixuan Wang

2026-11-05 1 min 150 words ARXIV CS HC

Summary: arXiv:2611.05844v1 Announce Type: new Abstract: Generative Artificial Intelligence (GenAI) is blurring the line between synthesized and recorded videos, but less is known about people's awareness of their discernment abilities and the consequences of miscalibration. This study examines three forms ...

🔗 **Read full article:**
https://arxiv.org/abs/2611.05844

Pairing Expert Critique with LLM-Driven Refinement: An Iterative Approach for Curating Quality AI Evaluation Datasets

Honson Ling, Kidus Admassu, Stephanie Rosenthal

2026-11-05 1 min 219 words

ARXIV CS HC

Summary: arXiv:2611.05749v1 Announce Type: new Abstract: Evaluating and improving domain-specific LLM responses requires high-quality reference data and rubrics, yet expert authoring is slow and expensive, and synthetic datasets may fail to capture practitioner preferences unless they are grounded in expert...

🔗 **Read full article:**
https://arxiv.org/abs/2611.05749

Personalized Visual Support After Automatic Speech Recognition

Beatriz Calero, Duarte Neves, Carlos Bruno, Luis Ricardo Santos, Mario Egea

2026-11-05

1 min

91 words

ARXIV CS HC

Summary: arXiv:2611.05625v1 Announce Type: new Abstract: Many spoken-language interfaces stop at producing a transcript. This project examines how a second layer of optional text and pictorial support can be adapted to individual needs. We specify four presentation levels and a behavior-based selection poli...

🔗 **Read full article:**
https://arxiv.org/abs/2611.05625

Just the Facts: How Dialogues with AI Reduce Conspiracy Beliefs

Thomas H. Costello, Gordon Pennycook, David G. Rand

2026-11-05

1 min

151 words

ARXIV CS HC

Summary: arXiv:2611.05572v1 Announce Type: new Abstract: Personalized dialogues with AI substantially reduce conspiracy beliefs. To identify why, we conducted two experiments with U.S. adults recruited online (total N = 2,193) that removed possible causes one at a time. The debunking effect was robust to di...

🔗 **Read full article:**
https://arxiv.org/abs/2611.05572

A Bayesian Optimized Biquad Filter-Burg Model for Real-Time EEG Phase Estimation

Ivan Martinez-Kay, Tatiana Kardashina, ShiNung Ching, Eric C. Leuthardt, Benjamin T. Acland

2026-11-05 1 min 184 words ARXIV QBIO NC

Summary: arXiv:2611.05374v2 Announce Type: replace-cross Abstract: Real-time phase estimation on stimulus-aligned transient EEG is critical for closed-loop neuromodulation systems targeting the theta-band, yet most real-time phase estimation methods have been developed and validated on resting-state, high-f...

Read full article:
https://arxiv.org/abs/2611.05374

Beauty is in the ELBO of the Beholder: A Variational Account of Processing Fluency in Face Perception

Francisco M. L\'opez, Jochen Triesch

2026-11-05 1 min 157 words ARXIV QBIO NC

Summary: arXiv:2608.24219v3 Announce Type: replace-cross Abstract: Facial attractiveness has been linked to statistical regularities such as symmetry and averageness, suggesting that beauty may depend on the ease with which a face is perceived. We empirically test this hypothesis by training variational aut...

Read full article:
https://arxiv.org/abs/2608.24219

Flow Matching with In-Context Priors for Out-of-Distribution Brain Dynamics

Sam Gijsen, Micha{\l} {\L}ukomski, Marc-Andr\'e Schulz, Kerstin Ritter

2026-11-05 1 min

186 words

ARXIV QBIO NC

Summary: arXiv:2606.11833v2 Announce Type: replace-cross Abstract: Flow matching and diffusion models enable conditional generation across domains ranging from images to proteins, with recent extensions to out-of-distribution contexts. Yet generative models of neural time series have largely remained restri...

Read full article:
https://arxiv.org/abs/2606.11833

Cell-type-specific modularity in E/I networks shapes dynamical regimes and computational capabilities

Emmanouil Giannakakis, Victor Buend\'ia, Oleg Vinogradov, Anna Levina

2026-11-05 1 min

149 words

ARXIV QBIO NC

Summary: arXiv:2611.06720v1 Announce Type: cross Abstract: Structural modularity is a ubiquitous feature of brain networks that shapes neural dynamics and information processing. Yet excitatory and inhibitory populations can differ in their modular organization, and it remains unclear how this asymmetry aff...

Read full article:
https://arxiv.org/abs/2611.06720

PINR-ESI: Physics-Informed Self-Supervised Implicit Neural Representation Framework for EEG Extended-Source Imaging

👤 Zhiyong Zhou, Sheng Liao, Youshen Xiao, Xiaolin Li, Yakang Dai

📅 2026-11-05 📖 1 min

📝 176 words

ARXIV QBIO NC

Summary: arXiv:2611.06700v1 Announce Type: cross Abstract: EEG extended-source imaging aims to localize active cortical regions and characterize their spatial extent, yet supervised inverse networks typically depend on synthetic EEG-source pairs and may be sensitive to synthetic-to-real mismatch. We propose...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06700

Time-Division Multiplexing: The Operational Rhythm and Temporal Organization Layer of Digital Life

👤 Haibin Tong, Luming Guo, Nana Cui, Jiang Yu

📅 2026-11-05 📖 1 min 📝 198 words

ARXIV QBIO NC

Summary: arXiv:2611.06603v1 Announce Type: cross Abstract: Continual agents must handle urgent interaction while retaining alternatives, replaying errors, and committing safe long-term changes. A previous digital-life framework distinguishes online (busy) and offline (idle) channels, but leaves open whether...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06603

Learning Time-Varying Connectomes for Stable Behavior Across Development

👤 Christof Fehrman, Jacob J. Morra, Eva A. Naumann

📅 2026-11-05 📖 1 min 📝 232 words

ARXIV QBIO NC

Summary: arXiv:2611.06895v1 Announce Type: new Abstract: Neural circuits must maintain functionality to drive effective behavior while their size, topology, geometry, and intrinsic dynamics change across development. To model developing visuomotor circuits and behavior, we investigated whether a single comp...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06895

Synchronized Hearts, Desynchronized Brains: A Nonlinear Dynamics Perspective for Function and Failure

👤 Irene Pellini, Simon Bauer, Seba Contreras, Ruben Fern\'andez Busnadiego, Anushka Deb, Sophia Hartmann, Sayedeh Hussaini, D\"orte M. Katschinski, Sarah K\"oster, Stephan E. Lehnart, Tobias Moser, Andreas Neef, Ulrich Parlitz, Silvio Rizzoli, Maria-Patapia Zafeiriou, Wolfram-Hubertus Zimmermann, Johannes Zierenberg, Philip Bittihn, Viola Priesemann

📅 2026-11-05 📖 1 min 📝 199 words

ARXIV QBIO NC

Summary: arXiv:2611.06618v1 Announce Type: new Abstract: Both the heart and the brain are built from electrically excitable cells, yet the emergent dynamics of these organs could not differ more. In the myocardium, a coherent traveling wave of action potentials ensures coordinated contraction; in the cerebr...

🔗 **Read full article:**
https://arxiv.org/abs/2611.06618

Mind Control of LLMs via Closed-Loop Stimulation: Out-of-Distribution Robustness

Kasra Fallah, Rudramani Singha, Hannah Ghanei, Paul Sajda, James Anderson, Erfan Zabeh

2026-11-05 1 min 263 words ARXIV QBIO NC

Summary: arXiv:2611.06418v1 Announce Type: new Abstract: Brains and large language models need not be considered equivalent systems to reveal an interpretable commonality: both exhibit high-dimensional, nonlinear dynamics whose behavior can only be modeled imperfectly. This creates a shared challenge whenev...

Read full article:
https://arxiv.org/abs/2611.06418

A minimal E/I circuit model explains differential recovery from amblyopia under TTX and reverse occlusion

Lawrence Udeigwe

2026-11-05 1 min 210 words ARXIV QBIO NC

Summary: arXiv:2611.06189v1 Announce Type: new Abstract: Recovery from amblyopia beyond the critical period remains a major challenge in visual neuroscience. Experimental studies show that transient inactivation of the fellow eye with tetrodotoxin (TTX) produces stronger and more persistent recovery than re...

Read full article:
https://arxiv.org/abs/2611.06189

Restorative urban soundscapes

📅 2026-11-05 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s44384-026-00080-7

Evaluation of an audio-based monitoring system for detecting retching and vomiting in gastroparesis: an observational cohort study

📅 2026-11-05 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s44355-026-00086-8

Psilocybin modulates nucleus accumbens activity via astrocytic transcriptional regulation

📅 2026-11-05 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s12276-026-01872-3

Modular control of gait under visual deprivation using muscle-mode UCM analysis of COM and foot trajectory

📅 2026-11-05 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-79052-6

Epigenetic modification of neurotransmitter pathways during fasting: a systematic review

👤 Majd A. AbuAlrob 📅 2026-11-05 📖 1 min 📝 293 words FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundDietary restriction, encompassing intermittent fasting and caloric restriction, represents a prominent non-pharmacological approach to modulating brain aging and neuronal resilience. Emerging evidence links metabolism to long-term neural plasticity through epigenetic mechanisms, yet findin...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1969439

Dynamical disorders: synchronization dynamics in complex emotional conditions

S. S. Høgenhaug 2026-11-05 1 min 207 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionMental health conditions present a particular challenge to categorical nosology. We propose that such conditions are better understood as Dynamical Disorders, defined not only by symptom cluster profiles but by underlying disrupted patterns of coordination across bio-psycho-social scales...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1878289

Path planning algorithm based on pulse-coupled neural network with dual constraints of heading-deviation and node penalty

Xiaoyan Wang 2026-11-05 1 min 321 words FRONTIERS NEUROROBOTICS

Summary: Path planning for mobile robots is a crucial aspect of robot navigation. Traditional search algorithms are limited by heuristic design and search strategies, often struggling to balance search efficiency and path quality in intricate obstacle environments. The Pulse-Coupled Neural Network (PCNN) off...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1907089

Research progress and prospects of cochlear implant technology

Xiang Yao 2026-11-05 1 min 251 words FRONTIERS NEUROSCIENCE

Summary: Traditional cochlear implant systems depend on external devices and face challenges related to convenience, esthetics, and auditory performance in complex soundscapes. Recently, research has shifted from “functional substitution” to “functional enhancement” and “biological fusion.” However, most eme...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2002803

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt 2025-01-09 1 min 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

> **Summary:** Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

> **Summary:** Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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https://pubmed.ncbi.nlm.nih.gov/41676644/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1BUB2BG5RbxOblm-hBbiJWEhGG43qlVrvGNHOTqBKva9wWrItM&fc=None&ff=20261105040318&v=2.20.1

Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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A review of deep learning applications on histopathologic and genomic data in focal segmental glomerulosclerosis

Sasanko Sekhar Gantayat 2026-11-03 1 min 66 words LOW VISION

Summary: Focal segmental glomerulosclerosis explains the histological pattern of injury seen through light microscopy in kidney biopsy specimens and is the end result of different injuries to podocytes. The clinical classification of patients by FSGS lesion is challenging because of a broad spectrum of etiol...

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Effects of balance training with electrical muscle stimulation on ankle joint muscle on sensory reweighting and motor strategy

Naoya Hasegawa 2026-11-03 1 min 60 words LOW VISION

Summary: CONCLUSIONS: EMS-based balance training was associated with the induction of visual reweighting dynamics and adaptive postural strategies involving multiple joints. Specifically, EMS-induced involuntary ankle assistance elicited hip-centered adjustments in postural control, contributing to upper-tru...

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Reproxalap 0.25% in dry eye disease: A meta-analysis of randomized controlled trials

Hashem Abu Serhan 2026-11-04 1 min 62 words LOW VISION

Summary: CONCLUSION: Reproxalap was associated with consistent improvements across key clinical outcomes in DED. Exploratory allergic conjunctivitis findings should be considered hypothesis-generating rather than clinically conclusive. Interpretation of the overall evidence is limited by exclusive industry s...

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State-of-the-art in automated image-based species identification systems

Adrian Egli 2026-11-04 1 min 64 words LOW VISION

Summary: Accurate and rapid species-level identification of bacteria and fungi from culture plates are longstanding challenges in diagnostic microbiology. Visual assessment of colony morphology lacks sufficient specificity for definitive identification, while biochemical profiling is often too slow for routi...

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What we can learn from nature to restore human vision

Boris Baziyan 2026-11-04 1 min 72 words LOW VISION

Summary: Irreversible vision loss, affecting millions of people worldwide, most often ends in the death of retinal neurons that the human eye cannot replace. Across the tree of life, however, evolution has repeatedly produced strategies that address the core problems of vision restoration: zebrafish and sala...

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Adverse Events Associated with Trabecular Meshwork Bypass Devices from 2020-2025

👤 Dilru C Amarasekera 📅 2026-11-04 📖 1 min 📝 54 words LOW VISION

Summary: CONCLUSIONS: A low number of AEs were submitted for HM, ISW, and ISI in MAUDE from 2020-2025. Common complications were stent malposition, inflammation, break, and bleeding, with more frequent inflammation with HM and injector breaks with ISW. Similar AE numbers were described for HM and ISW, with a...

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Treatment-induced changes in higher-order aberrations after ray-tracing-guided vs. wavefront-optimized laser in situ keratomileusis

👤 Ramin Khoramnia 📅 2026-11-04 📖 1 min 📝 43 words LOW VISION

Summary: CONCLUSIONS: Wavefront-optimized LASIK increased total HOAs, spherical aberration and coma, while no increase in HOAs was detected after wavelight plus. Spherical aberration decreased (negative change) after wavelight plus. The benefit of ray-tracing-guided LASIK tended to be larger in patients with...

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MindExplore+: Towards Long-Horizon Vision-Language-Action System

👤 Yunsong Li 📅 2026-11-04 📖 1 min 📝 59 words LOW VISION

Summary: Vision-Language-Action (VLA) is crucial for autonomous decision-making in embodied systems. While current methods have advanced single-skill abilities, their short-horizon capability limits applicability in real-world scenarios. To address this challenge, we propose MindExplore+, a hierarchical VLA ...

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Participant-Level Cross-Fit Ensemble Disagreement for Reliability-Aware Vision-Based Infant Respiratory-Rate Estimation

👤 Hanbeom Yoon 📅 2026-11-04 📖 1 min 📝 62 words LOW VISION

Summary: Contactless video-based respiratory-rate (RR) estimates may be unreliable for individual infant recordings even when overall accuracy is acceptable. We examined whether disagreement among participant-separated models could rank estimation reliability in public AIR-125/AIR-400 data (15 infants, 374 a...

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Retinoblastoma

Helen Dimaras 2026-11-04 1 min 69 words LOW VISION

Summary: Retinoblastoma, the most common primary intraocular malignancy of childhood, represents a substantial global health equity concern, exemplifying profound disparities in global burden and outcome despite being a largely curable cancer. Retinoblastoma is thought to arise from cone photoreceptor precur...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

Paul W Hodges 2026-07-30 1 min 53 words TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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What are the neural networks that have been modulated by anatomically targeted therapies to promote recovery of consciousness? A scoping review

👤 Konstantinos Margetis 📅 2026-11-03 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Disorders of Consciousness (DoC) arise from insults to neural networks responsible for awareness and arousal. DoC subtypes are frequently misdiagnosed, as mechanisms underlying consciousness remain poorly understood. Neurostimulation has been investigated as a potential therapy for DoC recovery; how...

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Effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field

Prinpida Sonthiphand 2026-11-03 1 min 68 words TDCS TACS TRNS

Summary: Due to climate change and increasing food demand, conventional agriculture relies heavily on chemical fertilizers. This study investigated the effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field. Metag...

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Transcranial Pulse Stimulation in Parkinson's Disease: A Systematic Review

Marcelo Lourenço da Silva 2026-11-03 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: TPS shows preliminary potential to modulate sensorimotor networks and improve selected motor outcomes in PD. However, evidence remains insufficient to establish overall clinical efficacy. Larger, rigorously controlled trials are required to determine sustained benefits, optimal stimulat...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

👤 Jinwu Wang 📅 2026-11-03 📖 1 min 📝 30 words TDCS TACS TRNS

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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First mitogenome-based phylogenetic analysis and divergence time estimation of deep-sea elasipodid sea cucumbers (Holothuroidea: Elasipodida)

👤 Haibin Zhang 📅 2026-11-03 📖 1 min 📝 29 words TDCS TACS TRNS

Summary: CONCLUSIONS: Collectively, this study establishes a foundational mitogenomic dataset and provide a robust scientific foundation for future studies on the temporal divergence and evolutionary history of elasipodid sea cucumbers.

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The application of transcranial magnetic stimulation in treatment-resistant depression: insights from omics

Tianhao Bao 2026-11-04 1 min 59 words TDCS TACS TRNS

Summary: Treatment-resistant depression (TRD) is a refractory form of major depressive disorder (MDD) for which conventional antidepressant strategies are often insufficient. Transcranial magnetic stimulation (TMS) has demonstrated clinical efficacy in TRD, but its biological mechanisms remain incompletely u...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

Xiaolong Liu 2026-11-04 1 min 34 words TDCS TACS TRNS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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A Neuro-endocrine-immune Network Perspective on Transcranial Direct Current Stimulation for Metabolic Dysregulation in Schizophrenia

Jianhua Chen
2026-11-04 1 min 61 words TDCS TACS TRNS

Summary: Individuals with schizophrenia (SCZ) frequently exhibit glycolipid metabolic disorders. This clinical issue cannot be fully explained by the high metabolic risk associated with antipsychotic medications alone. It is increasingly recognized to involve complex multi-system interactions, particularly t...

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Effects of Different Stimulation Parameters on the Efficacy of Transcranial Direct Current Stimulation for Post-Stroke Dysphagia: An Updated Systematic Review and Meta-Analysis of Randomized Controlled Trials

👤 Yuan Lin
📅 2026-11-04 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS yields favorable effects on swallowing function in patients with PSD when administered to the lesional hemisphere, the contralesional hemisphere, or bilaterally, and there are no differences in efficacy among the three stimulation modes. An intensity of 1.8 mA represents a potential...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 68 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Correction: EBA-related body-preferring responses in the occipitotemporal cortex measured with functional near-infrared spectroscopy

Min Wang 2026-11-03 1 min 6 words FNIRS

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1914488.].

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Supportive parenting moderates the association between neural synchrony and irritability: Evidence from mother-son dyads

👤 Xiuyun Lin 📅 2026-11-03 📖 1 min 📝 62 words FNIRS

Summary: While research on parent-child neural synchrony using fNIRS hyperscanning has expanded, its association with child psychopathology, especially transdiagnostic symptoms such as irritability, remains poorly understood. Moreover, no prior studies have explored the potential contextual factors that may ...

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An fNIRS investigation of cortical responses to expressed emotion in bipolar disorder

👤 Lam Ghai Lim 📅 2026-11-03 📖 1 min 📝 49 words FNIRS

Summary: CONCLUSION: The findings indicate region- and context-specific differences in cortical hemodynamic responses and functional coupling in individuals with BD during EE processing. These fNIRS-derived measures may serve as candidate features for further investigation in larger and independent studies, ...

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Training Interbrain Synchrony through Dyadic Neurofeedback Modulates Empathy-for-Pain

👤 Simone Shamay-Tsoory
📅 2026-11-03 📖 1 min 📝 63 words FNIRS

Summary: Dyadic neurofeedback offers a socially grounded approach for modulating neural processes that unfold between interacting individuals. While recent work demonstrates that dyadic neurofeedback can enhance inter-brain synchrony during interaction, whether its effects extend beyond the trained dyad rema...

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Familial risk for attention-deficit/hyperactivity disorder and related behaviors in toddlerhood are preliminarily associated with altered frontoparietal development in infancy measured by functional near-infrared spectroscopy

👤 Heather M Joseph
📅 2026-11-04 📖 1 min 📝 66 words FNIRS

Summary: INTRODUCTION: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common childhood neurodevelopmental disorder associated with impairments across the lifespan. Children with ADHD have been found to have delayed brain maturation. These delays may be present and detectable with functional near-infrar...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

👤 Xiaolong Liu
📅 2026-11-04 📖 1 min 📝 34 words FNIRS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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The application of brain-computer interface on cognitive function and activities of daily living in stroke patients: a systematic review and meta-analysis

Xuan Gong 2026-11-03 1 min 52 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-based training shows preliminary promise for improving cognitive function and ADL in stroke patients. Current evidence remains insufficient to recommend BCI for emotion or speech rehabilitation. However, the GRADE certainty of evidence was rated as low for cognitive function and very...

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Mapping Brain-Computer Interface Research in Depression (1995-2026): A Bibliometric Analysis

Xianlun Duan 2026-11-03 1 min 31 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The field has evolved from EEG-based characterization toward network-level and real-time neurofeedback research. These findings identify emerging research directions while highlighting the need for further methodological standardization and clinical validation.

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A comprehensive review of biosignal foundation models

Stefanos Zafeiriou 2026-11-03 1 min 71 words BRAIN COMPUTER INTERFACE

Summary: Foundation models are emerging as a promising approach to biosignal analysis, offering solutions to common challenges in learning from biosignal data, such as high variability, noise or limited annotations. This survey provides a comprehensive review of the state-of-the-art in this rapidly advancing...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

Jinwu Wang 2026-11-03 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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Learning-dependent hippocampal-striatal reorganization during sequential decision-making in pigeons

Zhigang Shang 2026-11-03 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Adaptive sequential decision-making requires animals to learn how actions shape future states and rewards while adjusting behavioral strategies as experience accumulates. Although the hippocampus is implicated in state-structure representation and the striatum in value learning, how learning strateg...

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taVNS for uterine contraction pain and early recovery after spontaneous vaginal birth: a three-arm randomized controlled trial protocol

Tai-Pin Guo 2026-11-03 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: BACKGROUND: Uterine contraction pain after spontaneous vaginal birth (SVB) hinders early maternal recovery. Transcutaneous auricular vagus nerve stimulation (taVNS), as a safe, and non-invasive alternative therapy, can alleviate pain and improve sleep and emotion. Although a recent trial has confirm...

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Engineering Neural Interface Materials for Brain-Computer Interfaces: Toward Reliable Neural Information and Functional Recovery

👤 Jie Wu 📅 2026-11-04 📖 1 min 📝 63 words BRAIN COMPUTER INTERFACE

Summary: Motor dysfunction is a major disabling outcome of neurological disorders, including stroke, spinal cord injury, and cerebral palsy. By establishing direct communication between neural activity and external effectors, brain-computer interfaces (BCIs) offer a promising strategy for neural functional r...

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A Comparison Between Movement-Related Brain Activity Modalities for Increasing Corticospinal Excitability Using an Associative Brain-Computer Interface

👤 Mads Jochumsen 📅 2026-11-04 📖 1 min 📝 39 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: An associative BCI using either motor execution or motor imagery effectively increases corticospinal excitability. These findings suggest that both forms of efferent activity are viable, allowing flexibility in selecting the modality based on user capability or clinical context.

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A high robustness decoding technique for ultrasound modulated electroencephalogram

👤 Dong Ming 📅 2026-11-04 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Ultrasound-modulated electroencephalography (USMEEG) has become a vital tool in neuroscience because it integrates millimeter-level spatial resolution with millisecond-level temporal resolution. However, under extremely complex scenarios, excessively strong or weak acoustoelectric signal coupling le...

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Augmented reality paradigm and EA-DGCNN decoding algorithm for motor imagery in stroke patients

👤 Tzyy-Ping Jung 📅 2026-11-04 📖 1 min 📝 64 words BRAIN COMPUTER INTERFACE

Summary: OBJECTIVE: Stroke represents one of the leading causes of long-term motor dysfunction on a global scale. Existing motor imagery (MI) brain-computer interface (BCI) rehabilitation training systems face critical challenges of suboptimal patient engagement and limited electroencephalography (EEG) featu...

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Lemmings CSS Animation

👤 MarkBlah 📅 2026-11-05 📖 1 min 📝 13 words HACKER NEWS

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Altera Announces Agilex 2X, a Complete FPGA and SoC Portfolio

👤 gmays 📅 2026-11-05 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.altera.com/newsroom/agilex-2x Comments URL: https://news.ycombinator.com/item?id=50375653 Points: 21 # Comments: 9

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Want to Rent an Internet Address at the White House?

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> **Summary:** Article URL: https://krebsonsecurity.com/2026/11/want-to-rent-an-internet-address-at-the-white-house/ Comments URL: <a href="https://news.ycombinator.com/item?id=50375728">https://news.yc...

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T-MRB Call for Nominations for Associate Editors

👤 Deidre Artis 📅 2026-04-22 📖 1 min 📝 16 words EMBS

> **Summary:** The post T-MRB Call for Nominations for Associate Editors appeared first on IEEE EMBS.

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https://www.embs.org/t-mrb-call-for-nominations-ae/#new_tab

2026 Adcom Petition is Open

👤 Deidre Artis 📅 2026-09-11 📖 1 min 📝 14 words EMBS

> **Summary:** The post 2026 Adcom Petition is Open appeared first on IEEE EMBS.

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https://www.embs.org/about/embs-officers/2026-embs-administrative-committee-adcom-election-2/#new_tab

Analysis of the dynamic evolution and immunosuppressive functions of myeloid cells in gliomas based on single-cell transcriptomics

📖 1 min 📝 23 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Hao Yuan, Jingyi Yang, Zili Bai, Hongyan Wang, Jiaxuan Tong, Peng Chen, Shouping Gong

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S000689932600466X?dgcid=rss_sd_all

A red fluorescent sensor for intensity-and lifetime imaging of cAMP in living cells

👤 Ivanicic, M., Tiroshi, L., Gourinchas, G., Adachi, K., Yasuda, R., Griesbeck, O., Dombeck, D. A.

📅 2026-11-04 📖 1 min 📝 123 words BIORXIV NEUROSCIENCE

> **Summary:** cAMP (cyclic adenosine monophosphate) is a crucial intracellular messenger involved in neuromodulation and plasticity, but more information on its spatiotemporal regulation is desirable. Here we constructed a red fluorescent indicator based on a truncated cAMP binding site derived from a regulatory ...

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https://www.biorxiv.org/content/10.64898/2026.11.04.764060v1?rss=1

Calcium permeability–enhancing mutations in GluRIIC suppress sustained presynaptic homeostatic plasticity triggered by loss of GluRIIA

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📅 2026-10-27 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAccumulating experimental evidence indicates that neurons adjust their synaptic strength, in response to genetic or environmental/ pharmacological perturbations, to ensure stability in neural circu...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2625954123?af=R

VAP organizes presynaptic Ca2+ signaling and synaptic vesicle mobilization through Kv2 membrane contact sites

👤 Cameron D. PatonIn Ha ChoMichael B. Hoppaahttps://ror.org/049s0rh22Division of Arts and Sciences, Department of Biological Sciences, Dartmouth College, Hanover, NH 03755

📅 2026-10-27 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceThe endoplasmic reticulum (ER) supports neuronal function by regulating Ca2+, lipid, and protein homeostasis, yet how ER–plasma membrane contact sites contribute to synaptic transmission remains p...

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https://www.pnas.org/doi/abs/10.1073/pnas.2602106123?af=R

Intraventricular clindamycin enhances interstitial brain fluid drainage and the blood–brain barrier after a traumatic brain injury

👤 Landon S. AshbyMegan G. CroweDaveraj ChakravartySanjeev MedamGabriel FramptonAyesha B. DasDavid A. HormuthThomas E. YankeelovVincent Y. C. WangLinda J. Noble-HaeussleinBenjamin J. Umlaufahttps://ror.org/00hj54h04Department of Neurosurgery, The University of Texas at Austin, Austin, TX 78712bhttps://ror.org/00hj54h04Department of Biomedical Engineering, The University of Texas at Austin, Austin, TX 78712chttps://ror.org/00hj54h04Oden Institute for Computational Engineering and Sciences, The University of Texas at Austin, Austin, TX 78712dhttps://ror.org/00hj54h04Cancer Research Center, The University of Texas at Austin, Austin, TX 78712ehttps://ror.org/00hj54h04Department of Diagnostic Medicine, The University of Texas at Austin, Austin, TX 78712fDepartment of Imaging Physics, The University of Texas M.D. Anderson Cancer Center, Houston, TX 77030ghttps://ror.org/00hj54h04Department of Neurology, The University of Texas at Austin, Austin, TX 78712hhttps://ror.org/00hj54h04Department of Psychology, The University of Texas at Austin, Austin, TX 78712ihttps://ror.org/00hj54h04Mulva Clinic for the Neurosciences, The University of Texas at Austin, Austin, TX 78712

📅 2026-10-27 📖 1 min 📝 44 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDespite advances in the acute clinical management of severe traumatic brain injuries, treating injury-derived secondary pathogenesis remains a major challenge. We describe an alternative strategy ...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2536755123?af=R

Digital health meets Parkinson’s disease: a bibliometric exploration of wearable sensors research and emerging frontiers (2005–2026)

Lan Xiong 2026-10-08 1 min 228 words FRONTIERS NEUROINFORMATICS

Summary: BackgroundWearable and sensor-based monitoring technologies are increasingly used to support objective, longitudinal assessment in Parkinson’s disease (PD). However, the global development, knowledge structure, and emerging research frontiers of this interdisciplinary field remain incompletely mappe...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1937585

Bite-block effects on native s/ʃ and English r/l in child speakers of a language with/without this contrast

F. Sayako Earle 2026-11-02 1 min 171 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Learning to produce non-native speech contrasts requires developing new phonological categories as well as new articulatory-perceptual mappings for these new categories. However, within the same speakers, it remains unclear how these new categories compare to contrasts that are already fully establi...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1949253

From biomechanical data to robotic execution: design and control of an intelligent Tuina system

Wei Yao
2026-11-02 1 min 181 words FRONTIERS NEUROROBOTICS

Summary: IntroductionTraditional Chinese medicine (TCM) Tuina (massage) relies heavily on practitioners' physical stamina and subjective experience, leading to operational variability that hinders therapeutic standardization and scalable rehabilitation service delivery. To address this limitation, this study...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1937295

Multimodal regulation of neurite extension by extracellular matrix composition and electrical stimulation on aligned collagen fiber scaffolds

Madalynn J Thompson, Joanna Borowska-Fielding, Ivy A Murphy, Ying Zhang, Victor Rafuse and John P Frampton

2026-10-26 1 min 279 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Peripheral nerve regeneration is regulated by structural, biochemical, and electrical cues, yet their combined effects on neurite outgrowth remain unclear. In this study, an aligned fibrous scaffold was developed to evaluate how extracellular matrix (ECM) composition and electrical stimul...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb32c

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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An open-source eye-tracking system adaptable to diverse terrestrial vertebrates reveals state-dependent visual acquisition in lizards

Mark Shein-Idelson 2026-11-03 1 min 68 words LOW VISION

Summary: Gaze shifts are fundamental to visual exploration and processing, yet empirical knowledge is largely confined to a small number of model species, limiting our understanding of general principles of visual acquisition across animals. Here, we introduce a versatile, low-cost, and open-source framework...

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Ocular Considerations for the Department of War's New Testosterone Initiative

Harkaran S Rana 2026-11-03 1 min 75 words LOW VISION

Summary: On 15 July 2026, the Secretary of War announced that the Department of War will begin annual testosterone deficiency screening for male Active Duty and Reserve Component personnel aged 30 and older as part of the periodic health assessment, with screening available on request to younger servicemembe...

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A review of deep learning applications on histopathologic and genomic data in focal segmental glomerulosclerosis

Sasanko Sekhar Gantayat 2026-11-03 1 min 66 words LOW VISION

> **Summary:** Focal segmental glomerulosclerosis explains the histological pattern of injury seen through light microscopy in kidney biopsy specimens and is the end result of different injuries to podocytes. The clinical classification of patients by FSGS lesion is challenging because of a broad spectrum of etiol...

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Effects of balance training with electrical muscle stimulation on ankle joint muscle on sensory reweighting and motor strategy

Naoya Hasegawa 2026-11-03 1 min 60 words LOW VISION

> **Summary:** CONCLUSIONS: EMS-based balance training was associated with the induction of visual reweighting dynamics and adaptive postural strategies involving multiple joints. Specifically, EMS-induced involuntary ankle assistance elicited hip-centered adjustments in postural control, contributing to upper-tru...

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Reproxalap 0.25% in dry eye disease: A meta-analysis of randomized controlled trials

Hashem Abu Serhan 2026-11-04 1 min 62 words LOW VISION

Summary: CONCLUSION: Reproxalap was associated with consistent improvements across key clinical outcomes in DED. Exploratory allergic conjunctivitis findings should be considered hypothesis-generating rather than clinically conclusive. Interpretation of the overall evidence is limited by exclusive industry s...

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State-of-the-art in automated image-based species identification systems

Adrian Egli 2026-11-04 1 min 64 words LOW VISION

Summary: Accurate and rapid species-level identification of bacteria and fungi from culture plates are longstanding challenges in diagnostic microbiology. Visual assessment of colony morphology lacks sufficient specificity for definitive identification, while biochemical profiling is often too slow for routi...

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What we can learn from nature to restore human vision

Boris Baziyan 2026-11-04 1 min 72 words LOW VISION

Summary: Irreversible vision loss, affecting millions of people worldwide, most often ends in the death of retinal neurons that the human eye cannot replace. Across the tree of life, however, evolution has repeatedly produced strategies that address the core problems of vision restoration: zebrafish and sala...

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Adverse Events Associated with Trabecular Meshwork Bypass Devices from 2020-2025

Dilru C Amarasekera 2026-11-04 1 min 54 words LOW VISION

Summary: CONCLUSIONS: A low number of AEs were submitted for HM, ISW, and ISI in MAUDE from 2020-2025. Common complications were stent malposition, inflammation, break, and bleeding, with more frequent inflammation with HM and injector breaks with ISW. Similar AE numbers were described for HM and ISW, with a...

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Treatment-induced changes in higher-order aberrations after ray-tracing-guided vs. wavefront-optimized laser in situ keratomileusis

Ramin Khoramnia 2026-11-04 1 min 43 words LOW VISION

Summary: CONCLUSIONS: Wavefront-optimized LASIK increased total HOAs, spherical aberration and coma, while no increase in HOAs was detected after wavelight plus. Spherical aberration decreased (negative change) after wavelight plus. The benefit of ray-tracing-guided LASIK tended to be larger in patients with...

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MindExplore+: Towards Long-Horizon Vision-Language-Action System

Yunsong Li 2026-11-04 1 min 59 words LOW VISION

Summary: Vision-Language-Action (VLA) is crucial for autonomous decision-making in embodied systems. While current methods have advanced single-skill abilities, their short-horizon capability limits applicability in real-world scenarios. To address this challenge, we propose MindExplore+, a hierarchical VLA ...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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What are the neural networks that have been modulated by anatomically targeted therapies to promote recovery of consciousness? A scoping review

👤 Konstantinos Margetis 📅 2026-11-03 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Disorders of Consciousness (DoC) arise from insults to neural networks responsible for awareness and arousal. DoC subtypes are frequently misdiagnosed, as mechanisms underlying consciousness remain poorly understood. Neurostimulation has been investigated as a potential therapy for DoC recovery; how...

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Effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field

Prinpida Sonthiphand 2026-11-03 1 min 68 words TDCS TACS TRNS

Summary: Due to climate change and increasing food demand, conventional agriculture relies heavily on chemical fertilizers. This study investigated the effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field. Metag...

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Transcranial Pulse Stimulation in Parkinson's Disease: A Systematic Review

Marcelo Lourenço da Silva 2026-11-03 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: TPS shows preliminary potential to modulate sensorimotor networks and improve selected motor outcomes in PD. However, evidence remains insufficient to establish overall clinical efficacy. Larger, rigorously controlled trials are required to determine sustained benefits, optimal stimulat...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

👤 Jinwu Wang 📅 2026-11-03 📖 1 min 📝 30 words TDCS TACS TRNS

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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First mitogenome-based phylogenetic analysis and divergence time estimation of deep-sea elasipodid sea cucumbers (Holothuroidea: Elasipodida)

👤 Haibin Zhang 📅 2026-11-03 📖 1 min 📝 29 words TDCS TACS TRNS

Summary: CONCLUSIONS: Collectively, this study establishes a foundational mitogenomic dataset and provide a robust scientific foundation for future studies on the temporal divergence and evolutionary history of elasipodid sea cucumbers.

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The application of transcranial magnetic stimulation in treatment-resistant depression: insights from omics

Tianhao Bao 2026-11-04 1 min 59 words TDCS TACS TRNS

Summary: Treatment-resistant depression (TRD) is a refractory form of major depressive disorder (MDD) for which conventional antidepressant strategies are often insufficient. Transcranial magnetic stimulation (TMS) has demonstrated clinical efficacy in TRD, but its biological mechanisms remain incompletely u...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

Xiaolong Liu 2026-11-04 1 min 34 words TDCS TACS TRNS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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A Neuro-endocrine-immune Network Perspective on Transcranial Direct Current Stimulation for Metabolic Dysregulation in Schizophrenia

Jianhua Chen
2026-11-04 1 min 61 words TDCS TACS TRNS

Summary: Individuals with schizophrenia (SCZ) frequently exhibit glycolipid metabolic disorders. This clinical issue cannot be fully explained by the high metabolic risk associated with antipsychotic medications alone. It is increasingly recognized to involve complex multi-system interactions, particularly t...

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Effects of Different Stimulation Parameters on the Efficacy of Transcranial Direct Current Stimulation for Post-Stroke Dysphagia: An Updated Systematic Review and Meta-Analysis of Randomized Controlled Trials

👤 Yuan Lin
📅 2026-11-04 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS yields favorable effects on swallowing function in patients with PSD when administered to the lesional hemisphere, the contralesional hemisphere, or bilaterally, and there are no differences in efficacy among the three stimulation modes. An intensity of 1.8 mA represents a potential...

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Targeting cognitive control and affective processing in aggression: neuromodulation approaches in CRC/TRR 379

👤 Jana Zweerings 📅 2026-11-04 📖 1 min 📝 63 words TDCS TACS TRNS

Summary: Interventions aimed at reducing aggression in mental disorders remain insufficiently addressed by current research, despite the substantial individual and societal burden associated with aggressive behavior. Consequently, strategies for modulating aggressive behavior are of high clinical relevance b...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 68 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Correction: EBA-related body-preferring responses in the occipitotemporal cortex measured with functional near-infrared spectroscopy

Min Wang 2026-11-03 1 min 6 words FNIRS

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1914488.].

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Supportive parenting moderates the association between neural synchrony and irritability: Evidence from mother-son dyads

Xiuyun Lin 2026-11-03 1 min 62 words FNIRS

Summary: While research on parent-child neural synchrony using fNIRS hyperscanning has expanded, its association with child psychopathology, especially transdiagnostic symptoms such as irritability, remains poorly understood. Moreover, no prior studies have explored the potential contextual factors that may ...

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An fNIRS investigation of cortical responses to expressed emotion in bipolar disorder

Lam Ghai Lim 2026-11-03 1 min 49 words FNIRS

Summary: CONCLUSION: The findings indicate region- and context-specific differences in cortical hemodynamic responses and functional coupling in individuals with BD during EE processing. These fNIRS-derived measures may serve as candidate features for further investigation in larger and independent studies, ...

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Training Interbrain Synchrony through Dyadic Neurofeedback Modulates Empathy-for-Pain

👤 Simone Shamay-Tsoory
📅 2026-11-03 📖 1 min 📝 63 words FNIRS

Summary: Dyadic neurofeedback offers a socially grounded approach for modulating neural processes that unfold between interacting individuals. While recent work demonstrates that dyadic neurofeedback can enhance inter-brain synchrony during interaction, whether its effects extend beyond the trained dyad rema...

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Familial risk for attention-deficit/hyperactivity disorder and related behaviors in toddlerhood are preliminarily associated with altered frontoparietal development in infancy measured by functional near-infrared spectroscopy

👤 Heather M Joseph
📅 2026-11-04 📖 1 min 📝 66 words FNIRS

Summary: INTRODUCTION: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common childhood neurodevelopmental disorder associated with impairments across the lifespan. Children with ADHD have been found to have delayed brain maturation. These delays may be present and detectable with functional near-infrar...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

Xiaolong Liu
2026-11-04 1 min 34 words FNIRS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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The application of brain-computer interface on cognitive function and activities of daily living in stroke patients: a systematic review and meta-analysis

Xuan Gong 2026-11-03 1 min 52 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-based training shows preliminary promise for improving cognitive function and ADL in stroke patients. Current evidence remains insufficient to recommend BCI for emotion or speech rehabilitation. However, the GRADE certainty of evidence was rated as low for cognitive function and very...

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Mapping Brain-Computer Interface Research in Depression (1995-2026): A Bibliometric Analysis

Xianlun Duan 2026-11-03 1 min 31 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The field has evolved from EEG-based characterization toward network-level and real-time neurofeedback research. These findings identify emerging research directions while highlighting the need for further methodological standardization and clinical validation.

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A comprehensive review of biosignal foundation models

Stefanos Zafeiriou 2026-11-03 1 min 71 words BRAIN COMPUTER INTERFACE

Summary: Foundation models are emerging as a promising approach to biosignal analysis, offering solutions to common challenges in learning from biosignal data, such as high variability, noise or limited annotations. This survey provides a comprehensive review of the state-of-the-art in this rapidly advancing...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

Jinwu Wang 2026-11-03 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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Learning-dependent hippocampal-striatal reorganization during sequential decision-making in pigeons

Zhigang Shang 2026-11-03 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Adaptive sequential decision-making requires animals to learn how actions shape future states and rewards while adjusting behavioral strategies as experience accumulates. Although the hippocampus is implicated in state-structure representation and the striatum in value learning, how learning strateg...

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taVNS for uterine contraction pain and early recovery after spontaneous vaginal birth: a three-arm randomized controlled trial protocol

Tai-Pin Guo 2026-11-03 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: BACKGROUND: Uterine contraction pain after spontaneous vaginal birth (SVB) hinders early maternal recovery. Transcutaneous auricular vagus nerve stimulation (taVNS), as a safe, and non-invasive alternative therapy, can alleviate pain and improve sleep and emotion. Although a recent trial has confirm...

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Engineering Neural Interface Materials for Brain-Computer Interfaces: Toward Reliable Neural Information and Functional Recovery

👤 Jie Wu 📅 2026-11-04 📖 1 min 📝 63 words BRAIN COMPUTER INTERFACE

Summary: Motor dysfunction is a major disabling outcome of neurological disorders, including stroke, spinal cord injury, and cerebral palsy. By establishing direct communication between neural activity and external effectors, brain-computer interfaces (BCIs) offer a promising strategy for neural functional r...

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A Comparison Between Movement-Related Brain Activity Modalities for Increasing Corticospinal Excitability Using an Associative Brain-Computer Interface

👤 Mads Jochumsen 📅 2026-11-04 📖 1 min 📝 39 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: An associative BCI using either motor execution or motor imagery effectively increases corticospinal excitability. These findings suggest that both forms of efferent activity are viable, allowing flexibility in selecting the modality based on user capability or clinical context.

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A high robustness decoding technique for ultrasound modulated electroencephalogram

👤 Dong Ming 📅 2026-11-04 📖 1 min 📝 60 words BRAIN COMPUTER INTERFACE

Summary: Ultrasound-modulated electroencephalography (USMEEG) has become a vital tool in neuroscience because it integrates millimeter-level spatial resolution with millisecond-level temporal resolution. However, under extremely complex scenarios, excessively strong or weak acoustoelectric signal coupling le...

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Implicit learning of melodic structure: A role for pitch?

📅 2024-01-22 📖 1 min 📝 180 words PSYCHOMUSICOLOGY

Summary: Growing evidence suggests that pitch influences musical processing, with melodic processing being enhanced in higher pitch ranges (e.g., Fujioka et al., 2005) and rhythmic processing being enhanced in lower pitches, and these effects may have a basis in elementary properties of the auditory system (...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000303

To those who feel unloved, misunderstood, or afraid

📅 2026-09-22 📖 3 min 📝 687 words FMHY

Summary: I understand the world can feel devoid of love, but I swear, there is a spark within humanity that has the capability to care deeply. To love unconditionally. It doesn't always shine through us all, but it exists, it's real, it's tangible, and no one is exempt from it. We often see it shin...

🔗 **Read full article:**
https://fmhy.net/posts/love

Thursday Daily Thread: Python Careers, Courses, and Furthering Education!

👤 /u/AutoModerator 📅 2026-11-05 📖 1 min 📝 219 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> <h1>Weekly Thread: Professional Use, Jobs, and Education 🏢</h1> <p>Welcome to this week's discussion on Python in the professional world! This is your spot to talk about job hunting, career growth, and educational resources in Python. Please note, this thread is <stron...

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https://www.reddit.com/r/Python/comments/1xo41r2/thursday_daily_thread_python_careers_courses_and/

Show HN: Gauge – Open-Source CLI to recursively improve your docs

👤 0x63_Problems 📅 2026-11-04 📖 2 min 📝 591 words HACKER NEWS

> **Summary:** <p>Hi HN, I’m Evan from Gauge. We’re open-sourcing our CLI to recursively improve your docs for coding agents. The CLI includes a local execution backend, so you don't need an account to try it. Explainer video here: <a href="https://youtu.be/zhLUllKB1Mg" rel="nofollow">https://youtu.be/zhLUllKB1Mg<...

🔗 **Read full article:**
https://github.com/gauge-sh/gauge-cli

Death Comes for XSLT

👤 colejohnson66 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://dbushell.com/2026/11/03/xslt-rip/ Comments URL: https://news.ycombinator.com/item?id=50371408 Points: 43 # Comments: 26

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https://dbushell.com/2026/11/03/xslt-rip/

Hacking the Xbox 360 Hypervisor Part 3: Bad Update Improvements

👤 muricula 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://icode4.coffee/?p=1746 Comments URL: https://news.ycombinator.com/item?id=50372164 Points: 61 # Comments: 2

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https://icode4.coffee/?p=1746

A thousand fewer paths to root

👤 dangelosaurus 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://openai.com/index/a-thousand-fewer-paths-to-root/ Comments URL: https://news.ycombinator.com/item?id=50372929 Points: 77 # Co...

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https://openai.com/index/a-thousand-fewer-paths-to-root/

Images 2.6

👤 davidbarker 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://openai.com/index/introducing-images-2-6/ Comments URL: https://news.ycombinator.com/item?id=50372945 Points: 79 # Comments: 68

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You'll need nerves of steel to work with AI agents

👤 theorchid 📅 2026-11-05 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://orchidfiles.com/youll-need-nerves-of-steel-to-work-with-ai-agents/ Comments URL: <a href="https://news.ycombinator.com/item?id=50373149">https://news.ycombinator.com/item?id=503731...

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Starfall – Affordable, routine access to microgravity

👤 gmays 📅 2026-11-05 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.spacex.com/starfall Comments URL: https://news.ycombinator.com/item?id=50373308 Points: 49 # Comments: 19

🔗 **Read full article:**
https://www.spacex.com/starfall

Type S and M errors as a “rhetorical tool”

noreply@blogger.com (Daniel Lakens)
2025-09-28 17 min 3572 words

TWENTY PERCENT STATISTICIAN

Summary: Update 30/09/2025: I have added a reply by Andrew Gelman below my original blog post. We recently posted a preprint criticizing the idea of Type S and M errors (https://osf.io/2phzb_v1). From our abstract: “While these concepts have been pr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/09/type-s-and-m-errors-as-rhetorical-tool.html

Why we should stop using statistical techniques that have not been adequately vetted by experts in psychology

noreply@blogger.com (Daniel Lakens)
2025-10-29 27 min 5516 words

TWENTY PERCENT STATISTICIAN

Summary: In a recent post on Bluesky, where Richard Morey reflects on a paper he published with Clintin Davis-Stober that points out concerns with the p-curve method (Morey & Davis-Stober, 2025), he writes: <p cla...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/10/why-we-should-stop-using-statistical.html

ERP Decoding for Everyone: Software and Webinar

👤 Steve Luck 📅 2023-06-23 📖 2 min 📝 420 words ERP BOOT CAMP

Summary: You can access the recording h strong>.
You can access the final PDF of the slides <a href="https://ucdavis.box.com/s/f...

🔗 **Read full article:**
https://erpinfo.org/blog/2023/6/23/decoding-webinar

Important Changes to the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-05 📖 2 min 📝 444 words ERP BOOT CAMP

Summary: We are disappointed to announce that we will not be holding a regular 10-day ERP Boot Camp this summer. We have held Boot Camps nearly every summer since 2007, supported by a series of generous grants from NIMH that allowed us to provide scholarships for all attendees. Unf...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/5/changes-to-the-2024-erp-boot-camp

Standardized Measurement Error now available in BrainVision Analyzer and MNE-Python

👤 Steve Luck 📅 2025-12-19 📖 1 min 📝 253 words ERP BOOT CAMP

Summary: <p class="">A few years ago, we developed a universal metric of data quality for averaged ERPs called the standardized measurement error (click here for more information or watch our <a href="https://www.youtube.com/watch?v=tEKsx0p53rs" t...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/12/19/sme-now-available-in-bva-and-mne-python

Transcranial Focused Ultrasound: A Transformative Tool for Intracranial Ablation, Drug Delivery, and Neuromodulation

📅 2025-11-11 📖 1 min 📝 136 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Transcranial focused ultrasound (tFUS) is an emerging neuromodulation and therapeutic technology offering noninvasive, submillimeter precision for targeting deep brain structures. Unlike transcranial magnetic stimulation (TMS) and transcranial electric stimulation (tES), which are limited by depth-f...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11241049

Network Models of Neurodegeneration: Bridging Neuronal Dynamics and Disease Progression

📅 2026-01-16 📖 1 min 📝 196 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Neurodegenerative diseases are characterized by the accumulation of misfolded proteins and widespread disruptions in brain function. Computational modeling has advanced our understanding of these processes, but efforts have traditionally focused on either neuronal dynamics or the biological processe...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11357276

Optogenetics: Pinpoint Light on Precise Neuromodulation

📅 2025-11-06 📖 1 min 📝 137 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Optogenetics has emerged as a pivotal tool in neuroscience, enabling intricate modulation of targeted neurons within the nervous system. Despite its transformative potential, achieving high spatiotemporal resolution in neuromodulation remains a significant challenge, particularly in free-behaving an...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11230544

Emerging Noninvasive Approaches for the Suppression of Pathological Tremor

📅 2025-12-16 📖 1 min 📝 215 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Pathological tremor affects over 40 million people worldwide, significantly impairing daily activities and quality of life. Pharmacological treatments show limited efficacy, with up to 30% discontinuation rates, while surgical interventions like deep brain stimulation achieve significant tremor redu...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11301615

Assistive Trajectory Planning for Lower Limb Exoskeletons: Strategies From Laboratory-Optimized Gait to Environmentally-Adaptive Locomotion Through Multimodal Parameter Awareness

📅 2025-12-30 📖 1 min 📝 182 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Lower limb assistive exoskeletons (LLEs) show great potential to reduce metabolic cost, improve walking performance, and correct abnormal gait patterns. Among their core control architectures, assistive trajectory planning is key to determining system responsiveness and effectiveness under varying l...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11319164

Join the Mailing List

👤 dziura 📅 2025-12-10 📖 1 min 📝 13 words EMBS

> **Summary:** The post Join the Mailing List appeared first on IEEE EMBS.

🔗 **Read full article:**
https://embc.embs.org/2026/#new_tab

EMBS MentorLink: Fostering Careers Together

👤 Deidre Artis 📅 2026-08-18 📖 1 min 📝 14 words EMBS

> **Summary:** The post EMBS MentorLink: Fostering Careers Together appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/embs-mentorlink-fostering-careers-together/

Do neuroimaging data have higher-order interactions?

📖 1 min 📝 16 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Rikkert Hindriks, Fernando Santos, Matteo Neri, Prejaas Tewarie

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006117?dgcid=rss_sd_all

Context-dependent effects of ketamine on emotional evidence integration

👤 Rascu, M., Grohn, J., Costi, S., Zangani, C., Agyapong, S., Smith, K. A., Cowen, P. J., Hunt, L. T., Klein-Flügge, M. C., Weber, L. A.

📅 2026-11-04 📖 1 min 📝 200 words BIORXIV NEUROSCIENCE

Summary: Understanding others' emotions requires continuously integrating noisy affective information as emotional states change over time. Research on emotion processing, however, has thus far focused on static environments or discrete events. Here, we developed a continuous emotional evidence integration t...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.30.762604v1?rss=1

Disrupted mannosidase enzymes in C9orf72 ALS/FTD

Ondatje, B. N., Gittings, L. M., Preller, K. R., Alsop, E., Garcia-Mansfield, K., Hansen, N., Lovell, B., Sharma, R., Piras, I. S., Martinez, M., Meechoovet, B., Bonfitto, A., Song, S., Levy, J., Huentelman, M., Pirrotte, P., Van Keuren-Jensen, K., Sattler, R.

2026-11-04 1 min 116 words BIORXIV NEUROSCIENCE

Summary: Amyotrophic lateral sclerosis (ALS) is an age-related neurodegenerative disorder resulting from the loss of motor neurons in the brain and spinal cord. Neuroinflammation, induced by glial cells including microglia (MG) and astrocytes, is an acknowledged aspect of this neurodegeneration. Generating i...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.30.763135v1?rss=1

Loneliness as a modulator of intersubject neural alignment across social and nonsocial contexts

Jeong, B., Kim, M. J.

2026-11-04 1 min 189 words BIORXIV NEUROSCIENCE

Summary: Loneliness is associated with cognitive biases that diverge from normative patterns of understanding and may be reflected in altered neural responses to the world. Here, we examined how loneliness modulates neural synchrony across individuals during the processing of social and nonsocial information...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.11.03.763905v1?rss=1

Disentangling visual and motor contributions to continuous psychophysics

👤 Pisu, V., Koselevs, A., Lam, C., Mamassian, P., Straub, D., Rothkopf, C. A., Maiello, G.

📅 2026-11-04

📖 1 min 📝 278 words

BIORXIV NEUROSCIENCE

Summary: Continuous psychophysics measures visual function by asking observers to track a randomly moving target. Because continuous responses provide more information than traditional discrete judgements, a few minutes of tracking can estimate visual sensitivity, making the approach attractive for testing c...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.03.763761v1?rss=1

Inosine Triphosphate Pyrophosphatase as a Candidate Associated with White Matter Abnormalities in Chronic Heart Failure

👤 Takeuchi, K., Suzuki, H., Sumiyoshi, A., Kurosawa, R., Sugano, A., Urushihata, T., Arai, M., Nochioka, K., Ito, K., Satoh, A., Nakazawa, T., Yasuda, S.

📅 2026-11-04 📖 1 min 📝 298 words BIORXIV NEUROSCIENCE

Summary: Background: White matter hyperintensities (WMHs) are common with normal brain aging and frequently observed also in chronic heart failure (CHF). However, the relationship between CHF and white matter abnormalities and the biological factors remain unclear. We investigated WMH and associated white ma...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.03.764123v1?rss=1

Monoaminergic changes are associated with impaired neurogenesis along the longitudinal axis of the hippocampus in ageing and dementia

Mathews, K. J., Abdeen, A. H., Mor, D., Cottam, V., Dalton, M. A., Kwok, J., Double, K. L.

2026-11-04 1 min 214 words BIORXIV NEUROSCIENCE

Summary: Neurogenesis persists in the adult human hippocampus and is thought to support memory and emotional regulation. Both normal ageing and Parkinson disease dementia (PDD) are associated with reduced hippocampal neurogenesis, but why this occurs and how this might change across the across the hippocampa...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.03.764096v1?rss=1

An Eyewear-Based EEG Interface for Temporal-Region Monitoring and Emotion Recognition

Li, X., Chai, L., Xu, S., Ni, J., So, K.-F.

2026-11-04 1 min 200 words BIORXIV NEUROSCIENCE

Summary: Conventional electroencephalography (EEG) devices are often impractical for ambulatory settings due to their reliance on specialized hardware and expert operation. To address this, we developed a user-friendly eyewear-based EEG system featuring dry titanium electrodes positioned approximately 5 mm i...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.02.763849v1?rss=1

Locomotor activity enhances environmental representation by boosting visuo-frontal communication

Milton, R., Slapik, M., Egranov, S., Parajuli, A., Franch, M., McConnell, A., Dragoi, V.

2026-11-04

1 min

188 words

BIORXIV NEUROSCIENCE

Summary: During environment exploration we rapidly acquire visual information to identify targets of interest. Here, we discovered that locomotion limits how long natural scenes are viewed dur- ing free exploration. Whether the brain has developed compensatory mechanisms to ensure that sensory information is...

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https://www.biorxiv.org/content/10.64898/2026.11.02.763783v1?rss=1

Unilateral naris occlusion with custom-made plugs for reversible olfactory deprivation in juvenile and adult mice

Dorrego-Rivas, A., Peedikayil-Kurien, S., Huang, L., Stanley, T. C., Galliano, E.

2026-11-04

1 min

138 words

BIORXIV NEUROSCIENCE

Summary: Deprivation paradigms that reduce environmental input have been successfully applied across multiple species and sensory modalities, including vision, audition, somatosensation, and olfaction in order to understand how neurons and circuits respond to loss of sensory input. One notable advantage of s...

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https://www.biorxiv.org/content/10.64898/2026.11.03.763032v1?rss=1

Naturalistic recall preserves temporal specificity despite cortical feature reorganization

👤 Peng, S., Levine, B., Buchsbaum, B. R.
📅 2026-11-04 📖 1 min 📝 142 words

BIORXIV NEUROSCIENCE

Summary: Although sensory input arrives in a largely continuous stream, recall is temporally compressed, discontinuous, and often structured as events. However, the temporal specificity of recall-related neural activity within events, and the separation and cortical organization of recalled features relative...

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Spider snooze stages

📅 2026-11-04 📖 1 min 📝 0 words

NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/d41586-026-03487-0

Aggression experience and observation drive shared behavioural and neural changes

📅 2026-11-04 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41586-026-11217-9

Physiological APP processing is required for synaptic plasticity and cognition

📅 2026-11-04 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41419-026-09378-6

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

Robert Oostenveld 2025-08-13 1 min 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

Suzy J Styles 2025-12-08 1 min 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Intrastromal Antibiotic Therapy for Recalcitrant Bacterial Keratitis

👤 Mark Chehade 📅 2026-11-03 📖 1 min 📝 52 words LOW VISION

Summary: CONCLUSIONS: Adjunctive intrastromal antibiotic therapy achieved a high rate of infection resolution in this cohort of recalcitrant bacterial keratitis, with a low complication rate. This approach may be considered in refractory cases or when unable to utilize conventional topical therapy. However, ...

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An open-source eye-tracking system adaptable to diverse terrestrial vertebrates reveals state-dependent visual acquisition in lizards

👤 Mark Shein-Idelson 📅 2026-11-03 📖 1 min 📝 68 words LOW VISION

Summary: Gaze shifts are fundamental to visual exploration and processing, yet empirical knowledge is largely confined to a small number of model species, limiting our understanding of general principles of visual acquisition across animals. Here, we introduce a versatile, low-cost, and open-source framework...

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Ocular Considerations for the Department of War's New Testosterone Initiative

Harkaran S Rana 2026-11-03 1 min 75 words LOW VISION

Summary: On 15 July 2026, the Secretary of War announced that the Department of War will begin annual testosterone deficiency screening for male Active Duty and Reserve Component personnel aged 30 and older as part of the periodic health assessment, with screening available on request to younger servicemembe...

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A review of deep learning applications on histopathologic and genomic data in focal segmental glomerulosclerosis

Sasanko Sekhar Gantayat 2026-11-03 1 min 66 words LOW VISION

Summary: Focal segmental glomerulosclerosis explains the histological pattern of injury seen through light microscopy in kidney biopsy specimens and is the end result of different injuries to podocytes. The clinical classification of patients by FSGS lesion is challenging because of a broad spectrum of etiol...

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Effects of balance training with electrical muscle stimulation on ankle joint muscle on sensory reweighting and motor strategy

Naoya Hasegawa 2026-11-03 1 min 60 words LOW VISION

Summary: CONCLUSIONS: EMS-based balance training was associated with the induction of visual reweighting dynamics and adaptive postural strategies involving multiple joints. Specifically, EMS-induced involuntary ankle assistance elicited hip-centered adjustments in postural control, contributing to upper-tru...

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Reproxalap 0.25% in dry eye disease: A meta-analysis of randomized controlled trials

Hashem Abu Serhan 2026-11-04 1 min 62 words LOW VISION

Summary: CONCLUSION: Reproxalap was associated with consistent improvements across key clinical outcomes in DED. Exploratory allergic conjunctivitis findings should be considered hypothesis-generating rather than clinically conclusive. Interpretation of the overall evidence is limited by exclusive industry s...

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State-of-the-art in automated image-based species identification systems

Adrian Egli 2026-11-04 1 min 64 words LOW VISION

Summary: Accurate and rapid species-level identification of bacteria and fungi from culture plates are longstanding challenges in diagnostic microbiology. Visual assessment of colony morphology lacks sufficient specificity for definitive identification, while biochemical profiling is often too slow for routi...

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What we can learn from nature to restore human vision

Boris Baziyan 2026-11-04 1 min 72 words LOW VISION

Summary: Irreversible vision loss, affecting millions of people worldwide, most often ends in the death of retinal neurons that the human eye cannot replace. Across the tree of life, however, evolution has repeatedly produced strategies that address the core problems of vision restoration: zebrafish and sala...

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Adverse Events Associated with Trabecular Meshwork Bypass Devices from 2020-2025

👤 Dilru C Amarasekera 📅 2026-11-04 📖 1 min 📝 54 words LOW VISION

Summary: CONCLUSIONS: A low number of AEs were submitted for HM, ISW, and ISI in MAUDE from 2020-2025. Common complications were stent malposition, inflammation, break, and bleeding, with more frequent inflammation with HM and injector breaks with ISW. Similar AE numbers were described for HM and ISW, with a...

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Treatment-induced changes in higher-order aberrations after ray-tracing-guided vs. wavefront-optimized laser in situ keratomileusis

👤 Ramin Khoramnia 📅 2026-11-04 📖 1 min 📝 43 words LOW VISION

Summary: CONCLUSIONS: Wavefront-optimized LASIK increased total HOAs, spherical aberration and coma, while no increase in HOAs was detected after wavelight plus. Spherical aberration decreased (negative change) after wavelight plus. The benefit of ray-tracing-guided LASIK tended to be larger in patients with...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao 📅 2026-11-03 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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What are the neural networks that have been modulated by anatomically targeted therapies to promote recovery of consciousness? A scoping review

👤 Konstantinos Margetis
📅 2026-11-03 📖 1 min 📝 63 words

TDCS TACS TRNS

Summary: Disorders of Consciousness (DoC) arise from insults to neural networks responsible for awareness and arousal. DoC subtypes are frequently misdiagnosed, as mechanisms underlying consciousness remain poorly understood. Neurostimulation has been investigated as a potential therapy for DoC recovery; how...

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Effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field

Prinpida Sonthiphand 2026-11-03 1 min 68 words TDCS TACS TRNS

Summary: Due to climate change and increasing food demand, conventional agriculture relies heavily on chemical fertilizers. This study investigated the effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field. Metag...

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Transcranial Pulse Stimulation in Parkinson's Disease: A Systematic Review

Marcelo Lourenço da Silva 2026-11-03 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: TPS shows preliminary potential to modulate sensorimotor networks and improve selected motor outcomes in PD. However, evidence remains insufficient to establish overall clinical efficacy. Larger, rigorously controlled trials are required to determine sustained benefits, optimal stimulat...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

👤 Jinwu Wang 📅 2026-11-03 📖 1 min 📝 30 words TDCS TACS TRNS

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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First mitogenome-based phylogenetic analysis and divergence time estimation of deep-sea elasipodid sea cucumbers (Holothuroidea: Elasipodida)

👤 Haibin Zhang 📅 2026-11-03 📖 1 min 📝 29 words TDCS TACS TRNS

Summary: CONCLUSIONS: Collectively, this study establishes a foundational mitogenomic dataset and provide a robust scientific foundation for future studies on the temporal divergence and evolutionary history of elasipodid sea cucumbers.

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The application of transcranial magnetic stimulation in treatment-resistant depression: insights from omics

Tianhao Bao 2026-11-04 1 min 59 words TDCS TACS TRNS

Summary: Treatment-resistant depression (TRD) is a refractory form of major depressive disorder (MDD) for which conventional antidepressant strategies are often insufficient. Transcranial magnetic stimulation (TMS) has demonstrated clinical efficacy in TRD, but its biological mechanisms remain incompletely u...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

Xiaolong Liu 2026-11-04 1 min 34 words TDCS TACS TRNS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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A Neuro-endocrine-immune Network Perspective on Transcranial Direct Current Stimulation for Metabolic Dysregulation in Schizophrenia

👤 Jianhua Chen
📅 2026-11-04 📖 1 min 📝 61 words

TDCS TACS TRNS

Summary: Individuals with schizophrenia (SCZ) frequently exhibit glycolipid metabolic disorders. This clinical issue cannot be fully explained by the high metabolic risk associated with antipsychotic medications alone. It is increasingly recognized to involve complex multi-system interactions, particularly t...

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Effects of Different Stimulation Parameters on the Efficacy of Transcranial Direct Current Stimulation for Post-Stroke Dysphagia: An Updated Systematic Review and Meta-Analysis of Randomized Controlled Trials

👤 Yuan Lin
📅 2026-11-04 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: tDCS yields favorable effects on swallowing function in patients with PSD when administered to the lesional hemisphere, the contralesional hemisphere, or bilaterally, and there are no differences in efficacy among the three stimulation modes. An intensity of 1.8 mA represents a potential...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

Summary: CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li
📅 2026-10-31 📖 1 min 📝 68 words FNIRS

Summary: The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Correction: EBA-related body-preferring responses in the occipitotemporal cortex measured with functional near-infrared spectroscopy

Min Wang 2026-11-03 1 min 6 words FNIRS

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1914488.].

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Supportive parenting moderates the association between neural synchrony and irritability: Evidence from mother-son dyads

👤 Xiuyun Lin 📅 2026-11-03 📖 1 min 📝 62 words FNIRS

Summary: While research on parent-child neural synchrony using fNIRS hyperscanning has expanded, its association with child psychopathology, especially transdiagnostic symptoms such as irritability, remains poorly understood. Moreover, no prior studies have explored the potential contextual factors that may ...

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An fNIRS investigation of cortical responses to expressed emotion in bipolar disorder

👤 Lam Ghai Lim 📅 2026-11-03 📖 1 min 📝 49 words FNIRS

Summary: CONCLUSION: The findings indicate region- and context-specific differences in cortical hemodynamic responses and functional coupling in individuals with BD during EE processing. These fNIRS-derived measures may serve as candidate features for further investigation in larger and independent studies, ...

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Training Interbrain Synchrony through Dyadic Neurofeedback Modulates Empathy-for-Pain

👤 Simone Shamay-Tsoory
📅 2026-11-03 📖 1 min 📝 63 words

FNIRS

Summary: Dyadic neurofeedback offers a socially grounded approach for modulating neural processes that unfold between interacting individuals. While recent work demonstrates that dyadic neurofeedback can enhance inter-brain synchrony during interaction, whether its effects extend beyond the trained dyad rema...

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Familial risk for attention-deficit/hyperactivity disorder and related behaviors in toddlerhood are preliminarily associated with altered frontoparietal development in infancy measured by functional near-infrared spectroscopy

👤 Heather M Joseph
📅 2026-11-04 📖 1 min 📝 66 words FNIRS

Summary: INTRODUCTION: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common childhood neurodevelopmental disorder associated with impairments across the lifespan. Children with ADHD have been found to have delayed brain maturation. These delays may be present and detectable with functional near-infrar...

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Neural Mechanisms of Smartphone Addiction: A Systematic Review of Neuroimaging and Related Neurophysiological/ Neurostimulation Studies

Xiaolong Liu 2026-11-04 1 min 34 words FNIRS

Summary: CONCLUSION: These findings collectively underscore the pressing need for further research into the neurobiological mechanisms of smartphone addiction and the development of targeted interventions to mitigate its adverse effects. The PROSPERO Registration: CRD420261369691, https://www.crd.york.ac.uk/...

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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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The application of brain-computer interface on cognitive function and activities of daily living in stroke patients: a systematic review and meta-analysis

Xuan Gong 2026-11-03 1 min 52 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-based training shows preliminary promise for improving cognitive function and ADL in stroke patients. Current evidence remains insufficient to recommend BCI for emotion or speech rehabilitation. However, the GRADE certainty of evidence was rated as low for cognitive function and very...

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Mapping Brain-Computer Interface Research in Depression (1995-2026): A Bibliometric Analysis

Xianlun Duan 2026-11-03 1 min 31 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The field has evolved from EEG-based characterization toward network-level and real-time neurofeedback research. These findings identify emerging research directions while highlighting the need for further methodological standardization and clinical validation.

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A comprehensive review of biosignal foundation models

Stefanos Zafeiriou 2026-11-03 1 min 71 words BRAIN COMPUTER INTERFACE

Summary: Foundation models are emerging as a promising approach to biosignal analysis, offering solutions to common challenges in learning from biosignal data, such as high variability, noise or limited annotations. This survey provides a comprehensive review of the state-of-the-art in this rapidly advancing...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

Jinwu Wang 2026-11-03 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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Learning-dependent hippocampal-striatal reorganization during sequential decision-making in pigeons

Zhigang Shang 2026-11-03 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Adaptive sequential decision-making requires animals to learn how actions shape future states and rewards while adjusting behavioral strategies as experience accumulates. Although the hippocampus is implicated in state-structure representation and the striatum in value learning, how learning strateg...

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taVNS for uterine contraction pain and early recovery after spontaneous vaginal birth: a three-arm randomized controlled trial protocol

Tai-Pin Guo 2026-11-03 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: BACKGROUND: Uterine contraction pain after spontaneous vaginal birth (SVB) hinders early maternal recovery. Transcutaneous auricular vagus nerve stimulation (taVNS), as a safe, and non-invasive alternative therapy, can alleviate pain and improve sleep and emotion. Although a recent trial has confirm...

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Engineering Neural Interface Materials for Brain-Computer Interfaces: Toward Reliable Neural Information and Functional Recovery

👤 Jie Wu 📅 2026-11-04 📖 1 min 📝 63 words BRAIN COMPUTER INTERFACE

Summary: Motor dysfunction is a major disabling outcome of neurological disorders, including stroke, spinal cord injury, and cerebral palsy. By establishing direct communication between neural activity and external effectors, brain-computer interfaces (BCIs) offer a promising strategy for neural functional r...

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A Comparison Between Movement-Related Brain Activity Modalities for Increasing Corticospinal Excitability Using an Associative Brain-Computer Interface

Mads Jochumsen 2026-11-04 1 min 39 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: An associative BCI using either motor execution or motor imagery effectively increases corticospinal excitability. These findings suggest that both forms of efferent activity are viable, allowing flexibility in selecting the modality based on user capability or clinical context.

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https://pubmed.ncbi.nlm.nih.gov/42980617/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261104162626&v=2.20.1

The sound of manufactured music: Reviewing the role of artificial stimuli in music cognition research.

2024-01-22 1 min 259 words PSYCHOMUSICOLOGY

Summary: Having participants listen and react to musical stimuli is one of music cognition's foundational methods. Whereas most researchers have used stimuli adapted from existing musical traditions in such work, others have incorporated artificial stimuli (i.e., stimuli generated specifically for research t...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000304

Music-evoked nostalgia and charitable giving: A cross-cultural study in the United States and Mexico.

📅 2024-01-22 📖 1 min 📝 192 words PSYCHOMUSICOLOGY

Summary: Nostalgia, a past-oriented emotion characterized by complex affective responses, is a pervasive and fundamental human experience. Prior research has demonstrated that nostalgia serves various socioemotional functions, such as promoting a sense of belonging, enhancing one's perception of meaning in l...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000302

Preferred music listening does not affect cognitive inhibition in young and older adults.

📅 2023-10-12 📖 1 min 📝 227 words PSYCHOMUSICOLOGY

Summary: Previous literature has found links between music listening and cognitive performance. Specifically, background music may play a role in modulating cognitive inhibition. However, determining what type of background music affects cognitive inhibition throughout the lifespan has not been studied. The ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000300

Absolute pitch: A literature review of underlying factors, with special regard to music pedagogy.

📅 2023-07-10 📖 1 min 📝 202 words PSYCHOMUSICOLOGY

Summary: Absolute pitch (AP) is a fairly rare and special phenomenon that has relevance for musicology, psychology, genetics, and neuroscience. AP possessors are able to identify the pitch of an isolated sound or to produce that sound without a reference point. The authors' aim is to review the literature on...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000298

Capturing coordination and intentionality in joint musical improvisation.

📅 2023-08-03 📖 1 min 📝 217 words PSYCHOMUSICOLOGY

Summary: Humans collaborate with each other on a wide variety of tasks that are often largely improvised and unscripted. In this study, we investigated the dynamics of coordination in a joint musical improvisation task, what the effect of intentions is on coordination, and how musicians propagate these inten...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000299

Continuous affect responses to a large diverse set of unfamiliar music: Bayesian time-series and cluster analyses.

📅 2023-04-20 📖 1 min 📝 252 words PSYCHOMUSICOLOGY

Summary: Sixty-nine participants made continuous response judgments of perceived arousal and valence while listening to 30-s extracts of 100 unfamiliar pieces within a novel recommender system. Our purpose was to take advantage of the relatively large number of participants and pieces studied (compared with ...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000295

Monthly Updates [July]

📅 2026-07-01 📖 3 min 📝 640 words FMHY

Summary: <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   <p>Added FMHY Plugin to <a href="https://www.flowlau...

🔗 **Read full article:**
https://fmhy.net/posts/july-2026

Showcase Thread

👤 /u/AutoModerator 📅 2026-11-04 📖 1 min 📝 33 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Post all of your code/projects/showcases/AI slop here. Recycles once a month. <!-- SC_ON -->   submitted by   /u/AutoModerator
 <a href="https://www.reddit.com/r/Python/c...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xntm2j/showcase_thread/

Conductor Is Joining Meta

👤 kenrose 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.conductor.build/announcement Comments URL: https://news.ycombinator.com/item?id=50369503 Points: 35 # Comments: 20

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https://www.conductor.build/announcement

From wild to mild: Friendly foxes show changes in brain region

👤 gmays 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://news.cornell.edu/stories/2026/11/wild-mild-friendly-foxes-show-changes-brain-region Comments URL: <a href="https://news.ycombinator.com/item?id=50369527">https://n...

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https://news.cornell.edu/stories/2026/11/wild-mild-friendly-foxes-show-changes-brain-region

Have I Befriended a Murderer?

👤 gmays 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://longreads.com/2026/11/03/murder-friendship-doubt/ Comments URL: https://news.ycombinator.com/item?id=50369754 Points: 66 # ...

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Show HN: After a decade of reversing, I got co-op working in DOS Wing Commander

👤 daniel_rh 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://fireflyk64.github.io/wcmulti/ Comments URL: https://news.ycombinator.com/item?id=50369895 Points: 26 # Comments: 4

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https://fireflyk64.github.io/wcmulti/

Ripple-65B Preview: U.S. Open-Weight SOTA on Agentic Coding

👤 Philpax 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.ricursive.com/ripple Comments URL: https://news.ycombinator.com/item?id=50369925 Points: 121 # Comments: 45

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Insects Don't Swarm at LED Lights

👤 speckx 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.birdsoutsidemywindow.org/2026/11/03/insects-dont-swarm-at-led-lights/ Comments URL: <a href="https://news.ycombinator.com/item?id=50370610">https://news.ycombinator.com/...

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Qualcomm Cloud AI 100 Ultra – 150W AI Accelerator Card with 128GB at 548GB/s

👤 peter_d_sherman 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

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Residential Proxies Detected on White House Network

👤 Rasbora 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://layer3intel.com/blog/residential-proxies-white-house-network Comments URL: https://news.ycombinator.com/item?id=50371246 <...

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AR guitar with just a laptop and scissors

👤 castlew 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.arguitar.com/ Comments URL: https://news.ycombinator.com/item?id=50371370 Points: 5 # Comments: 1

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Everything is an AI scam now

👤 speckx 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

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Automatic computational reproducibility checks with Metacheck

👤 noreply@blogger.com (Daniel Lakens) 📅 2026-10-30 📖 19 min 📝 3854 words

TWENTY PERCENT STATISTICIAN

> **Summary:** When authors share their data and analysis code, the idea is that anyone should be able to run that code on that data and get the results reported in the paper. But sharing data and code in a way that results are reproducible is difficult, and researchers have often not received tr...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/automatic-computational-reproducibility.html

Registration is now full for the 2024 ERP Boot Camp

👤 Steve Luck 📅 2024-03-16 📖 1 min 📝 106 words ERP BOOT CAMP

> **Summary:** The demand for the 2024 ERP Boot Camp was far beyond our expectations, and we reached our maximum registration of 30 people within one day. We already have a waiting list of over 30 people, so we have closed the registration site. <...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/3/15/registration-full

New Paper: Using Multivariate Pattern Analysis to Increase Effect Sizes for ERP Amplitude Comparisons

👤 Steve Luck 📅 2024-06-10 📖 2 min 📝 521 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Bahle, B., Simmons, A. M., & Luck, S. J. (2024). Using multivariate pattern analysis to increase effect sizes for event-related potential analyses. Psychophysiology, 61, e14570. https://doi.org/10.1111/psyp.14570 [<a h...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/10/erp-core-decoding-paper

Call for Student/YP Deputy Editor

Deidre Artis 2026-09-29 1 min 14 words EMBS

Summary: The post Call for Student/YP Deputy Editor appeared first on IEEE EMBS.

🔗 **Read full article:**
https://www.embs.org/featured-news/call-for-student-yp-deputy-editor/

Stage-specific exposure to an activity-permissive medium enhances neuronal activity in oligodendrocyte-enriched cortical organoids

1 min 34 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Clara Chung, Michelle J. Kim, Grace Field, Konstantina Pilarinos, Elizabeth K. Kharitonova, Natalie Baker Campbell, Marie Shi, Christopher V. Gabel, Joseph L. Orofino, Ella Zeldich

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0006899326004634?dgcid=rss_sd_all

Investigating fatigability and cortical activation during prolonged foot tapping in multiple sclerosis

📖 1 min 📝 28 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zuhal Abasiyanik, Lousin Moumdjian, Natalia Skorupa, Daphne Kos, Bart Van Wijmeersch, Miguel D'Haeseleer, Inge Zijdewind, Peter Feys, Felipe Balistieri Santinelli

🔗 **Read full article:**
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Neural dynamics of auditory attention: Interactions between top-down and bottom-up processes

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): S. Graceffo, M. Elhilali

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006002?dgcid=rss_sd_all

Neurovascular coupling in simultaneous EEG-fNIRS studies: A scoping review of analytical frameworks and clinical applications

📖 1 min 📝 18 words NEUROIMAGE

Summary: Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Yu Wu, Rajamanickam Yuvaraj, Stéphane Perrey, Hasan Ayaz, Wei-Peng Teo

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006130?dgcid=rss_sd_all

A single day of natural daylight exposure and physical activity does not conclusively change circadian phase: A Bayesian crossover experiment

👤 Loock, A.-S., Milanesi, M., Schwendinger, F., Garbazza, C., Schmidt-Trucksaess, A., Lang, U., Blume, C.

📅 2026-11-04 📖 1 min 📝 280 words BIORXIV NEUROSCIENCE

Summary: Among the environmental signals that shape human physiology, the daily light-dark cycle is the primary zeitgeber of the human circadian system. Physical activity may provide an additional non-photic time cue and frequently co-occurs with daylight in everyday life. However, their independent and comb...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.03.763886v1?rss=1

Recruitment of intrasegmental and propriospinal interneuron networks of forelimb motor circuitry in chronic cervical contusion spinal cord injury

👤 Crowther, K., Harbeck, A., Middlesworth, M., Bosch, M., Lyttle, M. A., MacGibbon, C., Jaffe, D. A., Michel-Flutot, P., Fitchett, E., Li, S., Lepore, A. C.

📅 2026-11-04 📖 1 min 📝 250 words BIORXIV NEUROSCIENCE

Summary: Cervical spinal cord injury (SCI) disrupts descending supraspinal input and spinal networks controlling upper extremity motor function, making recovery of arm and hand function a major priority for affected individuals. The vast majority of regenerative strategies have been studied during the acute ...

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Mechanisms of impaired cerebral hemodynamics and memory resilience differ by sex: A role for mineralocorticoid receptors in VCID

DeMarco, G. M., Bernstein, G. A., Chen, D., Bloete, J., Schellinger, E. D., Westbrook, M. E., Baumoel, E. J., Luchetta, J., Chapman, A. C.

2026-11-04 1 min 190 words BIORXIV NEUROSCIENCE

Summary: INTRODUCTION: Chronic hypertension is associated with cerebral hypoperfusion and is a primary risk factor for vascular contributions to cognitive impairment and dementia (VCID). Aldosterone, a key mediator of hypertension, causes cerebrovascular dysfunction via its mineralocorticoid receptor (MR). M...

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https://www.biorxiv.org/content/10.64898/2026.10.31.763340v1?rss=1

Immersion as inference: An active inference account of induced realities and its relation to the two-competitor model

Renz, L., GauthierDickey, K. L., Vibell, J., Francis, A. L., Corona-Strauss, F. I., Strauss, D. J.

2026-11-04 1 min 339 words BIORXIV NEUROSCIENCE

Summary: Immersive technologies are designed for a single target: the human perceptual system. Engineers of virtual and augmented reality have long recognized that a display does not have to replicate the physical world, it only has to provide the inputs that satisfy the senses at the level at which the brai...

🔗 **Read full article:**
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Differential modulation of the neurocomputational signature of early and late mismatch responses to spoken words by awareness and task relevance

Harenbrock, J., Straube, T., Dellert, T., Schlossmacher, I.

2026-11-04 1 min 249 words BIORXIV NEUROSCIENCE

Summary: Mismatch responses (MMRs), including the mismatch negativity (MMN) and the P3, are a hallmark of the brains capacity to detect unexpected events. They are discussed to be generated by two main processes: adaptation and prediction error (PE). However, MMR research on underlying mechanisms has mostly ...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.03.757859v1?rss=1

Physical sensorimotor contingencies facilitate learning in monkeys

An, S.-J., Gabel, J., Jazayeri, M., Sohn, H.

2026-11-04 1 min 150 words

BIORXIV NEUROSCIENCE

Summary: Animal studies in neuroscience routinely use virtual tasks that deviate significantly from the animals' natural behavioral repertoire. This choice is often necessary for conducting careful and controlled experimentation. However, this approach raises a fundamental question: do virtual tasks expose n...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.01.763380v1?rss=1

One digit too many: brain rhythms and autonomic responses at the limits of working memory capacity

Kotyusov, A. I., Kasanov, D. A., Zaitseva, A. I., Kotchoubey, B., Pavlov, Y. G.

2026-11-04 1 min 250 words

BIORXIV NEUROSCIENCE

Summary: Physiological measures vary with working memory load, but whether they identify when demands exceed a person's working memory capacity remains unclear. We assessed neural and autonomic responses to exceeding digit span by one digit. During an adaptive digit span task, we recorded EEG, pupillometry, ...

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Integrative Transcriptomic Meta-Analysis Reveals SIRT1-SNCA Correlation in Parkinson's Disease

Cimen, H., Sonmez, E., Hotaslier, F.
2026-11-04 1 min 169 words

BIORXIV NEUROSCIENCE

Summary: Parkinson's disease (PD) is a progressive neurodegenerative disorder that affects midbrain dopaminergic neurons without any approved therapy. Candidate molecular targets identified in one model are not always traceable across the postmortem, induced pluripotent stem cell (iPSC), and cell-based syste...

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The behaviour bends, the code holds: working-memory category information in the multiple-demand network across graded interruption

Leticevscaia, O., Feredoes, E.
2026-11-04 1 min 279 words

BIORXIV NEUROSCIENCE

Summary: Protecting working memory (WM) from interference is essential for goal-directed behaviour, yet its neural mechanisms remain unclear. The multiple-demand (MD) network - a set of frontoparietal regions recruited by diverse cognitive demands - is a candidate system for maintaining WM content under inte...

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Contributions of Site Heterogeneity in Multi-site Brain-Phenotype Predictions

👤 Chen, G., Jones, M., Gadiyar, I., Satterthwaite, T. D., Chen, A., Vandekar, S.

📅 2026-11-04 📖 1 min

📝 264 words

BIORXIV NEUROSCIENCE

Summary: Machine learning studies of brain-phenotype associations typically summarize accuracy with a single Pearson correlation, yet in harmonized multi-site data this metric is shaped by training-set size, feature and outcome variability, and true effect heterogeneity. We analyzed structural MRI data from ...

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Fine-tuning brain stimulation to fight Parkinson’s disease

📅 2026-11-04 📖 1 min 📝 0 words

NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/d41586-026-03168-y

How a once-controversial idea is advancing understanding of Parkinson’s disease

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https://www.nature.com/articles/d41586-026-03169-x

Reward-related memory benefits cannot be explained by post-encoding stimulus-related thought

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https://www.nature.com/articles/s41598-026-78089-x

The human claustrum tracks uncertainty and prediction error

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Could chemicals contribute to Parkinson’s?

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https://www.nature.com/articles/d41586-026-03170-4

Six highlights from Parkinson’s disease research

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https://www.nature.com/articles/d41586-026-03172-2

Progress accelerates in Parkinson’s treatments

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https://www.nature.com/articles/d41586-026-03165-1

How the brain protects itself from Parkinson’s disease

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https://www.nature.com/articles/d41586-026-03171-3

Single-neuron encoding of rapidly learned visual information reshapes human perception

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https://www.nature.com/articles/s41593-026-02496-6

Parkinson’s: beyond the brain

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https://www.nature.com/articles/d41586-026-03173-1

Single-neuron encoding of rapidly learned visual information reshapes human perception

Gabriel Kreiman 2026-11-04 1 min 38 words NATURE NEUROSCIENCE

> **Summary:** Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02496-6 The authors show that rapid learning modulates neuronal activity in the human visual cortex, requiring extra processing time to support recog...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02496-6

The human claustrum tracks uncertainty and prediction error

2026-11-04 1 min 54 words NATURE NEUROSCIENCE

> **Summary:** Nature Neuroscience, Published online: 04 November 2026; doi:10.1038/s41593-026-02471-1 Single-neuron recordings provide direct evidence that the human claustrum tracks uncertainty and prediction errors during adaptive behavior. ...

🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02471-1

C1ql2 gates stress vulnerability by regulating microglia-mediated synaptic pruning in the BLA-vHPC circuit

👤 Chun-Yan WangWei-Zhu LiuShuang-Mei LiaoJing NiuHan-Qing PanWen-Jie YouKe ZhaoBao MaMing-Fen LuJia-Xin ZouPing HuWen-Hua ZhangBing-Xing Panahttps://ror.org/042v6xz23Jiangxi Institute of Respiratory Diseases and Department of Pathology, The First Affiliated Hospital, School of Basic Medical Sciences and Institute of Biomedical Innovation, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinabhttps://ror.org/042v6xz23Jiangxi Provincial Key Laboratory of Brain Science and Brain Health, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinachttps://ror.org/042v6xz23Postdoctoral Research Station, The First Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinadhttps://ror.org/042v6xz23Institute of Translational Medicine, Jiangxi Medical College, Nanchang University, Nanchang 330006, Chinaehttps://ror.org/042v6xz23Department of Neurology, The Second Affiliated Hospital, Jiangxi Medical College, Nanchang University, Nanchang 330006, China

📅 2026-10-21 📖 1 min 📝 43 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 43, October 2026.
SignificanceStress-related emotional disorders involve maladaptive synaptic remodeling in the basolateral amygdala (BLA), which is highly heterogeneous across neurons projecting to distinct downstream targets....

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2600854123?af=R

BVLab-analyzer: a unified visual toolbox for terabyte-scale whole-brain cell detection, neuron reconstruction, and vascular network extraction

Tingwei Quan 2026-10-07 1 min 139 words FRONTIERS NEUROINFORMATICS

Summary: BVLab-Analyzer is a standalone graphical software tool for end-to-end analysis of terabyte-scale whole-brain microscopy images. Operating through a zero-code interface, it handles data import (TIFF, BioImageVision, OME-Zarr), deep learning-based semantic segmentation, 3D skeleton reconstruction, and...

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Natural language processing for suicidal behavior detection in short Spanish clinical notes from first-contact interviews

Ansel Y. Rodríguez-González 2026-10-07 1 min 267 words

FRONTIERS NEUROINFORMATICS

Summary: IntroductionSuicide is a multifactorial phenomenon influenced by social, cultural, biological, and psychological factors, whose early detection remains challenging. According to the World Health Organization, suicide is the third leading cause of death among individuals aged 15–29 years, representin...

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Transformers for fMRI analysis: from spatiotemporal modeling to brain foundation models

👤 Bocheng Wang 📅 2026-10-07 📖 1 min 📝 217 words

FRONTIERS NEUROINFORMATICS

Summary: Functional magnetic resonance imaging (fMRI) serves as a pivotal tool for investigating the functional organization and dynamic connectivity of the human brain, indirectly reflecting neural activity through blood oxygen level-dependent (BOLD) signals. However, the high-dimensional spatiotemporal str...

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Multi-receptor whole-brain modeling reveals affinity-driven chemoarchitectural mechanism of virtual atomoxetine action

👤 Gustavo Deco 📅 2026-11-04 📖 1 min 📝 219 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Linking molecular mechanisms with large-scale brain dynamics remains a fundamental challenge in computational psychiatry. Receptor-informed whole-brain models have provided mechanistic insights into pharmacological interventions, yet they have largely relied on single-receptor topographies that cann...

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Music with monaural beats for anxiety reduction and mood enhancement in individuals with moderate trait-anxiety

👤 Gianluca Micchi 📅 2026-11-04 📖 1 min 📝 279 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Monaural beats embedded in music are a popular anxiety-relief intervention, yet implementation parameters and efficacy remain poorly understood. Across two studies of online participants with moderate trait anxiety, we tested whether theta-frequency (4–7 Hz) monaural beats in music produce anxiolyti...

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Brain-inspired predictive visual attention for spatiotemporal semantic understanding of complex film and television videos

👤 Peng Han 📅 2026-10-22 📖 1 min 📝 275 words FRONTIERS NEUROROBOTICS

Summary: Due to the unique characteristics of film and television, such as montage editing and narrative structure jumps, semantically salient targets frequently experience temporal feature drift, making it difficult for existing models to maintain stable semantic understanding output when keyframes are lost...

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MS-3D-CNN with quantum-inspired chaotic hybrid adaptive wolf-particle swarm optimization for tactical pattern classification in basketball from player tracking data

Haoyan Liu 2026-10-30 1 min 229 words FRONTIERS NEUROROBOTICS

Summary: IntroductionAI sport analysis methods have been considerably improved by developments in knowledge of tactically information and its codification. Nevertheless, methods for capturing multi-levels spatio-temporal information for complex team-sport behaviors are still limited. In opposed to motion rec...

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Reliability-aware multimodal sensor fusion for human motion monitoring with safety-constrained neurorobotic feedback in sports training

Jiaoyu Wang 2026-11-04 1 min 247 words FRONTIERS NEUROROBOTICS

Summary: IntroductionEmbodied intelligent systems require reliable perception and safety-constrained feedback mechanisms for human motion understanding under heterogeneous sensing conditions. In this study, volleyball training is considered as a representative dynamic human-motion scenario rather than the so...

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Deep reinforcement learning–driven model-based adaptive PID strategy for six-wheeled rocker–bogie mobile platform

Hongxue Feng
2026-11-04 1 min 202 words FRONTIERS NEUROROBOTICS

Summary: IntroductionSix-wheel rocker–bogie rovers encounter nonlinear dynamics, variable traction, and wheel slip that can degrade the performance of fixed-gain proportional–integral–derivative (PID) controllers. This study develops a hybrid deep reinforcement learning–PID framework to improve steering and ...

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Retraction: Hypothermia impairs glymphatic drainage in traumatic brain injury as assessed by dynamic contrast-enhanced MRI with intrathecal contrast

2026-11-04 1 min 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2046766

TEMPO-taVNS: a multimodal platform optimizing thermal, electrical, and mechanical auricular stimulation

Matthew P. Ward 2026-11-04 1 min 176 words FRONTIERS NEUROSCIENCE

Summary: Transcutaneous auricular vagus nerve stimulation (taVNS) is typically delivered electrically, while alternative stimulation modalities and their potential for personalization remain underexplored. We present TEMPO-taVNS (Thermal, Electrical, and Mechanical Parameter Optimization for taVNS), a human-...

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Baseline cognition accounts for most predictive information on cognitive recovery during rehabilitation after carbon monoxide poisoning

Peihui Liu 2026-11-04 1 min 271 words FRONTIERS NEUROSCIENCE

Summary: BackgroundCognitive change during rehabilitation after carbon monoxide (CO) poisoning may differ from prognosis at acute presentation. We examined whether information recorded in a rehabilitation pathway added predictive information beyond baseline cognition.MethodsThis single-center retrospective c...

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https://www.frontiersin.org/articles/10.3389/fnins.2026.1975750

NeuroTest: a benchtop testbed for evaluating sensing-capable electrophysiology and neurostimulation systems

Morrow JK, H Cho, M Benjaber, Denison TJ, Widge AS and Herron JA

2026-11-04 1 min

247 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. The development of increasingly complex implantable neuromodulation devices requires extensive benchtop validation and testing prior to in vivo implementation. Our goal is to enable this in vitro testing by creating a benchtop system that is accessible, easy-to-use, and functions reliably...

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Towards semantic brain-computer interfacing: the graded inventory of semantic triggers (GIST) paradigm

Joshua James Podmore, Moein Radman, Riccardo Poli, Silke Paulmann and Ian Daly

2026-11-04

1 min 299 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Semantic brain-computer interfaces (sBCIs) aim to decode conceptual information directly from neural activity, enabling more efficient and natural communication for individuals with severe motor impairments. However, existing approaches are often confounded by stimulus-driven perceptual p...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb724

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 2025-06-26 1 min 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 2025-08-13 1 min 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Suprachoroidal hemorrhage and detachment in the non-laser photocoagulation posterior pole after intravitreal Anti-VEGF injection: A case report

Xionggao Huang 2026-11-03 1 min 63 words LOW VISION

Summary: Intravitreal anti-vascular endothelial growth factor (anti-VEGF) injections are widely used for diabetic macular edema (DME) and neovascular ocular diseases. Suprachoroidal hemorrhage and detachment (SCHD) in the posterior pole non-laser photocoagulation area is an extremely rare complication after ...

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Visual Perceptual Learning Based on Adaptive Contrast Detection Training Enhances Vision and Contrast Sensitivity in Diabetic Retinopathy

Lixia Feng 2026-11-03 1 min 37 words LOW VISION

Summary: CONCLUSIONS: The results reveal significant plasticity in the visual system in DR and suggest that contrast detection training may extend to motion perception tasks. Our study provides additional support for VPL as a potential treatment for DR.

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Reading disability in glaucoma: A neglected dimension of visual disability

Michele Figus 2026-11-03 1 min 62 words LOW VISION

Summary: Reading disability is a common and clinically relevant, yet under-recognized, functional impairment in glaucoma, with significant impact on independence, daily activities, and quality of life, particularly in an increasingly digital environment. Reading performance is influenced by a complex interpl...

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The Amsler Grid Use Study (AGUS): Amsler Grid Nonadherence is Driven by Forgetfulness and Health Beliefs But Not Age

Lisa Nivison-Smith 2026-11-03 1 min 33 words LOW VISION

Summary: CONCLUSIONS: Amsler grid nonadherence was primarily unintentional and mostly due to forgetfulness. Poor adherence was associated with greater concerns in Amsler grid use, indicating modifiable factors that can be addressed to improve adherence.

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Intrastromal Antibiotic Therapy for Recalcitrant Bacterial Keratitis

👤 Mark Chehade 📅 2026-11-03 📖 1 min 📝 52 words LOW VISION

Summary: CONCLUSIONS: Adjunctive intrastromal antibiotic therapy achieved a high rate of infection resolution in this cohort of recalcitrant bacterial keratitis, with a low complication rate. This approach may be considered in refractory cases or when unable to utilize conventional topical therapy. However, ...

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An open-source eye-tracking system adaptable to diverse terrestrial vertebrates reveals state-dependent visual acquisition in lizards

👤 Mark Shein-Idelson 📅 2026-11-03 📖 1 min 📝 68 words LOW VISION

Summary: Gaze shifts are fundamental to visual exploration and processing, yet empirical knowledge is largely confined to a small number of model species, limiting our understanding of general principles of visual acquisition across animals. Here, we introduce a versatile, low-cost, and open-source framework...

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Ocular Considerations for the Department of War's New Testosterone Initiative

Harkaran S Rana 2026-11-03 1 min 75 words LOW VISION

Summary: On 15 July 2026, the Secretary of War announced that the Department of War will begin annual testosterone deficiency screening for male Active Duty and Reserve Component personnel aged 30 and older as part of the periodic health assessment, with screening available on request to younger servicemembe...

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A review of deep learning applications on histopathologic and genomic data in focal segmental glomerulosclerosis

Sasanko Sekhar Gantayat 2026-11-03 1 min 66 words LOW VISION

Summary: Focal segmental glomerulosclerosis explains the histological pattern of injury seen through light microscopy in kidney biopsy specimens and is the end result of different injuries to podocytes. The clinical classification of patients by FSGS lesion is challenging because of a broad spectrum of etiol...

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Effects of balance training with electrical muscle stimulation on ankle joint muscle on sensory reweighting and motor strategy

Naoya Hasegawa 2026-11-03 1 min 60 words LOW VISION

Summary: CONCLUSIONS: EMS-based balance training was associated with the induction of visual reweighting dynamics and adaptive postural strategies involving multiple joints. Specifically, EMS-induced involuntary ankle assistance elicited hip-centered adjustments in postural control, contributing to upper-tru...

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Reproxalap 0.25% in dry eye disease: A meta-analysis of randomized controlled trials

Hashem Abu Serhan 2026-11-04 1 min 62 words LOW VISION

Summary: CONCLUSION: Reproxalap was associated with consistent improvements across key clinical outcomes in DED. Exploratory allergic conjunctivitis findings should be considered hypothesis-generating rather than clinically conclusive. Interpretation of the overall evidence is limited by exclusive industry s...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

👤 Ali Alhaj 📅 2026-09-23 📖 1 min 📝 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

👤 Haoran Sun 📅 2026-09-25 📖 1 min 📝 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

👤 Hu Zhao 📅 2026-11-01 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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A phase-calibrated combined electrical-sensory neuromodulation platform for short-Term frontoparietal network modulation and cognitive fatigue resistance

👤 Qianxiang Zhou
📅 2026-11-02 📖 1 min 📝 57 words

TDCS TACS TRNS

Summary: Non-invasive multimodal neuromodulation is often limited by cross-modal temporal misalignment caused by modality-specific conduction delays. To address this challenge, we developed a microcontroller-based multiphysics stimulation platform integrating transcranial alternating current stimulation (tAC...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

👤 Roi Cohen Kadosh
📅 2026-11-02 📖 1 min 📝 48 words

TDCS TACS TRNS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Repeated prefrontal transcranial direct current stimulation improves behavioral regulation and modulates resting-state EEG microstate dynamics in young adults with problematic smartphone use: a randomized sham-controlled longitudinal study

👤 Yingying Zeng
📅 2026-11-03 📖 1 min 📝 62 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Repeated prefrontal tDCS produced concurrent behavioral improvement and selective reorganization of resting-state EEG microstate dynamics in young adults with PSU. These findings provide preliminary evidence that repeated prefrontal tDCS is associated with concurrent behavioral changes ...

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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What are the neural networks that have been modulated by anatomically targeted therapies to promote recovery of consciousness? A scoping review

👤 Konstantinos Margetis
📅 2026-11-03 📖 1 min 📝 63 words

TDCS TACS TRNS

Summary: Disorders of Consciousness (DoC) arise from insults to neural networks responsible for awareness and arousal. DoC subtypes are frequently misdiagnosed, as mechanisms underlying consciousness remain poorly understood. Neurostimulation has been investigated as a potential therapy for DoC recovery; how...

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Effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field

Prinpida Sonthiphand 2026-11-03 1 min 68 words TDCS TACS TRNS

Summary: Due to climate change and increasing food demand, conventional agriculture relies heavily on chemical fertilizers. This study investigated the effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field. Metag...

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Transcranial Pulse Stimulation in Parkinson's Disease: A Systematic Review

Marcelo Lourenço da Silva 2026-11-03 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: TPS shows preliminary potential to modulate sensorimotor networks and improve selected motor outcomes in PD. However, evidence remains insufficient to establish overall clinical efficacy. Larger, rigorously controlled trials are required to determine sustained benefits, optimal stimulat...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

Jinwu Wang 2026-11-03 1 min 30 words TDCS TACS TRNS

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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First mitogenome-based phylogenetic analysis and divergence time estimation of deep-sea elasipodid sea cucumbers (Holothuroidea: Elasipodida)

Haibin Zhang 2026-11-03 1 min 29 words TDCS TACS TRNS

Summary: CONCLUSIONS: Collectively, this study establishes a foundational mitogenomic dataset and provide a robust scientific foundation for future studies on the temporal divergence and evolutionary history of elasipodid sea cucumbers.

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 68 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Correction: EBA-related body-preferring responses in the occipitotemporal cortex measured with functional near-infrared spectroscopy

Min Wang 2026-11-03 1 min 6 words FNIRS

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1914488.].

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Supportive parenting moderates the association between neural synchrony and irritability: Evidence from mother-son dyads

👤 Xiuyun Lin 📅 2026-11-03 📖 1 min 📝 62 words FNIRS

Summary: While research on parent-child neural synchrony using fNIRS hyperscanning has expanded, its association with child psychopathology, especially transdiagnostic symptoms such as irritability, remains poorly understood. Moreover, no prior studies have explored the potential contextual factors that may ...

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An fNIRS investigation of cortical responses to expressed emotion in bipolar disorder

👤 Lam Ghai Lim 📅 2026-11-03 📖 1 min 📝 49 words FNIRS

Summary: CONCLUSION: The findings indicate region- and context-specific differences in cortical hemodynamic responses and functional coupling in individuals with BD during EE processing. These fNIRS-derived measures may serve as candidate features for further investigation in larger and independent studies, ...

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Training Interbrain Synchrony through Dyadic Neurofeedback Modulates Empathy-for-Pain

👤 Simone Shamay-Tsoory
📅 2026-11-03 📖 1 min 📝 63 words FNIRS

Summary: Dyadic neurofeedback offers a socially grounded approach for modulating neural processes that unfold between interacting individuals. While recent work demonstrates that dyadic neurofeedback can enhance inter-brain synchrony during interaction, whether its effects extend beyond the trained dyad rema...

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Familial risk for attention-deficit/hyperactivity disorder and related behaviors in toddlerhood are preliminarily associated with altered frontoparietal development in infancy measured by functional near-infrared spectroscopy

Heather M Joseph
2026-11-04 1 min 66 words FNIRS

Summary: INTRODUCTION: Attention-Deficit/Hyperactivity Disorder (ADHD) is a common childhood neurodevelopmental disorder associated with impairments across the lifespan. Children with ADHD have been found to have delayed brain maturation. These delays may be present and detectable with functional near-infrar...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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The application of brain-computer interface on cognitive function and activities of daily living in stroke patients: a systematic review and meta-analysis

Xuan Gong 2026-11-03 1 min 52 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-based training shows preliminary promise for improving cognitive function and ADL in stroke patients. Current evidence remains insufficient to recommend BCI for emotion or speech rehabilitation. However, the GRADE certainty of evidence was rated as low for cognitive function and very...

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Mapping Brain-Computer Interface Research in Depression (1995-2026): A Bibliometric Analysis

Xianlun Duan 2026-11-03 1 min 31 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The field has evolved from EEG-based characterization toward network-level and real-time neurofeedback research. These findings identify emerging research directions while highlighting the need for further methodological standardization and clinical validation.

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A comprehensive review of biosignal foundation models

Stefanos Zafeiriou 2026-11-03 1 min 71 words BRAIN COMPUTER INTERFACE

Summary: Foundation models are emerging as a promising approach to biosignal analysis, offering solutions to common challenges in learning from biosignal data, such as high variability, noise or limited annotations. This survey provides a comprehensive review of the state-of-the-art in this rapidly advancing...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

Jinwu Wang 2026-11-03 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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Learning-dependent hippocampal-striatal reorganization during sequential decision-making in pigeons

Zhigang Shang 2026-11-03 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Adaptive sequential decision-making requires animals to learn how actions shape future states and rewards while adjusting behavioral strategies as experience accumulates. Although the hippocampus is implicated in state-structure representation and the striatum in value learning, how learning strateg...

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taVNS for uterine contraction pain and early recovery after spontaneous vaginal birth: a three-arm randomized controlled trial protocol

Tai-Pin Guo 2026-11-03 1 min 58 words BRAIN COMPUTER INTERFACE

Summary: BACKGROUND: Uterine contraction pain after spontaneous vaginal birth (SVB) hinders early maternal recovery. Transcutaneous auricular vagus nerve stimulation (taVNS), as a safe, and non-invasive alternative therapy, can alleviate pain and improve sleep and emotion. Although a recent trial has confirm...

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Engineering Neural Interface Materials for Brain-Computer Interfaces: Toward Reliable Neural Information and Functional Recovery

Jie Wu 2026-11-04 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Motor dysfunction is a major disabling outcome of neurological disorders, including stroke, spinal cord injury, and cerebral palsy. By establishing direct communication between neural activity and external effectors, brain-computer interfaces (BCIs) offer a promising strategy for neural functional r...

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Advancing neuropsychology training: Insights and advice from a qualitative analysis in 102 brain-behavior experts.

2026-07-30 1 min 266 words NEUROPSYCHOLOGY

Summary: Objective: Neuropsychologists possess knowledge that could benefit trainees as they navigate important career decisions within their unique training pathways. This wisdom is typically passed down in an individual mentorship model wherein supervisors and advisors relay advice directly. While this mod...

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http://doi.org/10.1037/neu0001116

Sex-specific differences in amygdala resting-state functional connectivity with trait anxiety.

📅 2026-06-04 📖 1 min 📝 267 words NEUROPSYCHOLOGY

Summary: Objective: Anxiety disorders are the most common mental health conditions in the world, disproportionately affecting females and associated with significant morbidity. Neuroimaging can help us to understand the underpinnings of anxiety behaviors. The amygdala and its subnuclei have been strongly imp...

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http://doi.org/10.1037/neu0001100

Effects of simulated hearing loss on cognitive task performance: A scoping review.

📅 2026-07-30 📖 1 min 📝 235 words NEUROPSYCHOLOGY

Summary: Objective: Hearing loss (HL) and cognitive decline often co-occur in older adults. Many cognitive tests are administered using spoken instructions or stimuli, potentially biasing test results. The current scoping review aimed to provide an overview of studies using simulated HL in cognitively health...

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Normative cognitive test performance and typical error patterns among American Indian older adults: The Strong Heart Study.

📅 2026-06-15 📖 1 min 📝 245 words NEUROPSYCHOLOGY

Summary: Objective: The Strong Heart Study, a large cohort of American Indians from three U.S. regions, collected standardized neuropsychological assessments from participants aged 65–95 years at two visits (2010–2019), with prior documentation of lower overall score distributions and higher prevalence of co...

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The unity and diversity of executive functions across adulthood in a diverse Brazilian sample with varying educational attainment.

📅 2026-04-16 📖 1 min 📝 261 words NEUROPSYCHOLOGY

Summary: Objective: Executive functions (EFs) are a set of cognitive abilities associated with many social and clinical outcomes that are believed to be sensitive to culture, age, and educational attainment. Yet, this was determined using EF theoretical models and tests developed for highly educated young ad...

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http://doi.org/10.1037/neu0001085

Everyday executive functions in children with neurofibromatosis type 1: Data from nine institutions.

📅 2026-04-16 📖 1 min 📝 278 words NEUROPSYCHOLOGY

Summary: Objective: Executive function (EF) difficulties are prevalent in neurogenetic syndromes such as Neurofibromatosis Type 1 (NF1), the most common autosomal dominant condition associated with nervous system tumors. However, age-related differences in EF problems in children with NF1 remain poorly under...

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http://doi.org/10.1037/neu0001087

Integrity in six key brain regions predicts numeric working memory performance in 30,000 middle-aged and older adults: A UK Biobank magnetic resonance imaging study.

📅 2026-06-18 📖 1 min 📝 262 words NEUROPSYCHOLOGY

Summary: Objective: This study investigated the neuroanatomical correlates of working memory using the UK Biobank Numeric Memory Test (NMT; a visual backward digit-span task) in a large sample of middle-aged and older adults. Volumes of brain regions previously implicated in digit-span tasks were hypothesize...

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http://doi.org/10.1037/neu0001096

What Python project taught you the most about real world development?

👤 /u/ Sonu_Jana 📅 2026-11-04 📖 1 min 📝 44 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I'm interested in hearing about projects that genuinely improved your programming skills. Not necessarily huge projects, but ones that taught you something valuable <!-- SC_ON -->   submitted by   <a href="https://www.reddit.com/user/Sonu_Jana"> /u...

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Tell HN: Gemini trains on chat history, even for subscribers (personal accounts)

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> **Summary:** If you have a personal Google AI subscription, all your saved chat history in Gemini can be used for training and there is no way to opt out unless you turn off chat history or use temporary chat sessions. These chat transcripts may even be read by human reviewers. Although Google claims to disco...

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Geoffrey Hinton: it might turn out we were better off without AI

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US voters roundly reject data centers in ballot measures

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> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Jing Wang, Xiaoya Li, Siqi Jian, Peng Li

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Transcutaneous auricular vagus nerve stimulation attenuates post-stroke dysphagia by modulating microglial polarization via the endothelial VWF/PI3K-Akt axis

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> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Gabriela Rodríguez-Manzo, Ana Canseco-Alba, Jorge Rojas-Hernández

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Machine learning of connectome-wide functional connectivity in spinal muscular atrophy

📖 1 min 📝 36 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Yufen Li, Huasong Cai, Mengyu Zhang, Mengzhen Yan, Shasha Lan, Huirong Nie, Wanqing Shen, Cui Yan, Boyan Xu, Yujian Liang, Wen Tang, Yijuan Li, Pei Xiang, Yingqian Chen

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https://www.sciencedirect.com/science/article/pii/S0306452226007037?dgcid=rss_sd_all

Spectral-spatial modeling of neural couplings in functional connectivity of motor networks

📖 1 min 📝 25 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Hamidreza Abbaspour, Melissa H. Wade, Christopher P. Hurt, Christopher L. Gonzalez, Harrison C. Walker, Zachary T. Irwin

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006294?dgcid=rss_sd_all

Impact of acoustic property uncertainty on transcranial ultrasound simulation

👤 Kosciessa, J. Q., Webb, T. D., Martin, E., Treeby, B., Butts Pauly, K., Verhagen, L., Pichardo, S.

📅 2026-11-03 📖 1 min 📝 241 words BIORXIV NEUROSCIENCE

> **Summary:** Subject-specific simulations of wave propagation are commonly used to plan and report acoustic and thermal exposure during transcranial ultrasound (TUS) interventions. Their accuracy depends on acoustic and thermal tissue property assumptions, including sound speed, density, and attenuation. Reporte...

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https://www.biorxiv.org/content/10.64898/2026.10.30.763065v1?rss=1

Facing the waves of negative mood: Trait mindfulness moderates the links between physiology and emotion

Campos, A. M., Batchalli Maruthy, G., Cumming, G., Rodriguez, S., Himes, L., Rypma, B.

2026-11-03 1 min 197 words BIORXIV NEUROSCIENCE

Summary: Trait mindfulness is a dispositional ability characterized by awareness, non-judgment, and acceptance of internal and external experiences. Stress is associated with numerous psychological and physical health concerns including depression and anxiety. Mindfulness-based interventions (MBIs) have cons...

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https://www.biorxiv.org/content/10.64898/2026.10.30.760637v1?rss=1

Epilepsy: epidemiology, risk factors, classification, pathophysiology, clinical features, diagnosis and management

2026-11-04 1 min 0 words NATURE NEUROSCIENCE SUBJECTS

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Editorial: Exploring the role of criticality in neural systems

Tiago L. Ribeiro 2026-11-04 1 min 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fncom.2026.2036015

Editorial: BCIs for monitoring and treating mental health conditions

Andrei Dragomir 2026-11-04 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2039121

The association between dietary protein intake and mental disorders in older adults: a scoping review

Aliza Haslinda Hamirudin 2026-11-04 1 min 280 words

FRONTIERS HUMAN NEUROSCIENCE

Summary: BackgroundMental disorders are increasingly prevalent among older adults, significantly affecting their quality of life and overall wellbeing. Recent evidence suggests that dietary factors, particularly protein intake, may play a role in mental health outcomes. However, the association between dieta...

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.1949231

Editorial: Women in speech and language 2025

Anastasia M. Raymer 2026-11-04 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2009604

Editorial: Understanding consciousness: neural, cognitive, and philosophical perspectives

Joachim Keppler 2026-11-04 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnhum.2026.2006518

MA-Pose and DMMS-GCN: skeleton-based martial-arts category recognition with edge-device deployment

Shilin Guo 2026-11-04 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: With the development of visual sensor technology and human pose estimation algorithms, skeleton-based action recognition has received extensive attention in sports analysis and training systems. The recognition of martial-arts actions is difficult because different styles can share similar poses whi...

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https://www.frontiersin.org/articles/10.3389/fnbot.2026.1967140

A purinergic receptor dominance model for microglial state transitions: balancing perturbation and restraint during neuroinflammation

Minke Tang 2026-11-04 1 min 183 words FRONTIERS NEUROSCIENCE

Summary: Microglia show substantial functional heterogeneity during neuroinflammatory responses, yet the upstream mechanisms through which they interpret extracellular perturbation-associated signals and transition between functional states remain incompletely understood. Purinergic signaling is well positio...

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The butyrate-m6A-BDNF axis: a hypothesis of gut-brain signal transduction in PSD

Dajun Gu 2026-11-04 1 min 151 words FRONTIERS NEUROSCIENCE

Summary: A reduction in brain-derived neurotrophic factor (BDNF) protein levels has been reported in post-stroke depression (PSD). However, whether transcript abundance and translation efficiency change in parallel or become uncoupled in PSD remains to be determined. In addition to classical transcriptional ...

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Delineating neural contributions to electroencephalogram-based speech decoding

Motoshige Sato, Eri Hatakeyama, Masakazu Inoue, Yasuo Kabe, Sensho Nobe, Akito Yoshida, Atsushi Yamamoto, Yuya Kita, Mayumi Shimizu, Kenichi Tomeoka, Michael X Cohen and Shuntaro Sasai

2026-10-30 1 min 249 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Speech brain–computer interfaces (BCIs) have emerged as a pivotal technology in facilitating communication for individuals with speech impairments. Utilizing electroencephalography (EEG) for noninvasive speech BCIs offers an accessible and affordable solution, potentially benefiting a bro...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb68d

Diff-ADN: a diffusion-guided artifact denoising network with deterministic residual refinement for EEG

Mudasir Jamil, Muhammad Zulkifal Aziz, Binwen Huang and Xiaojun Yu

2026-10-30 1 min 241 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Physiological artifacts degrade electroencephalographic (EEG) recordings and can affect downstream brain–computer interface (BCI) analysis. This study develops a compact single-channel framework for ocular, muscular, cardiac, and mixed-artifact removal. Approach. The Diffusion-Guided Arti...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Foveal Crowding and Reduced Binocular Advantage In Myopic Anisometropia Under Brief Stimulus Presentation

👤 Jianfeng Wu 📅 2026-11-03 📖 1 min 📝 32 words LOW VISION

> **Summary:** CONCLUSIONS: In optically corrected MA patients, brief stimulus presentation revealed abnormal foveal crowding in response time and a reduced binocular advantage, suggesting decreased efficiency in processing visual information under high perceptual demand.

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Blinded by inequity: a scoping review of income-related and structural disparities in eye care in the United States

👤 Carrie B Dolan 📅 2026-11-03 📖 1 min 📝 67 words LOW VISION

> **Summary:** CONCLUSION: Income-related disparities in eye care access are associated with structural and financial barriers linked to reduced access to timely services. Addressing these inequities may require greater alignment among recognized needs, promising interventions, and policy priorities, as well as en...

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Suprachoroidal hemorrhage and detachment in the non-laser photocoagulation posterior pole after intravitreal Anti-VEGF injection: A case report

Xionggao Huang 2026-11-03 1 min 63 words LOW VISION

Summary: Intravitreal anti-vascular endothelial growth factor (anti-VEGF) injections are widely used for diabetic macular edema (DME) and neovascular ocular diseases. Suprachoroidal hemorrhage and detachment (SCHD) in the posterior pole non-laser photocoagulation area is an extremely rare complication after ...

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Visual Perceptual Learning Based on Adaptive Contrast Detection Training Enhances Vision and Contrast Sensitivity in Diabetic Retinopathy

Lixia Feng 2026-11-03 1 min 37 words LOW VISION

Summary: CONCLUSIONS: The results reveal significant plasticity in the visual system in DR and suggest that contrast detection training may extend to motion perception tasks. Our study provides additional support for VPL as a potential treatment for DR.

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Reading disability in glaucoma: A neglected dimension of visual disability

Michele Figus 2026-11-03 1 min 62 words LOW VISION

Summary: Reading disability is a common and clinically relevant, yet under-recognized, functional impairment in glaucoma, with significant impact on independence, daily activities, and quality of life, particularly in an increasingly digital environment. Reading performance is influenced by a complex interpl...

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The Amsler Grid Use Study (AGUS): Amsler Grid Nonadherence is Driven by Forgetfulness and Health Beliefs But Not Age

Lisa Nivison-Smith 2026-11-03 1 min 33 words LOW VISION

Summary: CONCLUSIONS: Amsler grid nonadherence was primarily unintentional and mostly due to forgetfulness. Poor adherence was associated with greater concerns in Amsler grid use, indicating modifiable factors that can be addressed to improve adherence.

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Intrastromal Antibiotic Therapy for Recalcitrant Bacterial Keratitis

👤 Mark Chehade 📅 2026-11-03 📖 1 min 📝 52 words LOW VISION

Summary: CONCLUSIONS: Adjunctive intrastromal antibiotic therapy achieved a high rate of infection resolution in this cohort of recalcitrant bacterial keratitis, with a low complication rate. This approach may be considered in refractory cases or when unable to utilize conventional topical therapy. However, ...

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An open-source eye-tracking system adaptable to diverse terrestrial vertebrates reveals state-dependent visual acquisition in lizards

👤 Mark Shein-Idelson 📅 2026-11-03 📖 1 min 📝 68 words LOW VISION

Summary: Gaze shifts are fundamental to visual exploration and processing, yet empirical knowledge is largely confined to a small number of model species, limiting our understanding of general principles of visual acquisition across animals. Here, we introduce a versatile, low-cost, and open-source framework...

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Ocular Considerations for the Department of War's New Testosterone Initiative

Harkaran S Rana 2026-11-03 1 min 75 words LOW VISION

Summary: On 15 July 2026, the Secretary of War announced that the Department of War will begin annual testosterone deficiency screening for male Active Duty and Reserve Component personnel aged 30 and older as part of the periodic health assessment, with screening available on request to younger servicemembe...

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A review of deep learning applications on histopathologic and genomic data in focal segmental glomerulosclerosis

Sasanko Sekhar Gantayat 2026-11-03 1 min 66 words LOW VISION

Summary: Focal segmental glomerulosclerosis explains the histological pattern of injury seen through light microscopy in kidney biopsy specimens and is the end result of different injuries to podocytes. The clinical classification of patients by FSGS lesion is challenging because of a broad spectrum of etiol...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

Zhifei Xie
2026-10-24 1 min 66 words TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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A phase-calibrated combined electrical-sensory neuromodulation platform for short-Term frontoparietal network modulation and cognitive fatigue resistance

Qianxiang Zhou 2026-11-02 1 min 57 words TDCS TACS TRNS

Summary: Non-invasive multimodal neuromodulation is often limited by cross-modal temporal misalignment caused by modality-specific conduction delays. To address this challenge, we developed a microcontroller-based multiphysics stimulation platform integrating transcranial alternating current stimulation (tAC...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

👤 Roi Cohen Kadosh
📅 2026-11-02 📖 1 min 📝 48 words

TDCS TACS TRNS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Repeated prefrontal transcranial direct current stimulation improves behavioral regulation and modulates resting-state EEG microstate dynamics in young adults with problematic smartphone use: a randomized sham-controlled longitudinal study

👤 Yingying Zeng
📅 2026-11-03 📖 1 min 📝 62 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Repeated prefrontal tDCS produced concurrent behavioral improvement and selective reorganization of resting-state EEG microstate dynamics in young adults with PSU. These findings provide preliminary evidence that repeated prefrontal tDCS is associated with concurrent behavioral changes ...

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao 📅 2026-11-03 📖 1 min 📝 69 words TDCS TACS TRNS

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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What are the neural networks that have been modulated by anatomically targeted therapies to promote recovery of consciousness? A scoping review

Konstantinos Margetis
2026-11-03 1 min 63 words TDCS TACS TRNS

Summary: Disorders of Consciousness (DoC) arise from insults to neural networks responsible for awareness and arousal. DoC subtypes are frequently misdiagnosed, as mechanisms underlying consciousness remain poorly understood. Neurostimulation has been investigated as a potential therapy for DoC recovery; how...

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Effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field

Prinpida Sonthiphand 2026-11-03 1 min 68 words TDCS TACS TRNS

Summary: Due to climate change and increasing food demand, conventional agriculture relies heavily on chemical fertilizers. This study investigated the effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field. Metag...

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Transcranial Pulse Stimulation in Parkinson's Disease: A Systematic Review

Marcelo Lourenço da Silva 2026-11-03 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: TPS shows preliminary potential to modulate sensorimotor networks and improve selected motor outcomes in PD. However, evidence remains insufficient to establish overall clinical efficacy. Larger, rigorously controlled trials are required to determine sustained benefits, optimal stimulat...

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Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

👤 Jinwu Wang
📅 2026-11-03 📖 1 min 📝 30 words

TDCS TACS TRNS

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 68 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Correction: EBA-related body-preferring responses in the occipitotemporal cortex measured with functional near-infrared spectroscopy

Min Wang 2026-11-03 1 min 6 words FNIRS

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1914488.].

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Supportive parenting moderates the association between neural synchrony and irritability: Evidence from mother-son dyads

👤 Xiuyun Lin 📅 2026-11-03 📖 1 min 📝 62 words FNIRS

Summary: While research on parent-child neural synchrony using fNIRS hyperscanning has expanded, its association with child psychopathology, especially transdiagnostic symptoms such as irritability, remains poorly understood. Moreover, no prior studies have explored the potential contextual factors that may ...

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An fNIRS investigation of cortical responses to expressed emotion in bipolar disorder

👤 Lam Ghai Lim 📅 2026-11-03 📖 1 min 📝 49 words FNIRS

Summary: CONCLUSION: The findings indicate region- and context-specific differences in cortical hemodynamic responses and functional coupling in individuals with BD during EE processing. These fNIRS-derived measures may serve as candidate features for further investigation in larger and independent studies, ...

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Training Interbrain Synchrony through Dyadic Neurofeedback Modulates Empathy-for-Pain

👤 Simone Shamay-Tsoory
📅 2026-11-03 📖 1 min 📝 63 words

FNIRS

Summary: Dyadic neurofeedback offers a socially grounded approach for modulating neural processes that unfold between interacting individuals. While recent work demonstrates that dyadic neurofeedback can enhance inter-brain synchrony during interaction, whether its effects extend beyond the trained dyad rema...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

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https://pubmed.ncbi.nlm.nih.gov/42972338/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261104015249&v=2.20.1

Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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The application of brain-computer interface on cognitive function and activities of daily living in stroke patients: a systematic review and meta-analysis

Xuan Gong 2026-11-03 1 min 52 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-based training shows preliminary promise for improving cognitive function and ADL in stroke patients. Current evidence remains insufficient to recommend BCI for emotion or speech rehabilitation. However, the GRADE certainty of evidence was rated as low for cognitive function and very...

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https://pubmed.ncbi.nlm.nih.gov/42973952/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261104015249&v=2.20.1

Mapping Brain-Computer Interface Research in Depression (1995-2026): A Bibliometric Analysis

Xianlun Duan 2026-11-03 1 min 31 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The field has evolved from EEG-based characterization toward network-level and real-time neurofeedback research. These findings identify emerging research directions while highlighting the need for further methodological standardization and clinical validation.

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https://pubmed.ncbi.nlm.nih.gov/42975207/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261104015249&v=2.20.1

A comprehensive review of biosignal foundation models

Stefanos Zafeiriou 2026-11-03 1 min 71 words BRAIN COMPUTER INTERFACE

Summary: Foundation models are emerging as a promising approach to biosignal analysis, offering solutions to common challenges in learning from biosignal data, such as high variability, noise or limited annotations. This survey provides a comprehensive review of the state-of-the-art in this rapidly advancing...

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https://pubmed.ncbi.nlm.nih.gov/42976382/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261104015249&v=2.20.1

Comparative Effects of Neurostimulation- and Technology-Augmented Robot-Assisted Training on Upper-Limb Motor Recovery After Stroke: A Systematic Review and Network Meta-Analysis

Jinwu Wang 2026-11-03 1 min 30 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: rTMS plus RAT yielded the largest FMA-UE estimate, followed by LFES plus RAT. Sparse head-to-head evidence, heterogeneity, and very low-to-moderate certainty preclude a definitive hierarchy and warrant comparative trials.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42977063/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261104015249&v=2.20.1

Learning-dependent hippocampal-striatal reorganization during sequential decision-making in pigeons

Zhigang Shang 2026-11-03 1 min 62 words BRAIN COMPUTER INTERFACE

Summary: Adaptive sequential decision-making requires animals to learn how actions shape future states and rewards while adjusting behavioral strategies as experience accumulates. Although the hippocampus is implicated in state-structure representation and the striatum in value learning, how learning strateg...

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42977519/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261104015249&v=2.20.1

Psychomusicology: A resounding closing cadence.

2024-01-22 1 min 256 words PSYCHOMUSICOLOGY

Summary: From 2012 to 2023, the American Psychological Association served as publisher of Psychomusicology: Music, Mind, and Brain. Annabel Cohen and Mark Schmuckler were the successive editors-in-chiefs during this time. As the journal is ceasing publication, the two editors reflect on the developm...

🔗 **Read full article:**
http://doi.org/10.1037/pmu0000305

Suggestibility and psychedelics: From therapeutics to social context.

📅 2024-12-12 📖 1 min 📝 175 words CLINICAL NEUROSCIENCE

Summary: Suggestibility is an important aspect of the psychedelic experience that merits greater attention in the expanding field of psychedelic research. This article provides a comprehensive road map for exploring the critical questions and issues at the intersection of suggestion, suggestibility, and psyc...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000412

Evaluating dispositional emotion-related factors in the effect of acceptance on affect reactivity: A secondary analysis of an experimental study.

📅 2024-12-30 📖 1 min 📝 257 words CLINICAL NEUROSCIENCE

Summary: Acceptance as a self-regulation strategy has beneficial effects on affect reactivity in response to stressors via reducing negative affect and state anxiety as well as improving positive affect. However, dispositional emotion-related factors, namely, experiential avoidance, emotional dysregulation, ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000416

Open monitoring meditation defined as the nonreactive meta-awareness of mind wandering.

📅 2024-12-12 📖 1 min 📝 128 words CLINICAL NEUROSCIENCE

Summary: Open monitoring meditation is generally defined as the nonreactive, meta-aware observation of moment-by-moment phenomenal experiences associated with mental contents without focusing on predetermined mental or physical objects. However, in the current literature, neither general definitions nor more...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000411

Multisensory interplay and theories of consciousness.

📅 2024-11-25 📖 1 min 📝 174 words CLINICAL NEUROSCIENCE

Summary: Sensory interplay typically refers to cases where the content of one sensory modality affects the content of another sensory modality. Although there is much evidence today of such interplay, it is unknown how to interpret them. Is it a one-way causal function between two separate states (e.g., a vi...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000407

How ghosting and stress impact vulnerable narcissism and maladaptive daydreaming: The role of mindfulness and rumination.

📅 2024-09-12 📖 1 min 📝 204 words CLINICAL NEUROSCIENCE

Summary: The present study aims to explore the influence of ghosting and stress on maladaptive daydreaming and vulnerable narcissism within a multivariate framework, using mindfulness and rumination as mediators. We employed a cross-sectional design and conducted it among 301 university students (M<...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000403

The conscious experience of thinking aloud: Are positive feelings associated with better think-aloud performance?

📅 2024-12-30 📖 1 min 📝 216 words CLINICAL NEUROSCIENCE

Summary: The think-aloud interview is a psychological procedure used to measure an individual’s cognition in solving a task. However, very little is known about how an individual’s conscious experience of the interview might be associated with their performance during the interview. This is consequential as ...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000410

Mystical experiences in virtual reality.

📅 2024-04-04 📖 1 min 📝 176 words CLINICAL NEUROSCIENCE

Summary: Virtual reality (VR) has shown promise as a psychological means of inducing mystical experiences. Here, we compared the subjective responses of 48 undergraduate volunteers to script-guided multiuser perceptual experiences in VR (VR perception [VR-P]) with their responses to a control condit...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000396

Effect of a peer-led meditation club on adolescents' well-being, empathy, and compassion.

📅 2024-09-09 📖 1 min 📝 197 words CLINICAL NEUROSCIENCE

Summary: Many studies have examined the effect of meditation sessions led by experienced teachers on personal benefits for adults. The present study investigated the effect of broad-based mindfulness practices with an adolescent peer facilitator on prosocial outcomes in addition to individual gains. The 20 s...

🔗 **Read full article:**
http://doi.org/10.1037/cns0000400

Prefrontal cortex activity of adolescents with problematic smartphone use in response to gains and losses: A functional near-infrared spectroscopy study.

📅 2026-09-10 📖 1 min 📝 237 words BEHAVIORAL NEUROSCIENCE

Summary: Problematic smartphone use (PSU) has become a widespread problem that is highly prevalent in adolescent population. Previous studies suggest that PSU is associated with impairment in reward processing system. Thus, increased sensitivity to rewards could make individuals more vulnerable to abusive be...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000668

Topiramate and anterior cingulate cortex high-frequency stimulation mitigate aggression and promote sociability in socially isolated male mice.

📅 2026-08-10 📖 1 min 📝 230 words BEHAVIORAL NEUROSCIENCE

Summary: Excessive aggression induced by social isolation (SI) has been closely linked to dysfunction in prefrontal circuits, particularly the anterior cingulate cortex (ACC). Topiramate, an antiepileptic drug known to enhance GABAergic and inhibit glutamatergic transmission, has demonstrated antiaggressive ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000663

Is memory spared in monkeys with selective lesions of the hippocampus?

📅 2026-08-03 📖 1 min 📝 239 words BEHAVIORAL NEUROSCIENCE

Summary: Normal explicit/declarative memory requires a system of anatomically related structures in the medial temporal lobe that includes the hippocampus and the entorhinal, perirhinal, and parahippocampal cortex. Although extensive medial temporal lobe damage in primates causes robust anterograde amnesia, ...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000658

Distal pup cues attenuate defensive reactivity and fear-potentiated startle magnitude in postpartum rats without altering sensorimotor gating.

📅 2026-07-09 📖 1 min 📝 241 words BEHAVIORAL NEUROSCIENCE

Summary: Using a novel behavioral paradigm that integrates fear-potentiated startle (FPS) and prepulse inhibition testing, this study examined how motherhood and infant cues impact innate defensive reactivity (as measured by the magnitude of basal startle reflex), attentional processing (as measured by prepu...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000661

A novel rat spatial inference task reveals rapid deduction of transitive relationships using schemas and deliberation.

📅 2026-07-13 📖 1 min 📝 264 words BEHAVIORAL NEUROSCIENCE

Summary: Inferential reasoning is a vital cognitive ability that enables animals to navigate novel situations by leveraging existing relational knowledge of memory schema, with hypothesized roles of prefrontal cortical—hippocampal circuits. Transitive inference (TI) tasks test the ability of subjects to infe...

🔗 **Read full article:**
http://doi.org/10.1037/bne0000660

Monthly Updates [April]

📅 2026-04-01 📖 4 min 📝 885 words FMHY

Summary: INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/april-2026

Monthly Updates [May]

📅 2026-05-01 📖 4 min 📝 892 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/may-2026

Monthly Updates [November]

📅 2026-11-01 📖 5 min 📝 1034 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/nov-2026

Show HN: Uncracked City, detective game where you question LLM suspects that lie

👤 passkey1510 📅 2026-11-03 📖 1 min 📝 171 words HACKER NEWS

> **Summary:** To give you some context on the project. I'm a solo founder and I use AI heavily in my work. The idea came from wanting players to discover cities. Each city has its own history, so every case is set in a real city with real public places. The crime and the people are on other hand invented. The ...

🔗 **Read full article:**
https://uncracked.city

C++26 Disapproved

👤 thatxliner 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://andreasfertig.com/blog/2026/11/cpp26-dis-failed-how-does-cpp-survive/ Comments URL: <a href="https://news.ycombinator.com/item?id=50360164">https://news.ycombinator.com/item?id=...

🔗 **Read full article:**
https://andreasfertig.com/blog/2026/11/cpp26-dis-failed-how-does-cpp-survive/

Giant canyon on Mars is still tectonically active

👤 gmays 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://lamont.columbia.edu/news/giant-canyon-mars-still-tectonically-active-today Comments URL: <a href="https://news.ycombinator.com/item?id=50360573">https://news.ycombinator.co...

🔗 **Read full article:**
https://lamont.columbia.edu/news/giant-canyon-mars-still-tectonically-active-today

Arrows to arrows, categories to queries (2025)

👤 firstbase 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://reasonablypolymorphic.com/blog/arrows-to-arrows/ Comments URL: https://news.ycombinator.com/item?id=50361107 Points: 26 # Co...

🔗 **Read full article:**
https://reasonablypolymorphic.com/blog/arrows-to-arrows/

Distinct cellular and genetic differences in colorectal cancer in young patients

👤 gmays 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: <a href="https://www.dana-farber.org/newsroom/news-releases/2026/study-reveals-distinct-cellular-and-genetic-differences-in-colorectal-cancer-in-younger-patients">https://www.dana-farber.org/newsroom/news-releases/2026/study-reveals-distinct-cellular-and-genetic-differences-in-colore...

🔗 **Read full article:**
https://www.dana-farber.org/newsroom/news-releases/2026/study-reveals-distinct-cellular-and-genetic-differences-in-colorectal-cancer-in-younger-patients

Jelly Sheet

👤 memalign 📅 2026-11-04 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://mrdoob.github.io/codevember-2026/01/ Comments URL: https://news.ycombinator.com/item?id=50361268 Points: 102 # Comments: 20

🔗 **Read full article:**
https://mrdoob.github.io/codevember-2026/01/

New software package: ERPLAB Studio

Steve Luck 2024-06-12 2 min 449 words ERP BOOT CAMP

Summary: We are excited to announce the release of a new EEG/ERP analysis package, ERPLAB Studio. We think it's a huge improvement over the classic EEGLAB user interface. See our cheesy <a href="https://www.youtube.com/watch?v=lIaKVQ9DD6E">...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/11/erplab-studio

Recording and slides now available for ERPLAB Studio webinar

Steve Luck 2024-06-28 1 min 30 words ERP BOOT CAMP

Summary: We held a webinar to demonstration ERPLAB Studio on 28 June 2024. Click here to access a recording. Click here to access a PDF of the slides.<...

🔗 **Read full article:**
https://erpinfo.org/blog/2024/6/28/recording-and-slides-now-available-for-erplab-studio-webinar

New Paper: Does the P3b component reflect working memory updating?

👤 Steve Luck 📅 2025-03-21 📖 7 min 📝 1540 words ERP BOOT CAMP

> **Summary:** Carrasco, C. D., Simmons, A. M., Kiat, J. E., & Luck, S. J. (in press). Enhanced working memory representations for rare events. Psychophysiology. https://doi.org/10.1111/psyp.70038 [<a href="https://doi.org/10.1101/2024.03.20...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/3/20/new-paper-oddball

10-Day ERP Boot Camp to be held June 15-24, 2026 in Davis, California

👤 Steve Luck 📅 2025-08-20 📖 1 min 📝 125 words ERP BOOT CAMP

> **Summary:** We have received another 5 years of funding from the National Institute of Mental Health, so we plan to hold ERP Boot Camps in each of the next 5 summers. The next one will be June 15-24, 2026 in Davis, California. The application portal will open around January 1, 2026. A...

🔗 **Read full article:**
https://erpinfo.org/blog/2025/8/20/boot-camp-summer-2026

Call for Papers: IEEE Brain Special Issue

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 36 words BRAIN

Summary: In a unique interdisciplinary collaboration with the IEEE's Society on Social Implications of Technology (SSIT) and IEEE Brain, J-FLEX is joining forces to explore both the technology of the Internet-of-Medical-Things (IoMT) solutions and medical wearables/ implantables.

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-journal-on-flexible-electronics/

IEEE Brain Joins the American Brain Coalition

👤 ieeebrain 📅 2025-03-03 📖 1 min 📝 61 words BRAIN

Summary: IEEE Brain is pleased to announce its acceptance as a nonprofit member of the American Brain Coalition (ABC), a prestigious alliance of over 150 organizations dedicated to advancing brain research, advocacy, and improving treatments for individuals affected by brain conditions. The ABC Board has ent...

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/ieee-brain-joins-the-american-brain-coalition-as-a-nonprofit-member/

Call for Papers: IEEE Transactions on Human-Machine Systems

👤 Adriel Carridice 📅 2025-06-18 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/braininsight-articles/call-for-papers-ieee-transactions-on-human-machine-systems/

Q&A with Dr. Alain Dagher, Neurologist and Professor at McGill University and the Montreal Neurological Institute

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dr-alain-dagher-neurologist-and-professor-at-mcgill-university-and-the-montreal-neurological-institute/

IEEE Advances Responsible Neurotechnology Standard

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/news-and-insights/ieee-brain-news/ieee-advances-responsible-neurotechnology-standard/

Q&A with Dhruv Vaish, PhD student at UC Berkeley and Best Poster Award Winner at the 2025 IEEE Brain Discovery and Neurotechnology Workshop

👤 ieeebrain 📅 2026-02-25 📖 1 min 📝 0 words BRAIN

🔗 **Read full article:**
https://brain.ieee.org/podcasts/qa-with-dhruv-vaish-phd-student-at-uc-berkeley-and-best-poster-award-winner-at-the-2025-ieee-brain-discovery-and-neurotechnology-workshop/

Active Lubrication Brakes in Tabletop and Hand-Held Haptic Surgical Drilling Simulators

📅 2026-06-29 📖 1 min 📝 222 words TRANSACTIONS HAPTICS

Summary: Haptic user interfaces that provide both kinesthetic and vibrotactile feedback have strong potential to enhance a wide range of medical training simulations. Yet, designing such devices remains challenging due to constraints related to integration, workspace, cost, and the need for diverse feedback ...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11585967

Reconciling Reduced the Sense of Ownership and Agency Toward a Virtual Keypad With Enhanced Haptic Experience Using Visually Adjusted Self-Haptics for Key-Input Interaction

📅 2026-07-06 📖 1 min 📝 250 words TRANSACTIONS HAPTICS

Summary: Self-Haptics is a tactile feedback technique where users utilize their own body to generate tactile sensations. This method allows the body to act purely as a prop, making it useful for virtual objects and user interfaces not directly related to the body. However, any method for maintaining the sens...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11595768

VibGrasp: Spatiotemporal Vibration Based Multimodal Haptic Rendering With a Lightweight Exo-Glove for 3D Shape Perception

📅 2026-04-20 📖 1 min 📝 173 words TRANSACTIONS HAPTICS

Summary: Haptic feedback has been employed to enhance shape perception during grasp interactions, improving immersion and task performance in extended reality (XR). Although wearable devices offer precise multi-phalangeal haptic feedback, their bulkiness and complexity often hinder practicality. In addition,...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11488477

Propeller-Based Handheld Force-Feedback Device for Navigation

📅 2026-04-22 📖 1 min 📝 168 words TRANSACTIONS HAPTICS

Summary: This article presents an evaluation of CrazyJoystick, a propeller-based handheld force-feedback device, for spatial navigation tasks in real-world and virtual reality environments. Building upon prior work validating the device's on-demand aerial deployment and directional cue discriminability, we i...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11493624

A Wearable Shoulder Skin-Deformation Device Generating Illusory Forces for Posture Correction

📅 2026-04-22 📖 1 min 📝 208 words TRANSACTIONS HAPTICS

Summary: The hanger reflex is a haptic illusion in which skin-shear deformation elicits a directional force sensation. This study presents a wearable haptic device that uses four pneumatic airbags to apply controlled skin shear to both shoulders for posture guidance. Although previous work has shown that sho...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11489488

Sensitivity at the Edges of the Finger Enables Mixed Reality Force Feedback

📅 2026-04-06 📖 1 min 📝 155 words TRANSACTIONS HAPTICS

Summary: We introduce a new paradigm for rendering high-force sensations by stimulating the edges of the fingerpad rather than the central contact surface. This non-blocking approach preserves natural finger use, enabling simultaneous physical and virtual interactions. To investigate its feasibility, we prop...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11475177

ThermalTexture: Temperature-Integrated Surface Rendering Using Thermal Referral Induced by Texture Slip-Feedback

📅 2026-03-23 📖 1 min 📝 204 words TRANSACTIONS HAPTICS

Summary: Thermal–vibration feedback has been reported to elicit the thermal referral phenomenon, offering new possibilities for enhancing user experience. Yet, little research has explored whether thermal referral occurs in the presence of texture-slip feedback. To address this, we developed a custom haptic ...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11454471

Guest Editorial: World Haptics Conference 2025

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11716332

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📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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Front Cover

📅 2026-09-30 📖 1 min 📝 1 words TRANSACTIONS HAPTICS

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http://ieeexplore.ieee.org/document/11716326

Multi-Channel Electromagnetic Interference Elimination for Shielding-Free MRI Using Null Operations

📅 2026-01-23 📖 1 min 📝 184 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Emerging technologies for electromagnetic interference (EMI) elimination have enabled radio frequency (RF) shielding-free magnetic resonance imaging (MRI), significantly reduced costs and increased accessibility. Existing methods often rely on multiple external sensors for EMI elimination...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11361141

Double Banking on Knowledge: A Unified All-in-One Framework for Unpaired Multi-Modality Semi-Supervised Medical Image Segmentation

📅 2026-01-21 📖 1 min 📝 231 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Multi-modality (MM) semi-supervised learning (SSL) based medical image segmentation has recently gained increasing attention for its ability to utilize MM data and reduce reliance on labeled images. However, current methods face several challenges: (1) Multiple networks assignment hinder scalability...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11359738

Virtual Imaging-Guided Harmonization of Chest X-Rays Using a Generative Adversarial Neural Network

📅 2026-01-26 📖 1 min 📝 247 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: To reduce variability in chest radiography (CXR) from acquisition and post-processing, and assess whether harmonization improves image quality and downstream diagnostic performance. Methods: A generative adversarial network (GAN) was trained exclusively on virtual images produced by a Vir...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11364153

Self-Supervised Speech Representations for Sleep Apnea Severity Prediction

📅 2026-01-20 📖 1 min 📝 253 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Obstructive sleep apnea (OSA) is a prevalent yet underdiagnosed condition characterized by repetitive upper airway obstruction during sleep. Current gold standard diagnostic standards rely on polysomnography (PSG), which is resource-intensive. Since upper airway characteristics impact bot...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11359456

Rectification of Cornea Induced Distortions in Microscopic Images for Assisted Ophthalmic Surgery

📅 2026-01-20 📖 1 min 📝 240 words TRANSACTIONS BIOMEDICAL ENGINEERING

Summary: Objective: Image distortions induced by the high refractive power of the eye's optical components challenge the accuracy of geometric information derived from intra-operative sensor data in ophthalmic surgery. Correcting these distortions is vital for advancing surgical assistance systems that rely ...

🔗 **Read full article:**
http://ieeexplore.ieee.org/document/11359487

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http://ieeexplore.ieee.org/document/11743499

IEEE Transactions on Biomedical Engineering Handling Editors Information

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IEEE Engineering in Medicine and Biology Society Publication Information

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Technical Parameters and Feedback Control for Blood-Brain Barrier Permeability Enhancement by Focused Ultrasound

📅 2025-12-02 📖 1 min 📝 166 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Focused ultrasound combined with intravenously infused microbubbles has been shown to effectively enhance the permeability of the blood-brain barrier, facilitating drug delivery to the brain. A wide range of technical parameters has been evaluated through preclinical studies and clinical trials. Gen...

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Decoding Spikes From Multiunit Data

📅 2026-01-22 📖 1 min 📝 250 words REVIEWS BIOMEDICAL ENGINEERING

Summary: Communication and control in biological systems is mediated by the timing of discharges –spikes– from excitable cells such as neurons and muscle fibers. Each spike is associated to a characteristic waveform that can be captured by sensors. The waveform's characteristics depend on the cell's biophysi...

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Distinct Patterns of Syntactic Errors Doubly Dissociate in Chronic Poststroke Aphasia

📅 2026-10-20 📖 1 min 📝 195 words COGNITIVE NEUROSCIENCE

Summary: The lesion correlates of syntactic deficits in aphasia remain poorly understood. Previous studies have suggested that distinct error types in expressive syntax may be associated with damage to different brain regions, but the specific lesion correlates of these errors have not been fully delineated....

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Neural Distinctions between Social and Nonsocial Reward Anticipation: An Examination of Activation Magnitude, Representational Patterns, Individual Differences, and Functional Connectivity

📅 2026-10-19 📖 1 min 📝 235 words COGNITIVE NEUROSCIENCE

Summary: Prevailing theories emphasize a “common neural currency” for social and nonsocial rewards, particularly during reward anticipation. However, direct comparisons of social and nonsocial reward anticipation remain limited, and clinical observations suggest that social reward processing can be selective...

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Exploring Neural Mechanisms Underlying Error-related Impairments in Active Working Memory Suggests an Adaptive Shielding of Contents during Cognitive Control

📅 2026-10-19 📖 1 min 📝 302 words COGNITIVE NEUROSCIENCE

Summary: Goal-directed behavior relies on cognitive flexibility—the ability to rapidly adapt ongoing thoughts and behaviors while preserving task-relevant information. The performance monitoring system optimizes such behavior by detecting and evaluating errors, whereas the working memory (WM) system maintain...

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Slow Oscillations Modulate Overnight Brain Changes in Working Memory Function

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM), the transient storage and manipulation of information, is a cognitive function that can improve with training. Accumulating evidence suggests that sleep, particularly slow oscillations (SOs) that occur during nonrapid eye movement sleep, supports this improvement. Yet, how sleep...

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Transformer Mechanisms Mimic Frontostriatal Gating Operations When Trained on Human Working Memory Tasks

📅 2026-10-20 📖 1 min 📝 237 words COGNITIVE NEUROSCIENCE

Summary: Working memory (WM) is thought to rely on sophisticated frontostriatal mechanisms for selective gating, supporting updating and readout of information to and from distinct memory “addresses” (neural populations). According to computational models, capacity limitations in WM arise due to challenges i...

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Visual Perspective Shapes Subjective Experience: Dissociable Parietal Contributions to the Constructive Nature of Memory

📅 2026-10-19 📖 1 min 📝 223 words COGNITIVE NEUROSCIENCE

Summary: Subjective features of memory are often treated as secondary to the objective content of remembered events. However, growing evidence suggests that these features actively shape how memories are constructed, experienced, and used. Rather than treating visual perspective as a peripheral correlate of ...

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Memory Traces Reflect How They Were Last Accessed

📅 2026-10-19 📖 1 min 📝 216 words COGNITIVE NEUROSCIENCE

Summary: Episodic memories are known to change with each act of retrieval. We hypothesize that accessing semantic knowledge in different ways during episodic retrieval—from unique perceptual features to taxonomic and thematic context—shapes how those memories are subsequently represented during recognition. ...

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The Temporal Dynamics of Attentional Allocation during Counterfactual Learning

📅 2026-10-19 📖 1 min 📝 194 words COGNITIVE NEUROSCIENCE

Summary: Counterfactual learning, the ability to learn from what could have happened under different circumstances, is a key cognitive mechanism supporting behavioral adaptation. While its neural and computational underpinnings are increasingly understood, the temporal dynamics of attention toward factual an...

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Theta and High-alpha Phase-locking Dynamics Unveil Face Perception in Ambiguous Stimuli

📅 2026-10-19 📖 1 min 📝 227 words COGNITIVE NEUROSCIENCE

Summary: Previous studies emphasize phase synchronization as a fundamental mechanism for integrating local features into coherent percepts. We employed a novel paradigm and dynamic graph analysis based on EEG to track neural dynamics associated with perceiving a face (or not) while keeping the stimuli identi...

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Capturing the Attentional Trade-off between Speech Planning and Comprehension

📅 2026-10-19 📖 1 min 📝 255 words COGNITIVE NEUROSCIENCE

Summary: In conversation, future speakers often plan speech simultaneously with comprehension, which means that they must divide attentional resources between these processes. In this EEG study, we used responses to linguistic attention probes (i.e., syllable “BA” presented during spoken sentences) to track ...

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👤 dziura 📅 2025-12-10 📖 1 min 📝 18 words EMBS

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Welcoming & Congratulating Our Newly Elected EMBS ExCom Members

👤 Nancy Zimmerman 📅 2025-12-14 📖 1 min 📝 18 words EMBS

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👤 Nancy Zimmerman 📅 2026-05-04 📖 1 min 📝 51 words EMBS

> **Summary:** Nominate volunteers for terms starting Jan 1, 2027: Asia Pacific Rep (3 yr), Technical Rep (3 yr), Student Rep (2 yr), Young Professional Rep (2 yr). Help shape the future… <a class="continue" href="https://www.embs.org/uncategorized/adcom-nominations-2027/">Continue Reading EMBS AdCo...

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👤 Deidre Artis 📅 2026-08-28 📖 1 min 📝 17 words EMBS

> **Summary:** The post IEEE Pulse Editor-in-Chief Search – Submit an Application! appeared first on IEEE EMBS.

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Exploratory bone–brain transcriptomic associations and an ODE-based digital-twin framework for fracture healing

📖 1 min 📝 15 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Chao Ma, Wei Lu, Ruipeng Guo

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A Hierarchical-Dynamic integration model for the formation of interpersonal neural synchrony during spoken communication

📖 1 min 📝 13 words BRAIN RESEARCH

Summary: Publication date: 1 January 2027 Source: Brain Research, Volume 1894 Author(s): Yating Bai, Qingfang Zhang

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https://www.sciencedirect.com/science/article/pii/S0006899326004567?dgcid=rss_sd_all

Dynamic multi-scale topology learning and cross-modal graph fusion for autism spectrum disorder diagnosis

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Huadong Zhang, Junyong Zhao, Shaoqi Wu, Krisztián Szigeti, Kellermayer Miklós, Jiande Sun, Jianping Qiao

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Diagnostic classification of ASD improves with positively sparsified DTI inter-regional correlation and graph-theoretical artificial intelligence

📖 1 min 📝 22 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Ravi Ratnaik, Sriram Kumar P., Lavanya Selvaganesh, Sandeep Singh Sengar, Jac Fredo Agastinose Ronickom

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Fast rate coding in heterogeneous spiking networks

Smeets, K., Wang, S., Schmutz, V., Gerstner, W.

2026-11-03 1 min 165 words

BIORXIV NEUROSCIENCE

Summary: A network of spiking neurons is said to operate in the rate coding regime if its dynamics can be accurately described by a continuous rate network. But how can discrete spike-based communication between neurons give rise to continuous rate-based dynamics? Computational studies showing that heterogen...

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Self-Other Neural Representational Distinction is Associated with Distorted Social Predictions in Social Anxiety

Qi, Y., Chen, Y., Hu, Y., Fan, Q., Wang, L., Yang, G., Hein, G., Wu, H.

2026-11-03 1 min 250 words

BIORXIV NEUROSCIENCE

Summary: Background: Social anxiety (SA) is characterized by distorted negative expectations about the future. Because optimism bias is inherently comparative, the extent to which these predictions reflect a fundamental disruption in self- versus other-referential processing remains poorly understood. Method...

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Diagnose Harmonisation (DHARM): An open-source tool for the harmonisation and analysis of batch effects in MRI-derived measures.

👤 Turnbull, J., Bhalerao, G., Nichols, T. E., McCarthy, P., Gillis, G., Blane, J., Zhaoyi Mo, B., Warrington, S., Sotiropoulos, S. N., Besenczi, R., Alfaro-Almagro, F., Smith, S. M., Griffanti, L.

📅 2026-11-03 📖 1 min 📝 321 words BIORXIV NEUROSCIENCE

Summary: Technical differences between scanners, acquisition protocols, machine hardware and processing pipelines can introduce systematic variation into MRI-derived measures. Numerous harmonisation methods have been developed to reduce these effects. However, validation of these methods is often performed u...

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Strategy-dependent causal relationships between large-scale cortical activity and perceptual decisions

👤 Mohammadi, Z., Pillow, J. W., Pinto, L., Glaser, J. I.

📅 2026-11-03 📖 1 min 📝 178 words

BIORXIV NEUROSCIENCE

Summary: Animals can adopt different internally generated strategies to solve decision-making tasks, but whether cortical regions distinctly contribute to decisions under different strategies remains unknown. To address this problem, we analyzed three datasets comprising either cortical widefield imaging, or...

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Differential effects of chronic hypertension and sex on vascular density along the septotemporal axis of the hippocampus in a rat model of VCID

Luchetta, J., Schellinger, E. D., Bolton, Z., DeMarco, G. M., Chapman, A. C.

2026-11-03 1 min 207 words

BIORXIV NEUROSCIENCE

Summary: Background: Chronic hypertension is a major modifiable risk factor for vascular contributions to cognitive impairment and dementia (VCID), yet vascular changes that precede cognitive dysfunction remain poorly understood. This is particularly relevant for the hippocampus, a deep brain structure criti...

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Metabolic complexity in the human brain supports metastable network dynamics and declines in normative ageing

Deery, H., Liang, E., Moran, C., Egan, G. F., Jamadar, S. D.

2026-11-03 1 min 253 words

BIORXIV NEUROSCIENCE

Summary: The adaptive capacity of the brain is tied to the complexity of its neural activity. While entropy of brain signals has been used to characterise this complexity in haemodynamic and electrophysiological systems, it remains unknown whether entropy contributes to the temporal dynamics of the metabolic...

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Norepinephrine inhibits Afferent activation of NTS Catecholamine Neurons in Mice

Fernandes, L. G., Jara, J. I., Appleyard, S. M.

2026-11-03 1 min 180 words

BIORXIV NEUROSCIENCE

Summary: The nucleus tractus solitarius (NTS) integrates visceral sensory input to regulate feeding behavior, with catecholaminergic (CA) and neuropeptide Y (NPY)-expressing neurons playing key roles in energy homeostasis. Recent evidence suggests functional heterogeneity among CA-NTS neurons, including dist...

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A Preventive Effect of Exercise in a Mouse Model of Essential Tremor

Constantopoulos, S., Bergdahl, A., Courtemanche, R.

2026-11-03 1 min 222 words

BIORXIV NEUROSCIENCE

Summary: The effects of exercise have not been widely studied for potential benefits in essential tremor (ET), while this has been growing for other neurodegenerative disorders. This study examines the effects of two types of exercise [high-intensity interval training (HIIT) and resistance training (RT)] on ...

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Chronic hypertension disrupts the blood-brain barrier in the ventral hippocampus in females but not males in a rat model of vascular cognitive impairment

Luchetta, J., DeMarco, G. M., Thaqi, A., Bernstein, G. A., Chavara, S. E., Schellinger, E. D., Chen, D., Chapman, A. C.

2026-11-03 1 min 318 words BIORXIV NEUROSCIENCE

Summary: Background: Chronic hypertension is a leading cause of cerebrovascular disease, ischemic stroke, and cognitive impairment. In addition, anxiety and hypertension often co-exist, however, little is known about the relationship between the two. Understanding how chronic hypertension impacts limbic brai...

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Lesion-Derived Estimated Functional and Structural Connectomes Predict Post-Stroke Recovery Outcomes

Gillies, C., Jamison, K., Boes, A., Kuceyeski, A.

2026-11-03 1 min 278 words BIORXIV NEUROSCIENCE

Summary: Accurate prognosis after stroke remains challenging, limiting the ability to predict individual recovery. While advanced connectome MRI can provide further prognostically-useful information regarding affected brain networks, its impractical nature has given rise instead to lesion-based estimates of ...

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Different modes of stimulation at the Neiguan acupoint (PC6) regulate different EEG microstates under VR immersion conditions

Guangjun Wang
2026-10-22 1 min 302 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionAcupuncture at the Neiguan acupoint (PC6) alleviates postoperative and chemotherapy-induced nausea and vomiting. Nevertheless, the central regulatory effects of PC6 stimulation on virtual reality motion sickness (VRMS), a form of physiological vertigo triggered by visual-vestibular senso...

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt
2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

Suzy J Styles 2025-12-08 1 min 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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An integrated CT/MRI diagnostic system for pancreatic cystic lesions that supports pathological classification and risk stratification

Jihui Hao 2026-11-03 1 min 22 words LOW VISION

Summary: CONCLUSIONS: SIRS-PCL is a pathology-based, quantitative decision-support system for preoperative pathological subtype classification and pathological risk stratification of surgically evaluated neoplastic PCLs.

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Foveal Crowding and Reduced Binocular Advantage In Myopic Anisometropia Under Brief Stimulus Presentation

👤 Jianfeng Wu 📅 2026-11-03 📖 1 min 📝 32 words LOW VISION

> **Summary:** CONCLUSIONS: In optically corrected MA patients, brief stimulus presentation revealed abnormal foveal crowding in response time and a reduced binocular advantage, suggesting decreased efficiency in processing visual information under high perceptual demand.

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Blinded by inequity: a scoping review of income-related and structural disparities in eye care in the United States

👤 Carrie B Dolan 📅 2026-11-03 📖 1 min 📝 67 words LOW VISION

> **Summary:** CONCLUSION: Income-related disparities in eye care access are associated with structural and financial barriers linked to reduced access to timely services. Addressing these inequities may require greater alignment among recognized needs, promising interventions, and policy priorities, as well as en...

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Suprachoroidal hemorrhage and detachment in the non-laser photocoagulation posterior pole after intravitreal Anti-VEGF injection: A case report

Xionggao Huang 2026-11-03 1 min 63 words LOW VISION

Summary: Intravitreal anti-vascular endothelial growth factor (anti-VEGF) injections are widely used for diabetic macular edema (DME) and neovascular ocular diseases. Suprachoroidal hemorrhage and detachment (SCHD) in the posterior pole non-laser photocoagulation area is an extremely rare complication after ...

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Visual Perceptual Learning Based on Adaptive Contrast Detection Training Enhances Vision and Contrast Sensitivity in Diabetic Retinopathy

Lixia Feng 2026-11-03 1 min 37 words LOW VISION

Summary: CONCLUSIONS: The results reveal significant plasticity in the visual system in DR and suggest that contrast detection training may extend to motion perception tasks. Our study provides additional support for VPL as a potential treatment for DR.

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Reading disability in glaucoma: A neglected dimension of visual disability

Michele Figus 2026-11-03 1 min 62 words LOW VISION

Summary: Reading disability is a common and clinically relevant, yet under-recognized, functional impairment in glaucoma, with significant impact on independence, daily activities, and quality of life, particularly in an increasingly digital environment. Reading performance is influenced by a complex interpl...

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The Amsler Grid Use Study (AGUS): Amsler Grid Nonadherence is Driven by Forgetfulness and Health Beliefs But Not Age

Lisa Nivison-Smith 2026-11-03 1 min 33 words LOW VISION

Summary: CONCLUSIONS: Amsler grid nonadherence was primarily unintentional and mostly due to forgetfulness. Poor adherence was associated with greater concerns in Amsler grid use, indicating modifiable factors that can be addressed to improve adherence.

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Intrastromal Antibiotic Therapy for Recalcitrant Bacterial Keratitis

👤 Mark Chehade 📅 2026-11-03 📖 1 min 📝 52 words LOW VISION

Summary: CONCLUSIONS: Adjunctive intrastromal antibiotic therapy achieved a high rate of infection resolution in this cohort of recalcitrant bacterial keratitis, with a low complication rate. This approach may be considered in refractory cases or when unable to utilize conventional topical therapy. However, ...

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An open-source eye-tracking system adaptable to diverse terrestrial vertebrates reveals state-dependent visual acquisition in lizards

👤 Mark Shein-Idelson 📅 2026-11-03 📖 1 min 📝 68 words LOW VISION

Summary: Gaze shifts are fundamental to visual exploration and processing, yet empirical knowledge is largely confined to a small number of model species, limiting our understanding of general principles of visual acquisition across animals. Here, we introduce a versatile, low-cost, and open-source framework...

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Ocular Considerations for the Department of War's New Testosterone Initiative

👤 Harkaran S Rana 📅 2026-11-03 📖 1 min 📝 75 words LOW VISION

Summary: On 15 July 2026, the Secretary of War announced that the Department of War will begin annual testosterone deficiency screening for male Active Duty and Reserve Component personnel aged 30 and older as part of the periodic health assessment, with screening available on request to younger servicemembe...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

👤 Hu Zhao 📅 2026-11-01 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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A phase-calibrated combined electrical-sensory neuromodulation platform for short-Term frontoparietal network modulation and cognitive fatigue resistance

👤 Qianxiang Zhou
📅 2026-11-02 📖 1 min 📝 57 words

TDCS TACS TRNS

Summary: Non-invasive multimodal neuromodulation is often limited by cross-modal temporal misalignment caused by modality-specific conduction delays. To address this challenge, we developed a microcontroller-based multiphysics stimulation platform integrating transcranial alternating current stimulation (tAC...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh

2026-11-02 | 1 min | 48 words | TDCS TACS TRNS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Repeated prefrontal transcranial direct current stimulation improves behavioral regulation and modulates resting-state EEG microstate dynamics in young adults with problematic smartphone use: a randomized sham-controlled longitudinal study

Yingying Zeng
2026-11-03 1 min 62 words TDCS TACS TRNS

Summary: CONCLUSIONS: Repeated prefrontal tDCS produced concurrent behavioral improvement and selective reorganization of resting-state EEG microstate dynamics in young adults with PSU. These findings provide preliminary evidence that repeated prefrontal tDCS is associated with concurrent behavioral changes ...

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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What are the neural networks that have been modulated by anatomically targeted therapies to promote recovery of consciousness? A scoping review

👤 Konstantinos Margetis
📅 2026-11-03 📖 1 min 📝 63 words

TDCS TACS TRNS

Summary: Disorders of Consciousness (DoC) arise from insults to neural networks responsible for awareness and arousal. DoC subtypes are frequently misdiagnosed, as mechanisms underlying consciousness remain poorly understood. Neurostimulation has been investigated as a potential therapy for DoC recovery; how...

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Effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field

Prinpida Sonthiphand 2026-11-03 1 min 68 words TDCS TACS TRNS

Summary: Due to climate change and increasing food demand, conventional agriculture relies heavily on chemical fertilizers. This study investigated the effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field. Metag...

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Transcranial Pulse Stimulation in Parkinson's Disease: A Systematic Review

Marcelo Lourenço da Silva 2026-11-03 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: TPS shows preliminary potential to modulate sensorimotor networks and improve selected motor outcomes in PD. However, evidence remains insufficient to establish overall clinical efficacy. Larger, rigorously controlled trials are required to determine sustained benefits, optimal stimulat...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

Yulong Bai 2026-10-30 1 min 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

Teri Kim 2026-10-30 1 min 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

Dandan Wan 2026-10-31 1 min 17 words FNIRS

Summary: CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

Kanghui Li 2026-10-31 1 min 68 words FNIRS

Summary: The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Correction: EBA-related body-preferring responses in the occipitotemporal cortex measured with functional near-infrared spectroscopy

Min Wang 2026-11-03 1 min 6 words FNIRS

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1914488.].

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Supportive parenting moderates the association between neural synchrony and irritability: Evidence from mother-son dyads

Xiuyun Lin
2026-11-03 1 min 62 words FNIRS

Summary: While research on parent-child neural synchrony using fNIRS hyperscanning has expanded, its association with child psychopathology, especially transdiagnostic symptoms such as irritability, remains poorly understood. Moreover, no prior studies have explored the potential contextual factors that may ...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

👤 Blandina T Mmbaga 📅 2026-10-31 📖 1 min 📝 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

👤 Matias Valdenegro-Toro 📅 2026-10-31 📖 1 min 📝 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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The application of brain-computer interface on cognitive function and activities of daily living in stroke patients: a systematic review and meta-analysis

Xuan Gong 2026-11-03 1 min 52 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-based training shows preliminary promise for improving cognitive function and ADL in stroke patients. Current evidence remains insufficient to recommend BCI for emotion or speech rehabilitation. However, the GRADE certainty of evidence was rated as low for cognitive function and very...

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Mapping Brain-Computer Interface Research in Depression (1995-2026): A Bibliometric Analysis

Xianlun Duan 2026-11-03 1 min 31 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The field has evolved from EEG-based characterization toward network-level and real-time neurofeedback research. These findings identify emerging research directions while highlighting the need for further methodological standardization and clinical validation.

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Monthly Updates [October]

📅 2026-10-01 📖 5 min 📝 1097 words FMHY

> **Summary:** <h1 id="wiki-updates" tabindex="-1">Wiki Updates </h1>   Re-ordered Live Sports based on the <a href="https://challonge.com/ajqvt0wo/fullscreen" rel=...

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https://fmhy.net/posts/oct-2026

Designing a zero-dependency Python scraping SDK: sticking with pure stdlib vs adding httpx?

👤 /u/Solid-Bidl 📅 2026-11-03 📖 1 min 📝 331 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> Hi r/Python, I have been maintaining a lightweight Python SDK (parse-everything on PyPI) that wraps an extraction and anti-bot bypass pipeline (handling TLS JA3/JA4 fingerprint impersonation, Cloudflare Turnstile, DataDome, etc., along...

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https://www.reddit.com/r/Python/comments/1xn28q6/designing_a_zerodependency_python_scraping_sdk/

Introduction to Programming in ATS

👤 ibobev 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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macOS 27 does not support CoreStorage volumes

👤 colejohnson66 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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Preparing for the Flipper One Kickstarter and opening pre-orders

👤 user_7832 📅 2026-11-03 📖 1 min 📝 14 words HACKER NEWS

> **Summary:** https://docs.flipper.net/one <hr /> Comments URL: https://news.ycombinator.com/item?id=50356994 Points: 159 # Comments: 74

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https://flipper.net/pages/flipper-one

Pratt parsing is a black box

👤 ibobev 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://vinipsmaker.github.io/blog/article/pratt-parsing-is-a-black-box/ Comments URL: <a href="https://news.ycombinator.com/item?id=50357457">https://news.ycombinator.com/item?id=50357457</...

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https://vinipsmaker.github.io/blog/article/pratt-parsing-is-a-black-box/

Show HN: Kairos – Open-source risk and fraud decisioning platform

👤 ayanb9440 📅 2026-11-03 📖 1 min 📝 283 words HACKER NEWS

> **Summary:** Hey HN, Ayan here from Finic. We’re launching Kairos, an open source alternative to Sift, Sardine, and other real-time fraud decisioning platforms. Kairos addresses two of the biggest problems facing traditional decisioning tools: a lack of labeled data to train on and dealing with fraudsters tha...

🔗 **Read full article:**
https://github.com/finic-ai/kairos

US tech giants' hidden AI debt swells 90% to $3.2T in 6 months

👤 angry_impala 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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https://asia.nikkei.com/business/technology/artificial-intelligence/us-tech-giants-hidden-ai-debt-swells-90-to-3.2tn-in-6-months

Safari introduces Adaptive Dark Mode in 27.3

👤 paddingtone 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://webkit.org/blog/18541/introducing-adaptive-dark-mode/ Comments URL: https://news.ycombinator.com/item?id=50358562 Points: 203<...

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https://webkit.org/blog/18541/introducing-adaptive-dark-mode/

The Little Book of Generative AI Foundations: An Intuitive Mathematical Primer

👤 ibobev 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://arxiv.org/abs/2605.29713 Comments URL: https://news.ycombinator.com/item?id=50358582 Points: 148 # Comments: 18

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The Internals of PostgreSQL

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Plasma p-tau181, Aβ1-42, and GFAP across Alzheimer's disease, mixed-type Alzheimer's disease, mild cognitive impairment, anxiety disorders, and depressive disorders

📖 1 min 📝 17 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Haifeng Lou, Chenyun Huang, Ji Shen, Dake Wang

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https://www.sciencedirect.com/science/article/pii/S0006899326004683?dgcid=rss_sd_all

RAD51 expression in pan-cancer and its prognostic and immune mechanisms in glioma

📖 1 min 📝 35 words BRAIN RESEARCH

> **Summary:** Publication date: 15 January 2027 Source: Brain Research, Volume 1895 Author(s): Liubing Hou, Yu Wang, Xiang Zhan, Jiayuan Li, Zizhou Zhang, Weijing Cai, Linlin Su, Wei Han, Xuetao Han, Yue Han, Xiaolan Lv, Xiaoying Xue, Huandi Zhou

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Genetic variants in NEF2L2 rs35652124 and GABRB3 rs2081648 genes: association with autism spectrum disorder in Iraqi children

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Zainab Mohammed Burhan, Wisam Hindawi Hoidy

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Propofol suppresses intrinsic excitability and alters excitatory synaptic plasticity in ventral tegmental area dopamine neurons

📖 1 min 📝 14 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Min Hee Lee, Dong Kwan Kim

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https://www.sciencedirect.com/science/article/pii/S0306452226006986?dgcid=rss_sd_all

Improved motor learning in adenosine A1 receptor knockout mice

📖 1 min 📝 19 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Maxx Freed, Noah Raffone, Devin Wilke, R. Holly Fitch, Maxim Volgushev

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007177?dgcid=rss_sd_all

Feasibility of an 8-week lifestyle intervention combining exercise, diet, cognitive training, and social engagement in individuals with mild cognitive impairment: a pilot randomized controlled trial

📖 1 min 📝 26 words NEUROSCIENCE JOURNAL

Summary: Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Cai-Lian Wang, Hong Chen, Ruo-Yu Xu, Na Wei, Xiao-Qin Su, Jing Li, Ci Li, Jun-E Liu, Yu-Chen Qiao

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007219?dgcid=rss_sd_all

Characterization of the hemodynamic response function in default mode network subregions for task-evoked negative BOLD responses

📖 1 min 📝 29 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Hengda He, Bardiya G. Yazdi, Amirreza Sedaghat, Seyed Hani Hojjati, Sindy Ozoria, Peter Chernek, Jenseric Calimag, Farnia Feiz, Qolamreza R. Razlighi

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A multimodal MRI study of DTI-ALPS and cerebral blood flow changes across the Alzheimer's disease continuum

📖 1 min 📝 38 words NEUROIMAGE

Summary: Publication date: 1 December 2026 Source: NeuroImage, Volume 343 Author(s): Yian Gao, Manxi Xu, Peiling Ou, Haoyu Yang, Jia Wang, Wulong Liu, Yu Sun, Wenying Du, Yuli Wang, Tongxi Liu, Hui Xia, Liping Fu, Ni Shu, Zhen Yuan, Guolin Ma

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006373?dgcid=rss_sd_all

A phase decoder for reconstructing missing muscle activity

👤 Bahdasariants, S., Yakovenko, S. 📅 2026-11-03 📖 1 min 📝 140 words BIORXIV NEUROSCIENCE

Summary: Upper-limb assistive devices are commonly controlled with electromyographic (EMG) signals, but nerve injury, muscle loss, or practical limits on electrode placement can leave only a few usable signals, which constrains the decoding of intended movements. Here, we introduce a decoder that reconstruct...

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https://www.biorxiv.org/content/10.64898/2026.10.28.762227v1?rss=1

State-dependent emergence of coupling between oscillatory and aperiodic neural activity

Lu, Y., Li, Z., Mao, H., He, X., Liu, D., Chen, J., Tao, L., Xiao, Z.-C., Tian, X.

2026-11-03 1 min

150 words

BIORXIV NEUROSCIENCE

Summary: Neural oscillations and aperiodic activity are two quintessential mesoscopic-level features arising from shared circuits, yet what governs their relationship remains unclear. Here we show that coupling between oscillatory frequency and aperiodic exponent is sporadic and state-dependent. Bias-reduced...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.29.762761v1?rss=1

Pubertal timing is a critical dimension of adolescent brain vulnerability

Szakacs, H., Gombos, F., Kringelbach, M. L., Deco, G., Kovacs, I.

2026-11-03 1 min

200 words

BIORXIV NEUROSCIENCE

Summary: Adolescence is a period of rapid brain development, during which biological maturational timing varies substantially between individuals. This variability may drive specific developmental vulnerabilities to adverse experiences with long-term consequences, warranting early identification of at-risk i...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.02.763578v1?rss=1

Paraquat Inhalation Produces Male-Biased Dopaminergic Neuron Loss and Motor Deficits in Mice: Relevance to Parkinsons Disease

👤 Reina-Gonzalez, P. J., Boos, J. R., Abu-Salah, A., Anchan, A., Strazdins, R., Ay, M., Shah, A., Chalupa, D., Richardson, J. J., Sarkar, S., Cory-Slechta, D.

📅 2026-11-03 📖 1 min 📝 247 words BIORXIV NEUROSCIENCE

Summary: Long-term exposure to paraquat (PQ), a widely used herbicide, has been linked to increased risk of Parkinsons Disease (PD) in agricultural workers and rural communities in the United States (US), findings further supported by experimental animal studies. Although banned in many countries across the ...

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Imipramine binds to amyloid-β and extends the survival of Alzheimer's disease patients

Johnson, N. R., Beham, J., Sillau, S., Read, J., Wang, A. C.-J., Ahmed, M. M., Quang, D., Stephenson, D., Henen, M., Vogeli, B., Chial, H. J., Welty, R., Vugmeyster, L., Potter, H.

2026-11-03 1 min 288 words BIORXIV NEUROSCIENCE

Summary: Background Apolipoprotein E (apoE) plays numerous roles in the pathogenic pathway to Alzheimer's disease (AD). We previously performed a high-throughput screen and identified the antidepressant imipramine as an inhibitor of the interaction between apoE and amyloid-{beta} (A{beta}). We also found tha...

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NeuralFieldManifold: Uncovering the Manifold Geometry of Local Field Potentials

Fallah, K., Chen, H. N., Singha, R., Kong, E., Turi, G. F., Losonczy, A., Zabeh, E.

2026-11-03

1 min 283 words BIORXIV NEUROSCIENCE

Summary: Manifold geometry has become a powerful framework for understanding how the brain organizes high-dimensional neural activity into compact, interpretable structure. To date, however, most theoretical and methodological advances in neural manifold analysis have focused on spiking activity, leaving unr...

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Hmox1/CO regulates retina regeneration via modulation of Wnt/β-catenin signalling and induction of vaspb in zebrafish

Sherpa, R. D., Sen, A., Chaudhuri, A. G., Hui, S. P.

2026-11-03 1 min 258 words

BIORXIV NEUROSCIENCE

Summary: Muller glia (MG) in the zebrafish retina retain a latent stem cell capacity that is absent in mammals, yet the injury responsive signals that drive this reprogramming remain incompletely defined. Here, we identify heme oxygenase-1 (Hmox1), the stress-inducible, rate-limiting enzyme of heme catabolis...

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Brain-wide representations of olfactory behavior in C. elegans

Casademunt, H., Zhang, H., Laver, J., Park, C. F., Kim, S., Fredenberg, W., Hung, W., Calarco, J., Zhen, M., Samuel, A. D. T.

2026-11-03 1 min 196 words

BIORXIV NEUROSCIENCE

Summary: Olfactory navigation emerges from dynamic interactions between sensory input, brain activity state and the animal's own movements. In C. elegans, a compact nervous system enables brain-wide imaging in behaving animals exposed to controlled olfactory environments, with an undulatory locomotion that r...

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Graph neural network approach for inferring neuronal connectivity

Gupta, A., Neveu, C. L., Byrne, J. H.

2026-11-03 1 min 177 words

BIORXIV NEUROSCIENCE

Summary: Understanding the connectivity of neuronal networks is essential for understanding brain function. Current anatomical mapping techniques, such as serial section microscopy and viral tracing, are resource-intensive and cannot capture the efficacy of connectivity. A complementary approach is to infer ...

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Noise-referenced multi-shell diffusion-weighted imaging for regional cerebrospinal fluid thermometry

Behler, C., Breakspear, M., Ramadan, S., Dao, T. T., Close, T. G., Vos, S. B., Thienel, R., Giorgio, J., Behler, A., Fay, M.

2026-11-03 1 min 191 words

BIORXIV NEUROSCIENCE

Summary: We developed a multi-shell diffusion-weighted imaging approach for separate ventricular and subarachnoid cerebrospinal fluid thermometry. The approach uses measured image noise to assess agreement among temperature estimates. In 12 participants, temperature was estimated from four non-zero diffusion...

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Targeting MMP9: an antibody strategy for diabetic peripheral neuropathy

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https://www.nature.com/articles/s41574-026-01322-0

Empathy and compassion are comparable between individuals with cervical spinal cord injury and controls, but their association with interoceptive sensibility differs

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https://www.nature.com/articles/s41598-026-78229-3

Repetitive transcranial magnetic stimulation enhance memory in stress-disordered cynomolgus monkeys by altering the electrical activity of neurons

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https://www.nature.com/articles/s41598-026-78357-w

The brain lymphatic drainage system: structure, physiological functions, and pathological involvement in Parkinson’s disease

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https://www.nature.com/articles/s41531-026-01595-2

Directional blood flow within a shared capillary network links the nucleus of the solitary tract and area postrema in rats

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https://www.nature.com/articles/s42003-026-11128-5

Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

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🔗 **Read full article:**
https://www.nature.com/articles/s41593-026-02535-2

Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

👤 Yingjie Zhu
📅 2026-11-03 📖 1 min 📝 24 words NATURE NEUROSCIENCE

Summary: Nature Neuroscience, Published online: 03 November 2026; doi:10.1038/s41593-026-02535-2 Author Correction: Social rank modulates methamphetamine-seeking in dominant and subordinate male rodents via distinct dopaminergic pathways

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https://www.nature.com/articles/s41593-026-02535-2

FMRI-guided focal transcranial magnetic stimulation of anterior cingulate cortex reduces opioid seeking and restores connectivity

Zilu MaYing DuanHieu NguyenSabrina LinMd Mohaiminul HaqueDanni WangSamantha HoffmanAidan CarneyTaylor ScottEleni VarlasElliot SteinZheng-Xiong XiYavin ShahamHanbing LuYihong YangaNeuroimaging Research Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224bSchool of Electrical Engineering, International University, VNU-HCM, Ho Chi Minh City 71309, Vietnamchttps://ror.org/00waaqh38Viet Nam National University Ho Chi Minh City, Ho Chi Minh City 71309, Vietnamdhttps://ror.org/043mer456Department of Electrical and Electronic Engineering, University of Nebraska-Lincoln, Lincoln, NE 68588eMolecular Targets and Medication Discovery Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224fBehavioral Neuroscience Branch, Intramural Research Program, National Institute on Drug Abuse, National Institutes of Health, Baltimore, MD 21224

📅 2026-10-26 📖 1 min 📝 48 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceA major challenge in developing brain stimulation therapies for opioid addiction is identifying which neural circuits to target and tracking how stimulation reshapes relapse-related circuit functi...

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https://www.pnas.org/doi/abs/10.1073/pnas.2619013123?af=R

Editorial: Data-driven brain imaging and signal analysis: methods from acquisition to clinical insight

Zhiliang Wei

📅 2026-11-03 📖 1 min 📝 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2010682

Peripheral physiological decoding of arousal in a boundary-avoidance task: signal validity and individualized control

Anirudh Natarajan and Paul Sajda

2026-11-03 1 min 284 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Scoring well on a task-demand label does not show that a decoder measures arousal. We ask whether peripheral physiology decodes demand better than EEG, whether the index behaves like graded arousal rather than a clock, and what a controller could do with it. Approach. We re-analysed a pub...

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GROND: automated characterization of periodic and rhythmic EEG patterns

Jin Jing, Chenxi Sun, Tianyu Zhang, Matthew Byrd, Alexandra–Maria Tăuţan, Lara Basovic, Peter N Hadar, Marta P Fernandes, Daniel Goldenholz, Jennifer Kim, Aaron F Struck, Sahar F Zafar and M Brandon Westover

2026-11-03 2 min 469 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Periodic discharges (PDs) and rhythmic delta activity (RDA) are common electroencephalographic patterns in critically ill patients that require detailed characterization—including lateralization, spatial localization, and frequency estimation—according to the American Clinical Neurophysio...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb331

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

Robert Oostenveld 2026-02-12 1 min 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Social Determinants of Kidney and Cardiovascular Outcomes: Prioritizing Action to Effectively Manage Cardiovascular-Kidney-Metabolic Syndrome

👤 Tiffany M Powell-Wiley
📅 2026-11-02 📖 1 min 📝 70 words

LOW VISION

Summary: In 2005, the World Health Organization (WHO) created the inaugural Social Determinants of Health (SDOH) Commission to recognize how social and physical environments impact health for all populations. In doing so, they created an impetus for policy-making and governing bodies to improve the underlyin...

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Effectiveness of spectacle lenses with highly aspherical lenslets (HAL) in slowing axial elongation and spherical equivalent refraction progression in myopic and pre-myopic children: a systematic review and meta-analysis

👤 Jiaqi Yao
📅 2026-11-02 📖 1 min 📝 65 words

LOW VISION

Summary: CONCLUSION: Current evidence suggests that HAL spectacle lenses may slow axial elongation and SER progression in myopic children. In pre-myopic children, HAL lenses may offer a smaller benefit, particularly with high adherence, but the evidence is limited and of low quality, precluding a strong reco...

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Changes in self-concept and healthy behaviors following an audio instructional program on puberty among teenage girls with visual impairment: a quasi-experimental study

Ashraf Jehad Abuejheisheh

2026-11-02 1 min 51 words LOW VISION

Summary: CONCLUSION: The results of this investigation revealed that visually impaired adolescent girls had low knowledge, self-concept, and menstrual hygiene practices at puberty before the intervention. All measured outcome variables showed a significant improvement after the implementation of the audio in...

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FCM-Encoder: a novel interpretable framework with vision-language reasoning for coronary artery disease and solitary pulmonary nodule classification

Nikolaos I Papandrianos 2026-11-02 1 min 66 words LOW VISION

Summary: Although modern machine-learning models attain strong predictive performance, their clinical adoption remains limited by a lack of transparency. This concern is particularly relevant in multimodal diagnostic pipelines, where the contribution of each modality to the final prediction is more difficult...

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An integrated CT/MRI diagnostic system for pancreatic cystic lesions that supports pathological classification and risk stratification

Jihui Hao 2026-11-03 1 min 22 words LOW VISION

Summary: CONCLUSIONS: SIRS-PCL is a pathology-based, quantitative decision-support system for preoperative pathological subtype classification and pathological risk stratification of surgically evaluated neoplastic PCLs.

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Foveal Crowding and Reduced Binocular Advantage In Myopic Anisometropia Under Brief Stimulus Presentation

👤 Jianfeng Wu 📅 2026-11-03 📖 1 min 📝 32 words LOW VISION

> **Summary:** CONCLUSIONS: In optically corrected MA patients, brief stimulus presentation revealed abnormal foveal crowding in response time and a reduced binocular advantage, suggesting decreased efficiency in processing visual information under high perceptual demand.

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Blinded by inequity: a scoping review of income-related and structural disparities in eye care in the United States

👤 Carrie B Dolan 📅 2026-11-03 📖 1 min 📝 67 words LOW VISION

> **Summary:** CONCLUSION: Income-related disparities in eye care access are associated with structural and financial barriers linked to reduced access to timely services. Addressing these inequities may require greater alignment among recognized needs, promising interventions, and policy priorities, as well as en...

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Suprachoroidal hemorrhage and detachment in the non-laser photocoagulation posterior pole after intravitreal Anti-VEGF injection: A case report

Xionggao Huang 2026-11-03 1 min 63 words LOW VISION

Summary: Intravitreal anti-vascular endothelial growth factor (anti-VEGF) injections are widely used for diabetic macular edema (DME) and neovascular ocular diseases. Suprachoroidal hemorrhage and detachment (SCHD) in the posterior pole non-laser photocoagulation area is an extremely rare complication after ...

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Visual Perceptual Learning Based on Adaptive Contrast Detection Training Enhances Vision and Contrast Sensitivity in Diabetic Retinopathy

Lixia Feng 2026-11-03 1 min 37 words LOW VISION

Summary: CONCLUSIONS: The results reveal significant plasticity in the visual system in DR and suggest that contrast detection training may extend to motion perception tasks. Our study provides additional support for VPL as a potential treatment for DR.

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Reading disability in glaucoma: A neglected dimension of visual disability

Michele Figus 2026-11-03 1 min 62 words LOW VISION

Summary: Reading disability is a common and clinically relevant, yet under-recognized, functional impairment in glaucoma, with significant impact on independence, daily activities, and quality of life, particularly in an increasingly digital environment. Reading performance is influenced by a complex interpl...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

👤 Benjamin Ward-Cherrier 📅 2026-10-31 📖 1 min 📝 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

👤 Hu Zhao 📅 2026-11-01 📖 1 min 📝 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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A phase-calibrated combined electrical-sensory neuromodulation platform for short-Term frontoparietal network modulation and cognitive fatigue resistance

👤 Qianxiang Zhou
📅 2026-11-02 📖 1 min 📝 57 words

TDCS TACS TRNS

Summary: Non-invasive multimodal neuromodulation is often limited by cross-modal temporal misalignment caused by modality-specific conduction delays. To address this challenge, we developed a microcontroller-based multiphysics stimulation platform integrating transcranial alternating current stimulation (tAC...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh

2026-11-02 | 1 min | 48 words | TDCS TACS TRNS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Repeated prefrontal transcranial direct current stimulation improves behavioral regulation and modulates resting-state EEG microstate dynamics in young adults with problematic smartphone use: a randomized sham-controlled longitudinal study

👤 Yingying Zeng
📅 2026-11-03 📖 1 min 📝 62 words

TDCS TACS TRNS

Summary: CONCLUSIONS: Repeated prefrontal tDCS produced concurrent behavioral improvement and selective reorganization of resting-state EEG microstate dynamics in young adults with PSU. These findings provide preliminary evidence that repeated prefrontal tDCS is associated with concurrent behavioral changes ...

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

TDCS TACS TRNS

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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What are the neural networks that have been modulated by anatomically targeted therapies to promote recovery of consciousness? A scoping review

👤 Konstantinos Margetis
📅 2026-11-03 📖 1 min 📝 63 words

TDCS TACS TRNS

Summary: Disorders of Consciousness (DoC) arise from insults to neural networks responsible for awareness and arousal. DoC subtypes are frequently misdiagnosed, as mechanisms underlying consciousness remain poorly understood. Neurostimulation has been investigated as a potential therapy for DoC recovery; how...

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Effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field

Prinpida Sonthiphand 2026-11-03 1 min 68 words TDCS TACS TRNS

Summary: Due to climate change and increasing food demand, conventional agriculture relies heavily on chemical fertilizers. This study investigated the effects of chemical fertilizer application on soil properties, banana (Musa paradisiaca) yields, and soil microbiomes in a long-term cultivation field. Metag...

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Transcranial Pulse Stimulation in Parkinson's Disease: A Systematic Review

Marcelo Lourenço da Silva 2026-11-03 1 min 41 words TDCS TACS TRNS

Summary: CONCLUSIONS: TPS shows preliminary potential to modulate sensorimotor networks and improve selected motor outcomes in PD. However, evidence remains insufficient to establish overall clinical efficacy. Larger, rigorously controlled trials are required to determine sustained benefits, optimal stimulat...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

👤 Chen-Hui Pei
📅 2026-10-30 📖 1 min 📝 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 68 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Correction: EBA-related body-preferring responses in the occipitotemporal cortex measured with functional near-infrared spectroscopy

Min Wang 2026-11-03 1 min 6 words FNIRS

Summary: [This corrects the article DOI: 10.3389/fnhum.2026.1914488.].

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

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Effects of combined non-invasive brain stimulation and brain-computer interface on upper extremity motor function after stroke: a systematic review and meta-analysis

👤 Qiang Gao
📅 2026-11-03 📖 1 min 📝 69 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Current evidence does not demonstrate a clear additional benefit of NIBS as an adjunct to BCI for upper extremity rehabilitation after stroke. Given the limited number of studies and the instability of the pooled result in sensitivity analysis, these findings should be interpreted cautio...

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The application of brain-computer interface on cognitive function and activities of daily living in stroke patients: a systematic review and meta-analysis

Xuan Gong 2026-11-03 1 min 52 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-based training shows preliminary promise for improving cognitive function and ADL in stroke patients. Current evidence remains insufficient to recommend BCI for emotion or speech rehabilitation. However, the GRADE certainty of evidence was rated as low for cognitive function and very...

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Mapping Brain-Computer Interface Research in Depression (1995-2026): A Bibliometric Analysis

Xianlun Duan 2026-11-03 1 min 31 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: The field has evolved from EEG-based characterization toward network-level and real-time neurofeedback research. These findings identify emerging research directions while highlighting the need for further methodological standardization and clinical validation.

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Best hosting platform other than Render (Python) - Django project

👤 /u/Ramesh_9686 📅 2026-11-03 📖 1 min 📝 90 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I designed a personal portfolio ,, https://ramesh13.online/ ,, this is my own website ,, it is a heavy website . My HTTPs request bandwidth is greated than 5 GB as i am hosting in free ...

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https://www.reddit.com/r/Python/comments/1xmr1ap/best_hosting_platform_other_than_render_python/

Bear turns 10, and it still feels like day one

👤 psxuaw 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://blog.bear.app/2026/11/bear-turns-10-and-it-still-feels-like-day-one/ Comments URL: <a href="https://news.ycombinator.com/item?id=50353383">https://news.ycombinator.com/item?id=50...

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https://blog.bear.app/2026/11/bear-turns-10-and-it-still-feels-like-day-one/

Show HN: HackerChurn – which SaaS tools HN wants to cancel

👤 drewlem 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://hissingkitty.com/hacker-churn Comments URL: https://news.ycombinator.com/item?id=50353386 Points: 48 # Comments: 21

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USDB: Scaling Storage for Next-Generation Recommender Systems at ByteDance

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Gitbook Breached

👤 punchmesan 📅 2026-11-03 📖 2 min 📝 407 words HACKER NEWS

> **Summary:** We opened a ticket with support about Gitbook availability issues and got this in reply. We have yet to see any official customer communication or breach notification. Apologies for the formatting, did the best I could as it's a mobile paste. Update: GitBook security incident update Hey <$na...

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Stringman, an $800 open-source room scale gripper robot

👤 nharada 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

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Ashby (YC W19) Is Hiring EMs

👤 abhikp 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://www.ashbyhq.com/careers?ashby_jid=695b1ce6-c375-4fad-94ef-61b50a3fbd28&utm_source=hn Comments URL: <a href="https://news.ycombinator.com/item?id=50354582"...

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https://www.ashbyhq.com/careers?ashby_jid=695b1ce6-c375-4fad-94ef-61b50a3fbd28&utm_source=hn

Postman silently auto-installs AI plugins and attempts to share all API data

👤 sidgarimella 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/postmanlabs/postman-app-support/issues/14370 Comments URL: https://news.ycombinator.com/item?id=50354960 Points: ...

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Qwen 3.8 27B on my laptop can interpret model weights, too

👤 theanonymousone 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://simonwillison.net/2026/Nov/3/qwen-can-interpret-model-weights/ Comments URL: https://news.ycombinator.com/item?id=50355251</...

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How to write comparison content that doesn't age like milk

👤 rbaudibert 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://posthog.com/blog/comparison-content-guide Comments URL: https://news.ycombinator.com/item?id=50355365 Points: 12 # Comments: 3

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I use AI (Oct 2026)

👤 citelao 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://ben.stolovitz.com/posts/how-use-ai-oct-2026/ Comments URL: https://news.ycombinator.com/item?id=50355767 Points: 9 # Comments: 6...

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CAG repeat length shapes bidirectional functional connectome alterations in spinocerebellar ataxia type 3: a multimodal MRI study

📖 1 min 📝 32 words NEUROSCIENCE JOURNAL

> **Summary:** Publication date: 14 December 2026 Source: Neuroscience, Volume 619 Author(s): Xianfei Yang, Haishuo Xia, Yuwei Hu, Can Xu, Xingang Wang, Qingxin Deng, Qian Xie, Limeng Dai, Qifang Yang, Changying Li, Li Gui, Chen Liu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S0306452226007013?dgcid=rss_sd_all

Coarse motor-state structure across paired preoperative fMRI and intraoperative ECoG in patients with brain tumors

📖 1 min 📝 36 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Jingxi Hu, Jie Ma, Wanwan Guo, Yang Jiao, Jiaqi Yang, Hongwei Wang, Yiheng Yin, Renke He, Gang Cheng, Hongmin Bai, Hairong Zheng, Dong Liang, Jianning Zhang, Zhanli Hu

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006257?dgcid=rss_sd_all

Parietal structural and functional differentiation associated with visual tracking across motor participation profiles

📖 1 min 📝 26 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Keying Zhang, Meng Qi, Ling Jiang, Qing Yan, Yuyan Wang, Shuang Bai, Chunmei Cao, Dong Zhang, Siyuan Ma

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006178?dgcid=rss_sd_all

Brief Peppermint Odour Exposure and Immediate Free Recall of a Visual Object Array: A Three-Arm Randomised Experiment

Akar, B., Altıntas, T.
2026-11-02
1 min
270 words
BIORXIV NEUROSCIENCE

Summary: Background: Odours reach limbic and prefrontal targets through a uniquely direct pathway and are often claimed to enhance cognition, yet experimental evidence for specific essential oils is mixed and frequently confounded by expectancy. We tested whether a brief inhalation of peppermint or clove ess...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.31.763290v1?rss=1

Myelin Governs Hippocampal Network Function through the Extracellular Space it Occupies, Not Only the Axons it Insulates

Jia, X., Zeng, C., Liu, L., Gao, T., Bi, J., Zeng, Z., Liang, L., Qi, C., Lin, F., Cheng, F.
2026-11-03
1 min
250 words
BIORXIV NEUROSCIENCE

Summary: Myelin, the multilamellar sheath wrapping neuronal axons, enables saltatory conduction. Beyond electrical insulation, myelin occupies volume, and therefore shapes the extracellular space (ECS) through which interstitial fluid flows and neurotransmitter diffuses. How these two roles interact is unkno...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.02.763564v1?rss=1

Path knockout: an analysis tool for connectome-driven circuit discovery

McGrath, R., Hattori, D.
2026-11-03 1 min 207 words BIORXIV NEUROSCIENCE

Summary: Neural circuits in the brain transform sensory information into perceptual, emotional and motor output. The connectome -- a synaptic-resolution wiring diagram of all neurons in a nervous system -- provides a neuroanatomical blueprint for identifying different circuits that mediate specific computati...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.31.763128v1?rss=1

A medial entorhinal connectome reveals compartmentalized microcircuit architecture

Shi, T., Chen, Y., Jin, H., Jiang, H., Chen, X., Xiong, Z., Wu, F., Zhang, R.
2026-11-03 1 min 217 words BIORXIV NEUROSCIENCE

Summary: The medial entorhinal cortex (MEC) transforms multimodal inputs into spatial representations, but the synaptic architecture underlying its computations remains unclear. Here we generated a synapse-resolution connectomic dataset spanning all six layers of mouse MEC and the adjacent parasubiculum with...

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Wolffia globosa -derived nanovesicles as a natural source of bioactive metabolites with neuroprotective potential against methylglyoxal-induced oxidative stress

👤 Khongkla, E., Boonsnongcheep, P., Srisaisup, M., Charuraksa, T.

📅 2026-11-03 📖 1 min 📝 219 words

BIORXIV NEUROSCIENCE

Summary: Plant-derived nanovesicles (PDNVs) have attracted increasing attention for their potential biomedical applications, particularly in oxidative stress-related diseases and neuroprotection. Wolffia globosa, an emerging functional food and potential future food source, is rich in phytochemicals associat...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.31.762905v1?rss=1

Dissociable norepinephrine and dopamine representations during strategic decision-making and reward learning

👤 Aloor, J., Sit, T. P., Geuter, J., Tutt, J. O., Duan, C. A.

📅 2026-11-03 📖 1 min 📝 256 words

BIORXIV NEUROSCIENCE

Summary: Adaptive decision-making in competitive environments requires learning from immediate reward outcomes while tracking longer-timescale information to adjust behavioural strategies and remain unpredictable against others. Dopamine (DA) and norepinephrine (NE) have long been implicated in reward-guided...

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https://www.biorxiv.org/content/10.64898/2026.10.31.763226v1?rss=1

Oscillatory Phase Coding Emerges in Adaptive RNNS for Flexible Working Memory

Lu, K., Chu, T., TUERHONG, A., Sun, Y., Luo, H., Fang, F., WU, S.

2026-11-03 1 min

210 words

BIORXIV NEUROSCIENCE

Summary: Oscillatory phase coding (OPC) is widely observed in neural systems and has been implicated in cognitive functions such as memory and sequence processing. However, its computational role and neural substrates remain largely unclear. Here, we studied OPC by training recurrent neural networks (RNNs) w...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.02.763841v1?rss=1

Multi-scale place fields provide a predictive basis for hierarchical control

Yu, C., Maimon, S. R., Rich, P. D., Eliav, T., Las, L., Lee, A. K., Ulanovsky, N., Lengyel, M.

2026-11-03 1 min

150 words

BIORXIV NEUROSCIENCE

Summary: Hippocampal place cells typically exhibit unimodal spatial selectivity in small enclosures, but express multiple, irregularly spaced fields in environments of naturalistic scale. However, existing predictive-map theories of the hippocampus do not scale well to large environments and do not account f...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.28.762461v1?rss=1

BundleMorphometry: Spatially Resolved White Matter Bundle Shape Analysis Using Nonlinear Streamline-based Registration

Dasanayake, D., Amit, S. N., Chandio, B. Q.

2026-11-03 1 min 203 words

BIORXIV NEUROSCIENCE

Summary: Diffusion MRI tractometry quantifies microstructural properties along white matter (WM) bundles but does not directly measure their geometry, potentially overlooking localized morphological alterations. Here, we introduce BundleMorphometry, a framework for quantifying local WM bundle shape using the...

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https://www.biorxiv.org/content/10.64898/2026.11.02.763786v1?rss=1

Circuit regulation along an activity continuum shapes flexible and stable representations during learning

Yusufi, D., Ma, Y., Chen, L., Wang, G., Malone, T., Zhang, K., Kim, J. H., Shahid, F., Cekada, K., Barcelone, E., Gu, Y.

2026-11-03 1 min 207 words

BIORXIV NEUROSCIENCE

Summary: Learning requires neural representations to flexibly incorporate new information while progressively stabilizing to support enduring memories. How neural circuits balance these demands remains unclear. Here, we identify complementary mechanisms supporting representational flexibility and stability i...

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RIPK1/RIPK3 and MLKL differentially control mitophagy and mitochondrial homeostasis in hypoxic-ischaemic encephalopathy

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41419-026-09382-w

Large-scale identification of the regulatory potential of SNPs in human neural stem cells

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41467-026-79051-1

Advancing human iPSC-derived motor neuron models using glutamatergic modulators for synaptic function studies

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41419-026-09381-x

Integrative analysis of brain-age acceleration: health associations and multi-omics profiles

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https://www.nature.com/articles/s41514-026-00557-5

Subacute carbon monoxide exposure alters AgNOR number and area ratio in the cerebral cortex of rats

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41598-026-77555-w

Aging-related cellular and molecular changes reveal a role for Bglap3+ neurons in MEC spatial cognition

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

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https://www.nature.com/articles/s41467-026-78673-9

Early-life adversity induces social anxiety and cognitive deficits alongside neurobiological changes in rats

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s42003-026-11279-5

Orexin and pupillary modulation for hunger control

👤 Seher KosarEva BraceyNikola GrujicYihui DuDaria Peleg-RaibsteinDenis Burdakovahttps://ror.org/02q2nz798ETH Zürich, Department of Health Sciences and Technology, Neuroscience Center Zürich, Institute for Neuroscience, Schwerzenbach 8603, Switzerland

📅 2026-10-28 📖 1 min 📝 49 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceThis study reveals a genetically defined arousal-hunger circuit that enables mental engagement or external sensory input to naturally interrupt the urge to eat. These findings in mice suggest that...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2619759123?af=R

Dlx1&2 directly promote expression of Cxcl14 to control synapse development and interneuron survival

👤 Teng GuoSusan LindtnerJia Sheng HuSiavash Fazel DarbandiYurong WangEmily Ling-Lin PaiAthéna R. YpsilantiCindy T. J. van VelthovenAlex S. NordJohn L. R. RubensteinaNina Ireland Laboratory of Developmental Neurobiology, Department of Psychiatry and Behavioral Sciences, University of California San Francisco, San Francisco, CA 94143bDepartment of Neurobiology, Physiology, and Behavior, Center for Neuroscience, University of California Davis, Davis, CA 95618cDepartment of Psychiatry and Behavioral Sciences, Center for Neuroscience, University of California Davis, Davis, CA 95618dhttps://ror.org/03cpe7c52Allen Institute for Brain Science, Seattle, WA 98109

📅 2026-10-28 📖 1 min 📝 46 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDlxgenes control multiple steps in the development of cortical interneurons (CIN), including synapse development and cell survival. Here we explore molecular mechanisms how theDlxgenes promote neo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2616055123?af=R

Lysine crotonylation acts as an epigenetic switch for glutamate neurotransmission and spatial memory

👤 Utsav MukherjeeBudhaditya BasuPratyush SuryavanshiStacy E. BeyerSaaman GhodsiNathan RobillardYann VanrobaeysEric B. TaylorJoseph GlykysTed AbelSnehajyoti Chatterjeeahttps://ror.org/036jqmy94Department of Neuroscience and Pharmacology, Carver College of Medicine, University of Iowa, Iowa City, IA 52242bhttps://ror.org/036jqmy94Iowa Neuroscience Institute, University of Iowa, Iowa City, IA 52242chttps://ror.org/036jqmy94Interdisciplinary Graduate Program in Neuroscience, University of Iowa, Iowa City, IA 52242dhttps://ror.org/036jqmy94Stead Family Department of Pediatrics, Carver College of Medicine, University of Iowa, Iowa City, IA 52242ehttps://ror.org/036jqmy94Interdisciplinary Graduate Program in Genetics, University of Iowa, Iowa City, IA 52242fhttps://ror.org/036jqmy94Fraternal Order of Eagles Diabetes Research Center, University of Iowa, Iowa City, IA 52242ghttps://ror.org/036jqmy94Department of Molecular Physiology and Biophysics, Carver College of Medicine, University of Iowa, Iowa City, IA 52242

📅 2026-10-28 📖 1 min 📝 47 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceAlthough histone acetylation has long been considered a principal regulator of gene expression events underlying memory storage, the role of non-acetyl histone acyl modifications on long-term memo...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2608460123?af=R

Forniceal deep brain stimulation improves memory in Rett syndrome mice by promoting the maturation of newborn hippocampal neurons

Qi WangBin TangZhenyu WuHuda Y. ZoghbiJianrong Tangahttps://ror.org/05cz92x43Jan and Dan Duncan Neurological Research Institute at Texas Children's Hospital, Houston, TX 77030bhttps://ror.org/02pttbw34Department of Pediatrics, Division of Neurology, Baylor College of Medicine, Houston, TX 77030chttps://ror.org/02pttbw34Department of Molecular and Human Genetics, Baylor College of Medicine, Houston, TX 77030dhttps://ror.org/02pttbw34Program in Developmental Biology, Baylor College of Medicine, Houston, TX 77030ehttps://ror.org/02pttbw34Department of Neuroscience, Baylor College of Medicine, Houston, TX 77030fhttps://ror.org/02pttbw34HHMI, Baylor College of Medicine, Houston, TX 77030

📅 2026-10-28 📖 1 min 📝 52 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceDeep brain stimulation (DBS) has long been known for its efficacy in mitigating movement disorders. More recently, it has shown promise for addressing disorders of learning and memory: for example...

🔗 **Read full article:**
https://www.pnas.org/doi/abs/10.1073/pnas.2535887123?af=R

Defining and measuring proximity to criticality based on univariate time series

J. Samuel SooterAntonio J. FonteneleAndrea K. BarreiroCheng LyKeith B. HengenWoodrow L. ShewaDepartment of Physics, University of Arkansas, Fayetteville, AR 72701bhttps://ror.org/042tdr378Department of Mathematics, Southern Methodist University, Dallas, TX 75205chttps://ror.org/02nkdxk79Department of Mathematics and Applied Mathematics, Virginia Commonwealth University, Richmond, VA 23284dhttps://ror.org/01yc7t268Department of Biology, Washington University, St. Louis, MO 63130

2026-10-28 1 min 56 words PNAS NEUROSCIENCE

Summary: Proceedings of the National Academy of Sciences, Volume 123, Issue 44, November 2026.
SignificanceKnowledge of many complex systems comes from time series measurements of one or a few parts of the system. These dynamics depend on many detailed properties; must we keep track every detail to und...

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Computational approaches to deep learning–driven EEG analysis for neurological disease diagnosis: a systematic review

Jisu Elsa Jacob
2026-10-13 1 min 229 words

FRONTIERS NEUROINFORMATICS

Summary: Electroencephalography (EEG) is a widely used non-invasive neurophysiological technique for analyzing brain dynamics and diagnosing neurological diseases. However, the non-stationary nature, low signal-to-noise ratio, and inter-subject variability of EEG signals cause significant challenges to tradi...

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Short excitatory projections promote burst activity and modular functional organization in neural networks

Claus C. Hilgetag
2026-11-03 1 min 165 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Physical constraints in the brain favor short, distance-dependent connections. However, short-range projections are not merely a structural compromise; they also serve essential functional roles. Here, we explore networks with balanced excitation and inhibition, demonstrating that short excitatory p...

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Persistent memory through triple-loop consolidation under stochastic unit turnover

👤 Jianwei Lou 📅 2026-11-03 📖 1 min 📝 366 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Dissipative cognitive architectures maintain computation through continuous energy expenditure, where units that exhaust their energy are stochastically replaced with fresh random state. This creates a fundamental challenge: how can persistent, context-specific memory survive when all learnable stat...

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Editorial: Advances in perceptual decision making and brain oscillations

👤 Francesco Di Gregorio 📅 2026-11-03 📖 1 min 📝 0 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fncom.2026.2023605

Editorial: Machine learning integration in computational neuroscience: enhancing neural data decoding and prediction

George Giakos 2026-11-03 1 min 0 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

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Exploratory evaluation of electroencephalogram-derived features for classification of mild cognitive impairment in two independent datasets

Makoto Nakane 2026-11-03 1 min 347 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: A proper diagnosis of mild cognitive impairment (MCI) is one of the important challenges in the early detection of dementia. For the classification of MCI patients and healthy controls, some electroencephalogram (EEG)-derived indices were calculated and compared. Some studies have been conducted on ...

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Correction: Post-stroke spasticity: advances in network mechanisms and therapeutics

Zhiqi Lin 2026-10-29 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

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Embodied multimodal human-computer interaction for English language learning: integrating speech, gestures, emotion, and adaptive feedback

Yao Tang 2026-10-14 1 min 238 words FRONTIERS NEUROROBOTICS

Summary: Current English language learning systems are often limited by single-dimensional learning-state perception, insufficient modeling of affective-cognitive characteristics, and delayed or inflexible feedback regulation. To address these limitations, this study proposes an affective-cognitive gated cro...

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Effectiveness of bimanual kinesthetic teaching through different controller settings

Andrej Gams 2026-10-21 1 min 94 words FRONTIERS NEUROROBOTICS

Summary: Physical human-robot interaction (pHRI) through kinesthetic teaching enables intuitive robot programming, but extending this approach to bimanual systems introduces challenges in synchronized control. Thus, the best way to control bimanual robot systems for kinesthetic teaching remains unclear. In t...

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Editorial: Expanding the therapeutic dose range of cannabinoid-based therapy

Babak Baban 2026-11-03 1 min 0 words FRONTIERS NEUROSCIENCE

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https://www.frontiersin.org/articles/10.3389/fnins.2026.2028326

From beat to syntax: evolutionary and developmental paths of a shared neural resource for hierarchical processing in music and language

Tao Gong 2026-11-03 1 min 268 words FRONTIERS NEUROSCIENCE

Summary: Children who track a musical beat well tend to have stronger grammatical skills, and brief exposure to a regular rhythm can transiently improve grammatical processing. These findings raise the possibility of a shared underlying mechanism, and with it two questions: where did the link come from, and ...

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Vagus nerve stimulation in obesity: a systematic review and meta-analysis

Matthew A. Schiefer 2026-11-03 1 min 333 words FRONTIERS NEUROSCIENCE

Summary: BackgroundVagus nerve stimulation (VNS) has been investigated as a potential therapy for obesity because of the vagus nerve’s role in satiety, gastrointestinal signaling, autonomic regulation, and metabolic homeostasis. However, studies vary substantially in stimulation site, interface, parameters, ...

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Analysis and improvement of surface electromyography-based American Sign Language hand gesture recognition

Yasuharu Koike 2026-11-03 1 min 243 words FRONTIERS NEUROSCIENCE

Summary: Surface electromyography (sEMG) allows sign language gestures to be recognized from forearm muscle activity without visual access to the hand. Gestures that produce similar forearm activation patterns, however, remain difficult to distinguish. This study examined how these similarities relate to cla...

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Cybertissue neuroprosthetics: a new implantable neuroprosthetics paradigm to achieve seamless lifetime integration

Liang Guo 2026-10-27 1 min 234 words JOURNAL NEURAL ENGINEERING

Summary: Objective. Despite decades of advances in neural interface technology, the long-term instability of implantable neuroprostheses remains a major barrier to achieving reliable lifetime performance. Chronic foreign-body reactions and fibrotic encapsulation progressively degrade the electrode-tissue int...

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Combined voltammetry and optical brain imaging: toward high resolution neurochemical measurements

Vladislav Zavialov, Vassiliy Tsytsarev, Anna B Volnova, Raul R Gainetdinov and Mikhail Inyushin

2026-10-27 1 min 316 words JOURNAL NEURAL ENGINEERING

Summary: The question of the mechanisms by which cellular interactions in the brain are converted into integrative behavioral phenomena is closely related to functional localization, and also to the relationship between spatial and temporal resolution when interpreting experimental data. Classical electrophy...

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https://iopscience.iop.org/article/10.1088/1741-2552/aeb32e

CORRIGENDUM: In silico development and validation of Bayesian methods for optimizing deep brain stimulation parameters (2023 J. Neural Eng. 20 036015)

Sumedh S Nagrale, Ali Yousefi, Theoden I Netoff and Alik S Widge

2026-10-29 1 min 0 words JOURNAL NEURAL ENGINEERING

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Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

👤 Fanny Quandt 📅 2025-01-09 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

Summary: Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

Summary: Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

Summary: Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

Julia-Katharina Pfarr 2026-02-12 1 min 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

Robert Oostenveld 2026-02-12 1 min 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

Vitória Piai 2026-02-17 1 min 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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HiCP2GAN: a plug and play foundation model-based GAN for Hi-C enhancement

Oluwatosin Oluwadare 2026-11-02 1 min 65 words LOW VISION

Summary: The 3D organization of chromatin shapes gene regulation and cellular function, and Hi-C has become the primary technique for mapping chromatin interactions genome-wide. However, high-resolution data remain costly and scarce, leaving many studies with sparse contact maps that limit downstream analysi...

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Exploring Spatial Inequities and Livability: A Mixed-Methods Study Using Artificial Intelligence and Community Insights

Rodrigo Siqueira Reis 2026-11-02 1 min 62 words LOW VISION

Summary: Rapid urbanization has intensified health and social inequities through the uneven distribution of infrastructure, services, and public space. In St. Louis, Missouri (USA), these disparities remain geographically concentrated and rooted in long-standing patterns of racial segregation and disinvestme...

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Association Between Quality of Life and Visual Function in Inherited Retinal Disease: A Cross-sectional Study in a Retinal Genetics Centre

Robert E MacLaren
2026-11-02 1 min 27 words LOW VISION

Summary: CONCLUSIONS: Microperimetry is a patient-relevant measure in inherited retinal diseases. This study demonstrates to stakeholders, such as regulatory bodies, how visual function measures translate to patients' quality-of-life.

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Social Determinants of Kidney and Cardiovascular Outcomes: Prioritizing Action to Effectively Manage Cardiovascular-Kidney-Metabolic Syndrome

Tiffany M Powell-Wiley
2026-11-02 1 min 70 words LOW VISION

Summary: In 2005, the World Health Organization (WHO) created the inaugural Social Determinants of Health (SDOH) Commission to recognize how social and physical environments impact health for all populations. In doing so, they created an impetus for policy-making and governing bodies to improve the underlyin...

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Effectiveness of spectacle lenses with highly aspherical lenslets (HAL) in slowing axial elongation and spherical equivalent refraction progression in myopic and pre-myopic children: a systematic review and meta-analysis

👤 Jiaqi Yao
📅 2026-11-02 📖 1 min 📝 65 words

LOW VISION

Summary: CONCLUSION: Current evidence suggests that HAL spectacle lenses may slow axial elongation and SER progression in myopic children. In pre-myopic children, HAL lenses may offer a smaller benefit, particularly with high adherence, but the evidence is limited and of low quality, precluding a strong reco...

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Changes in self-concept and healthy behaviors following an audio instructional program on puberty among teenage girls with visual impairment: a quasi-experimental study

Ashraf Jehad Abuejheisheh

2026-11-02 1 min 51 words LOW VISION

Summary: CONCLUSION: The results of this investigation revealed that visually impaired adolescent girls had low knowledge, self-concept, and menstrual hygiene practices at puberty before the intervention. All measured outcome variables showed a significant improvement after the implementation of the audio in...

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FCM-Encoder: a novel interpretable framework with vision-language reasoning for coronary artery disease and solitary pulmonary nodule classification

Nikolaos I Papandrianos 2026-11-02 1 min 66 words LOW VISION

Summary: Although modern machine-learning models attain strong predictive performance, their clinical adoption remains limited by a lack of transparency. This concern is particularly relevant in multimodal diagnostic pipelines, where the contribution of each modality to the final prediction is more difficult...

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An integrated CT/MRI diagnostic system for pancreatic cystic lesions that supports pathological classification and risk stratification

Jihui Hao 2026-11-03 1 min 22 words LOW VISION

Summary: CONCLUSIONS: SIRS-PCL is a pathology-based, quantitative decision-support system for preoperative pathological subtype classification and pathological risk stratification of surgically evaluated neoplastic PCLs.

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Foveal Crowding and Reduced Binocular Advantage In Myopic Anisometropia Under Brief Stimulus Presentation

👤 Jianfeng Wu 📅 2026-11-03 📖 1 min 📝 32 words LOW VISION

> **Summary:** CONCLUSIONS: In optically corrected MA patients, brief stimulus presentation revealed abnormal foveal crowding in response time and a reduced binocular advantage, suggesting decreased efficiency in processing visual information under high perceptual demand.

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Blinded by inequity: a scoping review of income-related and structural disparities in eye care in the United States

👤 Carrie B Dolan 📅 2026-11-03 📖 1 min 📝 67 words LOW VISION

> **Summary:** CONCLUSION: Income-related disparities in eye care access are associated with structural and financial barriers linked to reduced access to timely services. Addressing these inequities may require greater alignment among recognized needs, promising interventions, and policy priorities, as well as en...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

👤 Martina C Meinke 📅 2026-08-13 📖 1 min 📝 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

👤 Senanur Akıncı 📅 2026-08-14 📖 1 min 📝 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

👤 Hannes P Saal 📅 2026-09-25 📖 1 min 📝 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

👤 Walter S Carr 📅 2026-10-01 📖 1 min 📝 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

Alan J Emanuel 2026-10-25 1 min 67 words TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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The Relationship Between Occlusal Tactile Acuity, Bite Force and Tooth Wear

Birgitta Häggman-Henrikson 2026-11-03 1 min 52 words TACTILE ACUITY

Summary: CONCLUSIONS: Bite force may be a clinically relevant factor associated with accumulated tooth wear, whereas OTA was not independently associated with tooth wear. Longitudinal studies are needed to determine whether bite force can predict tooth wear progression and whether more comprehensive assessme...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

James E Thornton 2026-10-08 1 min 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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Effects of multisensory tactile-auditory integration teaching on systematic anatomy learning in visually impaired medical students

Fanzhen Kong 2026-10-08 1 min 50 words BRAILLE

Summary: CONCLUSION: The tactile-auditory integrated training system significantly improves objective summative knowledge acquisition and subjective learning experiences in visually impaired medical students. This approach provides a scalable, cost-constrained, and replicable pedagogical model for inclusive ...

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Physical and Environmental Accessibility of Dental Services for Persons with Disabilities in Saudi Primary Health Care: A Cross-Sectional Study

👤 Estie Kruger 📅 2026-10-16 📖 1 min 📝 66 words BRAILLE

Summary: Background/Objectives: Persons with disabilities (PWDs) may experience barriers when accessing oral health care, including those related to the physical and sensory environment of the dental-care setting. In this study we assessed the accessibility of primary health care centres (PHCs) providing den...

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Visible Light Driven Diselenide Chemistry for Simultaneous 3D Patterning and Mechanical Reinforcement

👤 Huaping Xu 📅 2026-10-16 📖 1 min 📝 63 words BRAILLE

Summary: Three-dimensional (3D) patterning on polymer surfaces is typically achieved by photolithography or related techniques, which often involve complex procedures, high-energy ultraviolet light, and risk of material damage. Here, we report a visible light-driven strategy that exploits the dual functional...

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An accessible pregnancy test for blind and low-vision users

Vanja Garaj 2026-10-16 1 min 63 words BRAILLE

Summary: CONCLUSIONS: The findings suggest that accessible pregnancy testing must prioritise not only non-visual result reading but also privacy, dignity, reproductive autonomy and independent health management. These results represent the preliminary evidence of acceptability rather than a conclusive test o...

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Genome-Informed Loop-Mediated Isothermal Amplification (LAMP) Assays for Detection and Differentiation of Ralstonia pseudosolanacearum Phylotypes I and III

Jarred Yasuhara-Bell 2026-10-27 1 min 76 words BRAILLE

Summary: The Ralstonia solanacearum species complex (RSSC) causes bacterial wilt in a broad range of economically important hosts. The RSSC has been reclassified into three species, of which R. solanacearum sequevars 1 and 2 in phylotype IIB are designated as Select Agents in the U.S. Many rapid tests for "R...

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Encoding tactile stimuli for Braille recognition with organoids

Benjamin Ward-Cherrier 2026-10-31 1 min 62 words BRAILLE

Summary: This study proposes a transferable encoding strategy that maps tactile sensor data to electrical stimulation patterns, enabling neural organoids to perform an open-loop artificial tactile Braille classification task. Human forebrain organoids cultured on a low-density microelectrode array (MEA) are ...

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Effect of electrical stimulation on remyelination in preclinical models of multiple sclerosis: A systematic review

Hamdan Hamdan 2026-10-30 1 min 33 words TDCS TACS TRNS

Summary: CONCLUSION: Limited but convergent preclinical evidence suggests ES may enhance remyelination by modulating neuronal and inflammatory pathways. However, standardized protocols of stimulation and multimodal imaging studies are needed for clinical translation in MS.

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Dosage and safety of non-invasive brain stimulation for chronic musculoskeletal pain management: a scoping review

👤 Liming Jiang
📅 2026-10-30 📖 1 min 📝 64 words

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Summary: CONCLUSION: Current NIBS dosage for management of CMP remains heterogeneous and driven by empirical experience. Safety reporting is inadequate. Although reported adverse events were predominantly mild and transient, inconsistent adverse event ascertainment and reporting preclude reliable conclusion ...

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Effects of primary somatosensory cortex high-definition transcranial direct current stimulation combined with core stability training on pain and glutamate levels and neuromuscular function in chronic low back pain: a randomized controlled trial

👤 Chuhuai Wang
📅 2026-10-31 📖 1 min 📝 47 words

TDCS TACS TRNS

Summary: CONCLUSIONS: S1-targeted HD-tDCS combined with core stability training may improve short-term pain intensity and pain-related disability in adults with CLBP. The accompanying muscular, electrophysiological, and metabolomic findings were consistent with the clinical changes. These findings support S1...

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Dual-site multifocal HD-tDCS modulates bilingual language control via brain activity and connectivity

Taomei Guo 2026-10-31 1 min 67 words TDCS TACS TRNS

Summary: Bilinguals rely on language control mechanisms to manage cross-language interference. Interactions between the pre-supplementary motor area (pre-SMA) and the left cerebellum play an important role in bilingual language control. However, our previous brain stimulation study targeting the cerebellum a...

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Case Report: Electroencephalography-informed low-frequency repetitive transcranial magnetic stimulation targeting the supplementary motor area in severe functional movement disorder

Isao Matsushita 2026-10-31 1 min 26 words TDCS TACS TRNS

Summary: CONCLUSION: This case highlights the potential utility of combining EEG-guided neuromodulation with staged rehabilitation for the treatment of FMD and supports further investigation of this approach.

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Decoding Parkinson's disease: a systematic review of qEEG biomarkers across clinical domains

👤 Ehsan Rezayat 📅 2026-10-31 📖 1 min 📝 27 words TDCS TACS TRNS

Summary: CONCLUSIONS: qEEG is a promising biomarker for PD, with distinct electrophysiological correlates across cognitive, motor, pharmacological, and neuromodulatory domains. However, methodological standardization is needed before clinical use.

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Intensity-Dependent Effects of 4-Week Transcranial Direct Current Stimulation on Motor Learning in Healthy Young Adults

👤 Fengxue Qi 📅 2026-10-31 📖 1 min 📝 2 words TDCS TACS TRNS

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Spatial multi-omics of mouse incisor mesenchyme reveals metabolic remodeling during the MSC-to-TAC transition

Hu Zhao 2026-11-01 1 min 61 words TDCS TACS TRNS

Summary: The continuously growing mouse incisor provides a useful model for investigating stem cell niche organization and the MSC-to-TAC transition because it contains spatially distinct mesenchymal stem/stromal cell (MSC)-enriched and transit-amplifying cell (TAC)-enriched compartments and abundant TAC pop...

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A phase-calibrated combined electrical-sensory neuromodulation platform for short-Term frontoparietal network modulation and cognitive fatigue resistance

Qianxiang Zhou 2026-11-02 1 min 57 words TDCS TACS TRNS

Summary: Non-invasive multimodal neuromodulation is often limited by cross-modal temporal misalignment caused by modality-specific conduction delays. To address this challenge, we developed a microcontroller-based multiphysics stimulation platform integrating transcranial alternating current stimulation (tAC...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

👤 Roi Cohen Kadosh 📅 2026-11-02 📖 1 min 📝 48 words TDCS TACS TRNS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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Default mode network connectivity and temperament are associated with internalizing symptoms in early childhood

👤 Tobias Grossmann 📅 2026-10-29 📖 1 min 📝 61 words FNIRS

Summary: This study investigates neural and temperament precursors to internalizing psychopathology. Increased default mode network (DMN) functional connectivity and decreased frontoparietal network (FPN) connectivity are established neural signatures of internalizing disorders in adults, but it is unknown i...

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Prefrontal and sensorimotor cortices showed opposite patterns of hemodynamic activity changes between eyes-open and eyes-closed conditions in blepharospasm

👤 Chen-Hui Pei
📅 2026-10-30 📖 1 min 📝 66 words FNIRS

Summary: CONCLUSION: The BSP group and the HCs group showed opposite patterns of EO-EC differences in the prefrontal cortex and sensorimotor cortex. The delta EO-EC difference between the right S2 and the right FPA showed promising has excellent differential diagnostic value for BSP in this discovery cohort ...

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Naturalistic language assessment and brain-network biomarkers in post-stroke aphasia for precision neurorehabilitation

👤 Yulong Bai 📅 2026-10-30 📖 1 min 📝 55 words FNIRS

Summary: Advances in naturalistic language assessment, portable neuroimaging, electrophysiology, and brain-network modeling are creating new opportunities to evaluate post-stroke aphasia under conditions closer to real-world communication. Conventional aphasia batteries and isolated linguistic tasks remain e...

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Corrigendum to "Resilience and motor skill learning: An exploratory fNIRS study of behavioral and prefrontal activation patterns" [Physiology & Behavior 313 (2026) 115364]

👤 Teri Kim 📅 2026-10-30 📖 1 min 📝 2 words FNIRS

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words FNIRS

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Rapid estimation of photon measurement density functions using a deep convolutional neural network for functional near-infrared spectroscopy

👤 Lei Zhang 📅 2026-10-31 📖 1 min 📝 63 words FNIRS

Summary: The photon measurement density function (PMDF) plays a fundamental role in both pre-experimental optode arrangement and post-experimental data analysis in functional near-infrared spectroscopy (fNIRS). Conventionally, PMDFs are derived from structural MR images through tissue segmentation and photon...

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Interbrain synchrony analysis for fNIRS: A signal processing primer

👤 Kathrin Kostorz 📅 2026-10-31 📖 1 min 📝 65 words FNIRS

Summary: Interbrain synchrony (IBS), also referred to as interpersonal neural synchrony (INS), has become a central construct in social neuroscience and beyond. It describes the temporal alignment of neural activity across individuals and has been studied in a range of paradigms, including sequential interac...

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Differences in Near-Infrared Light Spectral Characteristics and Cranial MRI Parameters in Patients with Major Depressive Disorder

👤 Dandan Wan 📅 2026-10-31 📖 1 min 📝 17 words FNIRS

> **Summary:** CONCLUSION: Near-infrared light spectroscopy imaging parameters and cranial MRI cerebral blood flow differ in patients with MDD.

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Age-related patterns in local spontaneous activity measured by fNIRS: Greater unpredictability in infancy, local reorganization in childhood

👤 Kanghui Li 📅 2026-10-31 📖 1 min 📝 68 words FNIRS

> **Summary:** The maturation of the brain is typically described at the whole-brain scale, whereas age-related changes in spontaneous activity within local brain regions have received comparatively little attention. Resting-state regional dynamics (RSRD) are defined as the temporal variability characteristics of ...

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State-Dependent EEG Markers of Excitation/Inhibition and Hemodynamic State Are Associated with Random Noise Stimulation Effects

Roi Cohen Kadosh 2026-11-02 1 min 48 words FNIRS

Summary: CONCLUSIONS: The findings argue against a simple tonic-shift account of tRNS and support a state-dependent framework in which stimulation effects depend on ongoing electrophysiological and hemodynamic context. EEG and fNIRS provided complementary information about variability in learning and stimula...

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EEG feature-level integration for robust command generation in SSVEP-based brain-computer interfaces

Aishi Zhou 2026-10-30 1 min 20 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: Incorporating SSVEP-BR into SSVEP-based BCI frameworks improved flexibility and robustness, enhancing the practicality of asynchronous brain computer interaction systems.

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Retinal microvasculature alterations are associated with mild behavioral impairment in Southeast Asian dementia-free older adults

Christopher Chen 2026-10-30 1 min 26 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: Retinal microvascular alterations are associated with MBI in the pre-dementia stage, with exploratory findings suggesting related vascular and inflammatory biological correlates that warrant further investigation.

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Governing brain-computer interfaces as embodied autonomous neural systems: an ethical framework for neurorobotic governance in education

Jianfeng Hu 2026-10-31 1 min 59 words BRAIN COMPUTER INTERFACE

Summary: The rapid development of neurorobotic systems, particularly non-invasive brain-computer interfaces (BCIs) deployed in educational contexts, raises urgent ethical questions about child protection and societal impact. Neurorobotic systems acquire biological signals through embodied sensors, process th...

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Offline and simulated-online fNIRS decoding of ankle movements for prospective exo-ankle control: a pilot study in healthy adults and ankle-impaired patients

None Kai-Yu Tong
2026-10-31 1 min 64 words BRAIN COMPUTER INTERFACE

Summary: This study developed and evaluated a non-invasive brain-computer interface (BCI) framework that decodes functional near-infrared spectroscopy (fNIRS) signals from the motor cortex to generate offline and simulated-online exo-ankle control commands, as a proof-of-concept step toward future prospectiv...

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Comparison of two Crimean-Congo haemorrhagic fever virus serological tests in Tanzanian febrile patients using a Bayesian latent class approach

Blandina T Mmbaga 2026-10-31 1 min 57 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: In the absence of a reference standard and access to BSL-4 neutralisation testing, both the IDvet ELISA and the Luminex assay demonstrated high diagnostic performance with comparable sensitivity in detecting prior exposure to CCHFV among febrile patients in northern Tanzania. Furthermore...

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Uncertainty Quantification in Machine Learning for Biosignal Applications - A Review

Matias Valdenegro-Toro 2026-10-31 1 min 63 words BRAIN COMPUTER INTERFACE

Summary: Uncertainty Quantification (UQ) has gained traction in an attempt to improve the interpretability and robustness of machine learning predictions. Specifically (medical) biosignals such as electroencephalography (EEG), electrocardiography (ECG), electrooculography (EOG), and electromyography (EMG) co...

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https://pubmed.ncbi.nlm.nih.gov/42964313/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261103064109&v=2.20.1

Comparative efficacy of brain-computer interface training with different feedback modalities for post-stroke upper-limb motor recovery: a systematic review and network meta-analysis of randomized controlled trials

👤 Lixia Deng
📅 2026-10-31 📖 1 min 📝 46 words

BRAIN COMPUTER INTERFACE

Summary: CONCLUSION: BCI-Robot and BCI-MM showed statistically significant FMA-UE benefits versus UR, although certainty was low. As non-BCI robotic training also showed benefit, the BCI-Robot effect cannot be attributed to BCI alone. No BCI feedback modality was statistically superior to another. Further hi...

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https://pubmed.ncbi.nlm.nih.gov/42964381/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261103064109&v=2.20.1

Enhanced P300-based brain-computer interface decoding via region-weighted multiscale temporal-spatial representation learning method

👤 Jing Jin 📅 2026-10-31 📖 1 min 📝 61 words BRAIN COMPUTER INTERFACE

Summary: P300-based brain-computer interface (BCI) systems provide a vital communication channel for individuals with severe physical disabilities. While deep learning has achieved significant progress in signal processing, P300 recognition remains hindered by low signal-to-noise ratios and non-stationary dy...

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https://pubmed.ncbi.nlm.nih.gov/42966015/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261103064109&v=2.20.1

State-dependent effects of rTMS on cognitive function in bipolar II disorder: A randomized, double-blind, sham-controlled trial

Shaohua Hu 2026-11-01 1 min 46 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: This exploratory trial did not provide statistically robust evidence that concurrent complex cognitive engagement enhanced the acute working-memory effects of IPL-targeted 10-Hz rTMS in euthymic bipolar II disorder, suggesting that task type, cognitive load, and stimulation-task couplin...

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Functional electrical stimulation and brain-machine interfaces for simultaneous control of wrist and finger flexion

Cynthia A Chestek 2026-11-02 1 min 22 words BRAIN COMPUTER INTERFACE

Summary: CONCLUSIONS: We outline a method using an artificial brain-to-body interface that could restore continuous wrist and finger movements after spinal cord injury.

🔗 **Read full article:**
https://pubmed.ncbi.nlm.nih.gov/42972338/?utm_source=BucketBot&utm_medium=rss&utm_campaign=None&utm_content=1rSUu--tbw4049Wgf_RdKXdtNCvGW0lVBZFpHe7zaN4k4DwoD5&fc=None&ff=20261103064109&v=2.20.1

A research on the typical images of animals in the dreams of Naxi.

📅 2026-09-10 📖 1 min 📝 157 words DREAMING

Summary: Animal images in dreams have long been studied, yet existing research is not systematic or in-depth. The Naxi, a small Chinese ethnic group (about 320,000), possess a rich cultural heritage. In their belief, animals mediate between humans, deities, and nature, while dreams are key communication chan...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000353

Philosophical and medical analysis of dreams in avicenna’s perspective: Nature, mechanism, and functions.

📅 2026-06-15 📖 1 min 📝 218 words DREAMING

Summary: Avicenna was an eminent Islamic philosopher and physician. Drawing primarily on The Book of Healing and The Canon of Medicine, this article presents a systematic account of his theory of dreaming within the framework of his psychology and natural philosophy. Avicenna conceives drea...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000341

Threatening events and other themes in COVID-19-related dreams: Examining pandemic dreams from the perspective of current dream simulation theories.

📅 2026-06-25 📖 1 min 📝 218 words DREAMING

Summary: In this study on pandemic dream experiences in rural Nepal, 54 interviews were collected 1 year into the COVID-19 pandemic and compared with 62 interviews collected some years earlier. As in comparable studies, the dream narratives were categorized and analyzed according to their content, employing ...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000348

Correction to “Big-five personality traits and dream experiences: A systematic review and meta-analysis” by Marin et al. (2026).

📅 2026-09-10 📖 1 min 📝 399 words DREAMING

Summary: Reports an error in "Big-Five personality traits and dream experiences: A systematic review and meta-analysis" by Jai Marin, Noah Thomas, Lan Nguyen and Jared Walters (Dreaming, Advanced Online Publication, Mar 26, 2026, np; see record 2027-47158-001). The article is being made available op...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000349

When the body suffers, the mind dreams: The lived experience of cancer.

📅 2026-09-10 📖 1 min 📝 194 words DREAMING

Summary: Dreams have intrigued humanity across centuries, serving as portals to the unconscious, reflections of lived experiences, and sources of spiritual guidance. The study of dreaming spans neuroscience, psychology, anthropology, sociology, and religious thoughts, with each discipline offering unique ins...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000355

Big-Five personality traits and dream experiences: A systematic review and meta-analysis.

📅 2026-03-26 📖 1 min 📝 369 words DREAMING

Summary: [Correction Notice: An Erratum for this article was reported in Vol 36(3) of Dreaming (see record 2028-23196-002). The article is being made available open access under the CC-BY license following the Griffith University opt-in to the CAUL/ APA Read and Publish agreement. The correct copyrig...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000336

The dream insight framework: Harnessing the power of dreams in leadership.

📅 2026-08-20 📖 1 min 📝 80 words DREAMING

Summary: This article introduces the Dream Insight Framework, an approach for integrating dreamwork into leadership development. It explores how dreams, often dismissed in existing leadership approaches, can be powerful tools for enhancing creativity, decision-making, and problem-solving. By reviewing existi...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000354

Interpretability of dreams elicited by progressive-interval rapid-eye-movement and fixed-schedule awakening methods: Trajectories of dream affect, themes, and insight across the night.

📅 2026-07-30 📖 1 min 📝 201 words DREAMING

Summary: This study examined whether interpreting REM dream reports elicited using a REM-cycle-accommodating awakening protocol yielded more associations and greater insight than interpreting REM and non-REM dream reports obtained through a fixed-schedule awakening method. It also explored how interpretabili...

🔗 **Read full article:**
http://doi.org/10.1037/drm0000350

Monthly Updates [March]

📅 2026-03-01 📖 2 min 📝 571 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/mar-2026

Monthly Updates [June]

📅 2026-06-01 📖 4 min 📝 954 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/june-2026

Monthly Updates [August]

📅 2026-08-01 📖 3 min 📝 700 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/aug-2026

Monthly Updates [Sept]

📅 2026-08-31 📖 5 min 📝 1079 words FMHY

> **Summary:** INFO These update threads only contain major updates. If you're interested in seeing all minor changes you can follow our Commits Page on G...

🔗 **Read full article:**
https://fmhy.net/posts/sept-2026

What is the closest equivalent to AstroJS in the Python ecosystem?

👤 /u/sentientanus69 📅 2026-11-02 📖 1 min 📝 75 words REDDIT PYTHON

> **Summary:** <!-- SC_OFF --> I am aware a full blown clone wouldn't be possible (hard to imagine how islands would work with Python). But the part where Astro is a free form static site framework is pretty neat and I don't think the established players in this space within Python world are as m...

🔗 **Read full article:**
https://www.reddit.com/r/Python/comments/1xmcpk7/what_is_the_closest_equivalent_to_astrojs_in_the/

Open-Source 5G Ran Platforms: A Dual Perspective on Performance and Capabilities

👤 teleforce 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://arxiv.org/abs/2608.25820 Comments URL: https://news.ycombinator.com/item?id=50347464 Points: 39 # Comments: 4

🔗 **Read full article:**
https://arxiv.org/abs/2608.25820

Show HN: I built a little browser game to practice Vim commands

👤 dh1011 📅 2026-11-03 📖 1 min 📝 62 words HACKER NEWS

> **Summary:** I made hjkl to help myself remember Vim commands. It’s a little browser game you can play when you want to unwind and get a bit of Vim practice in. There are still a lot of commands I want to add, but I hope someone finds it useful in the meantime :) <hr /> Comments URL: <a href="https:/...

🔗 **Read full article:**
https://hjkl.duy.run

Nvidia – open-source auto-ontology

👤 campervans 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://github.com/NVIDIA/auto-ontology Comments URL: https://news.ycombinator.com/item?id=50348962 Points: 91 # Comments: 30

🔗 **Read full article:**
https://github.com/NVIDIA/auto-ontology

uBlock Origin filter to block autoplay videos in TechCrunch

👤 cornfieldlabs 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://cornfieldlabs.github.io/posts/ublock-stop-techcrunch-autoplay-video/post.html Comments URL: <a href="https://news.ycombinator.com/item?id=50349155">https://news.ycombina...

🔗 **Read full article:**
https://cornfieldlabs.github.io/posts/ublock-stop-techcrunch-autoplay-video/post.html

Revolut moves to self-hosted GLM

👤 pretext 📅 2026-11-03 📖 1 min 📝 13 words HACKER NEWS

> **Summary:** Article URL: https://twitter.com/guilleflorvs/status/2117379073785696395 Comments URL: https://news.ycombinator.com/item?id=50350004 Points: 87 <...

🔗 **Read full article:**
https://twitter.com/guilleflorvs/status/2117379073785696395

Easily download files from the Open Science Framework with Papercheck

noreply@blogger.com (Daniel Lakens)
2025-07-22 3 min 765 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Researchers increasingly use the Open Science Framework (OSF) to share files, such as data and code underlying scientific publications, or presentations and materials for scientific workshops. The OSF is an amazing service that has contributed immensely to a changed ...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2025/07/easily-download-files-from-open-science.html

How I use AI in my work

noreply@blogger.com (Daniel Lakens)
2026-10-11 21 min 4292 words

TWENTY PERCENT STATISTICIAN

Summary: <p>Tomorrow we will have a discussion in our lab about how to use AI as a scientist. I have increasingly started to use AI, having moved from Copilot Chat that our university provides last year, to some free credit on GitHub Copilot under their educational program, to Claude Pro integrated in Visual...

🔗 **Read full article:**
http://daniellakens.blogspot.com/2026/10/how-i-use-ai-in-my-work.html

Visuospatial attention modulates audiovisual temporal perception: Context-dependent associations between prestimulus alpha lateralization and perceptual reports

📖 1 min 📝 30 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Zeliang Jiang, Yuqing Zhao, Weicong Ren, Jie Zhang, Xinyuan Liu, Fei Yu, Xingwei An, Wenxiao Zhong, Di Wu, Pan Zhang, Zhijie Zhang

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600635X?dgcid=rss_sd_all

The visual cortex and beyond: investigating long-term cortical plasticity following monocular deprivation

📖 1 min 📝 29 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Franziska Wagner, Lukas Grajewski, Alexander Schmidt, Ulrike Teschner, Markus Ullsperger, Kathrin Finke, Christian Gaser, Robert Dahnke, Carsten M. Klingner, Stefan Brodoehl

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S105381192600604X?dgcid=rss_sd_all

Distinct neural patterns of valence and arousal in the developing brain: A representational similarity analysis of MEG

📖 1 min 📝 12 words NEUROIMAGE

> **Summary:** Publication date: 15 November 2026 Source: NeuroImage, Volume 342 Author(s): Ioannis Ntoumanis, Christos Papadelis

🔗 **Read full article:**
https://www.sciencedirect.com/science/article/pii/S1053811926006324?dgcid=rss_sd_all

Single-molecule bouncing at a liquid interface governs presynaptic enrichment

👤 Kudo, K., Longfield, S. F., McCann, A. J., Schoovaerts, N., Sanchez-Velasquez, J., Parra-Rivas, L. A., Roy, S., Verstreken, P., Hinde, E., Millard, S. S., Bademosi, A. T., Meunier, F. A.

📅 2026-11-02 📖 1 min 📝 187 words BIORXIV NEUROSCIENCE

> **Summary:** Many cellular processes rely on accumulating highly diffusive cytoplasmic proteins to a specific location within the cell. This spatial enrichment is generally thought to rely on locally immobilizing proteins or slowing diffusion through binding and molecular crowding. Paradoxically, cytoplasmic pro...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.11.01.763503v1?rss=1

Prolactin Receptor Long Isoform ΔS1 Expression is Reduced in the Dorsal Root Ganglia of Humans with Chronic Pain

He, L., Natarajan, K., Shiers, S. I., Arendt-Tranholm, A., Sayani, R., Jagan, S., Hossain, A., Rosenfield, J., Rodriguez Concepcion, A. J., Dussor, G., Akopian, A. N., Price, T. J.

2026-11-02 1 min 301 words BIORXIV NEUROSCIENCE

Summary: Prolactin receptor (PRLR) isoforms, specifically the short (PRLR-S) and long (PRLR-L), are differentially expressed in pathological conditions and may be involved in the induction and maintenance of pain in females. PRLR sex-dependent actions are well studied in rodent models of chronic pain, but th...

Read full article:
https://www.biorxiv.org/content/10.64898/2026.10.27.762326v1?rss=1

EigenBrain maps cortical molecular architecture from functional, geometric, and structural eigenmodes

Wan, B., Du, P., Ferrari, A., Saberi, A., John, A., Yan, J., Liu, X., Hansen, J. Y., Liu, Z.-Q., Hong, S.-J., Carruzzo, F., Kaliuzhna, M., Geffen, T., Katthagen, T., Misic, B., Thompson, P. M., Van De Ville, D., Margulies, D. S., Warrier, V., Bethlehem, R. A. I., Schlagenhauf, F., Kaiser, S., Bernhardt, B., Valk, S. L., Kirschner, M.

📅 2026-11-02 📖 1 min 📝 232 words BIORXIV NEUROSCIENCE

Summary: Neurotransmitter receptor and transporter distributions shape neural communication across the cortex and underpin mechanistic interpretations of brain function. Magnetic resonance imaging (MRI)-derived eigenmodes, low-dimensional spatial patterns that capture the principal axes of cortical organizat...

🔗 **Read full article:**
https://www.biorxiv.org/content/10.64898/2026.10.27.758642v1?rss=1

Bone transport induces CGRP release from sensory nerves to promote angiogenesis and heal diabetic foot wounds in rats

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-73891-z

Proapoptotic RHG proteins and mitochondria are required for sensory dendrite pruning in Drosophila

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41598-026-77081-9

An exploratory remote pilot study of environmental risk factors for Lewy body dementia considering sex differences

📅 2026-11-03 📖 1 min 📝 0 words NATURE NEUROSCIENCE SUBJECTS

🔗 **Read full article:**
https://www.nature.com/articles/s41531-026-01603-5

Safety of lecanemab in real-world patients with Alzheimer’s disease: a prospective, observational case series supported by FAERS signal analysis

Erhao Bao 2026-10-02 1 min 354 words FRONTIERS NEUROINFORMATICS

Summary: ObjectiveLecanemab, a humanized immunoglobulin monoclonal antibody, has been approved for the treatment of Alzheimer’s disease (AD). Although clinical trials have demonstrated its favorable safety profile, real-world studies are crucial for identifying rare adverse events (AEs) and evaluating post-m...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1977226

Correction: The PREP pipeline: standardized preprocessing for large-scale EEG analysis

Kay A. Robbins 2026-10-12 1 min 0 words FRONTIERS NEUROINFORMATICS

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.2015651

EEG emotion-recognition datasets for neuroarchitecture: an appraisal of openness, AI-readiness and ontology maturity

Han Jong Jun 2026-10-20 1 min 341 words FRONTIERS NEUROINFORMATICS

Summary: IntroductionNeuroarchitecture studies how designed space shapes emotion, often with electroencephalography (EEG), and computational models that decode emotion from EEG must be trained on labelled recordings. Yet none of the widely shared EEG emotion datasets was recorded while people moved through d...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fninf.2026.1971441

Are cortical microcircuits optimized for information flux?—A simulation-based reverse engineering study

Patrick Krauss 2026-10-20 1 min 271 words

FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: IntroductionA sufficiently large information flux in recurrent neural networks, quantified by the mutual information between successive network states, has been proposed as a marker of rich information-processing capabilities. This raises the question of whether biological neural networks, such as c...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1886601

Efficient neuroanalytic local–global learning with dual-branch CNN–lightweight Swin Transformer for Alzheimer’s disease analysis

👤 Arshad Ali 📅 2026-10-29 📖 1 min 📝 287 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Alzheimer’s disease (AD) is one of the main causes of dementia, where early detection is critical for postponing the progression of the disease and timely treatment. Recent computer-aided diagnosis using deep learning with structural MRI has shown optimal performance; however, the rising number of m...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1972923

Virtual brain models: a roadmap towards effective brain digital twins

👤 Viktor Jirsa 📅 2026-10-29 📖 1 min 📝 205 words FRONTIERS COMPUTATIONAL NEUROSCIENCE

Summary: Virtual Brain Models (VBMs) are emerging computational tools promising to revolutionize personalised and precision medicine. VBMs reproduce brain functional dynamics through a generative model that can be constructed and optimized against single subject data, thereby generating what is called a brai...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fncom.2026.1973956

A unified fixed-wavelet transformer decodes eight EEG classes across motor imagery, P300, and SSVEP paradigms

Hong-Gi Yeom 2026-10-29 1 min 314 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionMotor imagery (MI) can support self-initiated control, whereas brain–computer interfaces (BCIs) based on the P300 or the steady-state visual evoked potential (SSVEP) rely on ongoing visual stimulation. Repetitive visual stimulation can cause visual fatigue; when the stimuli are suspended...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1973838

Resolving the apparent paradox of multisensory expertise: coupling priors, common-cause priors, and how to distinguish among them

Pier Luigi Sacco 2026-10-29 1 min 348 words FRONTIERS HUMAN NEUROSCIENCE

Summary: Perceptual expertise is widely assumed to refine multisensory perception, yet the empirical literature appears inconsistent. Trained musicians show narrower audiovisual temporal integration windows and quicker segregation, whereas experienced wine tasters seem more influenced by a wine's color in th...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1960361

Editorial: Emerging talents in human neuroscience: psychophysiology and cognitive neuroscience 2025

Gaia Risso 2026-10-29 1 min 0 words FRONTIERS HUMAN NEUROSCIENCE

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.2016840

Simulation-based neural dynamics of stress-related negative cognitive bias in depressive symptoms: a dynamical EEG/ERP framework

Jianping Liu 2026-10-29 1 min 192 words FRONTIERS HUMAN NEUROSCIENCE

Summary: IntroductionThis simulation-based computational study develops a non-linear state-space framework for stress-related negative cognitive bias and multivariate EEG/ERP observations.MethodsA three-state model represents negative bias, cognitive control, and arousal, while symptom load and normalized mo...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnhum.2026.1972914

Effect of robotic assistance on upper-limb sensorimotor recovery: a systematic review and meta-analysis

👤 Sivakumar Balasubramanian
📅 2026-10-20 📖 1 min 📝 359 words FRONTIERS NEUROROBOTICS

Summary: IntroductionRobot-assisted upper-limb neurorehabilitation enables several beneficial therapy components, including repetitive task-specific practice, high training intensity, real-time feedback, gamification, and continuous physical interaction with the patient’s limb. However, conventional comparis...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1860365

Dolphin-inspired adaptive coordination for online multi-robot path planning in dynamic environments

👤 Lijuan Chen
📅 2026-10-29 📖 1 min 📝 200 words FRONTIERS NEUROROBOTICS

Summary: IntroductionOnline multi-robot path planning requires repeated joint motion decisions amid static obstacles, moving obstacles, and other robots.MethodsThis study introduces the Dolphin Cooperative Foraging Optimizer (DCFO), a population-based method that adapts its search strategy within each replan...

🔗 **Read full article:**
https://www.frontiersin.org/articles/10.3389/fnbot.2026.1993109

From confidence to caution: conformal gating for cross-session motor imagery brain–computer interfaces

Sixue Xing and Aarthy Nagarajan
2026-10-22 1 min 278 words

JOURNAL NEURAL ENGINEERING

Summary: Objective. Erroneous motor imagery brain–computer interface (MI-BCI) commands can cause unintended device actions, yet standard classifiers provide no mechanism to withhold uncertain predictions. We evaluate whether conformal prediction (CP) can serve as a command gate under cross-session EEG shift,...

🔗 **Read full article:**
https://iopscience.iop.org/article/10.1088/1741-2552/aead26

Human cortical high-gamma power scales with movement rate in healthy participants and stroke survivors

Fanny Quandt
2025-01-09 1 min 65 words

OOSTENVELD ROBERT

Summary: Motor cortical high-gamma oscillations (60-90 Hz) occur at movement onset and are spatially focused over the contralateral primary motor cortex. Although high-gamma oscillations are widely recognized for their significance in human motor control, their precise function on a cortical level remains el...

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NIRS-BIDS: Brain Imaging Data Structure Extended to Near-Infrared Spectroscopy

👤 Luca Pollonini 📅 2025-01-27 📖 1 min 📝 70 words OOSTENVELD ROBERT

> **Summary:** Functional near-infrared spectroscopy (fNIRS) is an increasingly popular neuroimaging technique that measures cortical hemodynamic activity in a non-invasive and portable fashion. Although the fNIRS community has been successful in disseminating open-source processing tools and a standard file forma...

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Pseudonymisation of neuroimages and data protection: Increasing access to data while retaining scientific utility

👤 Lyuba Zehl 📅 2025-06-26 📖 1 min 📝 67 words OOSTENVELD ROBERT

> **Summary:** For a number of years, facial features removal techniques such as 'defacing', 'skull stripping' and 'face masking/blurring', were considered adequate privacy preserving tools to openly share brain images. Scientifically, these measures were already a compromise between data protection requirements a...

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Cycling on the Freeway: The perilous state of open-source neuroscience software

👤 Tim M Tierney 📅 2025-08-13 📖 1 min 📝 74 words OOSTENVELD ROBERT

> **Summary:** Most scientists need software to perform their research (Barker et al., 2020;Carver et al., 2022;Hettrick, 2014;Hettrick et al., 2014;Switters & Osimo, 2019), and neuroscientists are no exception. Whether we work with reaction times, electrophysiological signals, or magnetic resonance imaging data, ...

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Optimal configuration of on-scalp OPMs with fixed channel counts

👤 Robert Oostenveld 📅 2025-08-13 📖 1 min 📝 69 words OOSTENVELD ROBERT

> **Summary:** Recent technological developments have brought optically pumped magnetometers (OPMs) within reach of the larger neuroscientific community. The current state-of-the-art consists of whole-head systems that measure the magnetic field at >100 locations. OPM sensors can be constructed to measure the fiel...

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ARTEM-IS for ERP: Agreed Reporting Template for EEG Methodology-International Standard for Event-Related Potential Experiments

👤 Suzy J Styles 📅 2025-12-08 📖 1 min 📝 73 words OOSTENVELD ROBERT

Summary: The choices we make during the recording, preprocessing and analysis of event-related potentials (ERP) data can affect study outcomes. As such, it is critical that they are transparently reported to allow for reproducibility. Yet, systematic reviews of reporting practices in the field have shown tha...

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Eye-Tracking-BIDS: the Brain Imaging Data Structure extended to gaze position and pupil data

👤 Julia-Katharina Pfarr 📅 2026-02-12 📖 1 min 📝 68 words OOSTENVELD ROBERT

Summary: The Brain Imaging Data Structure (BIDS) is a widely adopted, community-driven standard to organize neuroimaging data and metadata. Although numerous extensions have been developed to incrementally extend coverage to new modalities and data types, an unambiguous, granular specification for eye-tracki...

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Facilitating unusual bodily experiences and out-of-body experiences across wakefulness and sleep: A high-density EEG and neurophenomenology study

👤 Robert Oostenveld 📅 2026-02-12 📖 1 min 📝 67 words OOSTENVELD ROBERT

Summary: Unusual bodily experiences (UBEs) are illusory bodily perceptions that are not coherent with typical wakeful experiences, including flying or floating sensations, body distortions, and out-of-body experiences. This study examines UBEs in a controlled sleep laboratory using meditation and light stimu...

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The Impact of Brain Tumors and Craniotomy Lesions on Scalp EEG

👤 Vitória Piai 📅 2026-02-17 📖 1 min 📝 65 words OOSTENVELD ROBERT

Summary: Electroencephalography (EEG) is widely used in both research and clinical settings, yet its accuracy can be significantly impacted by subject-specific anatomical anomalies such as brain lesions and skull defects. This study investigates the effects of glioma-related brain lesions and craniotomy-indu...

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EMG-BIDS: an extension to the Brain Imaging Data Structure for electromyography

Arnaud Delorme 2026-08-01 1 min 64 words OOSTENVELD ROBERT

Summary: Electromyography (EMG) is fundamental to clinical assessment, rehabilitation, neuromuscular research, and human-machine interfaces. Despite decades of use, no widely adopted standard exists for organizing and sharing EMG data, limiting reusability and large-scale data aggregation. We present EMG-BID...

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Short-term visual and ocular responses to a myopia control contact lens: influence of optical zones and hydration techniques

César Antonio Albarrán-Diego 2026-10-31 1 min 40 words LOW VISION

Summary: CONCLUSIONS: The evaluated contact lenses showed good short-term performance and tolerance. A smaller central optical zone was associated with modest reductions in contrast sensitivity and perceived visual quality, without clear advantages in other outcomes. Hydration modality did not influence perf...

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Pediatric Retinal Disorders

Shyam S Chaurasia 2026-10-31 1 min 67 words LOW VISION

Summary: Pediatric retinal disorders consist of a plethora of pathophysiologies that directly impact the retinal structure and function, thereby compromising children's vision and quality of life. These disorders can affect children's vision in a wide range of diseases, from minor visual acuity deficits to t...

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All-optical 2D perovskite photodetector arrays for pixelated long-wave infrared imaging

Peijun Guo 2026-11-01 1 min 55 words LOW VISION

Summary: Infrared detection and imaging are important for applications including thermal monitoring, night vision, environmental sensing, and molecular identification. All-optical mid-wave infrared (MWIR) and long-wave infrared (LWIR) detection offers a compelling alternative to conventional infrared detecto...

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Inequalities in the distribution of the family physician workforce in the kingdom of Saudi Arabia: a regional analysis

Mohammed Khaled Al-Hanawi 2026-11-01 1 min 46 words LOW VISION

Summary: CONCLUSIONS: Although the national distribution of family physicians appears balanced, Lorenz curve analysis reveals significant inequalities across regions, sectors, and demographic groups. These findings have implications for workforce planning and monitoring, particularly in regions exhibiting co...

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Assessing effects of low-dose CT attenuation correction on quantification and image quality on a LAFOV PET system with a modified NEMA-IQ phantom

Paul K Marsden 2026-11-02 1 min 77 words LOW VISION

Summary: CONCLUSIONS: Even for low-dose CT protocols, PET quantification remained consistent in all NEMA-IQ spheres for many materials. Beam hardening and photon starvation in the CT image are likely to occur in specific body regions such as hips, shoulders or the abdomen for an arms-down scan and these effe...

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Functional and gain-based vestibulo-ocular measures across serum 25(OH)D categories: a dissociation between fHIT performance and vHIT gain

Alper Köycü 2026-11-02 1 min 30 words LOW VISION

Summary: CONCLUSIONS AND SIGNIFICANCE: Low 25(OH)D was accompanied by reduced functional vestibulo-ocular performance in the vertical canal planes despite normal gains. fHIT may capture subclinical change that vHIT gain does not.

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HiCP2GAN: a plug and play foundation model-based GAN for Hi-C enhancement

Oluwatosin Oluwadare 2026-11-02 1 min 65 words LOW VISION

Summary: The 3D organization of chromatin shapes gene regulation and cellular function, and Hi-C has become the primary technique for mapping chromatin interactions genome-wide. However, high-resolution data remain costly and scarce, leaving many studies with sparse contact maps that limit downstream analysi...

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Exploring Spatial Inequities and Livability: A Mixed-Methods Study Using Artificial Intelligence and Community Insights

Rodrigo Siqueira Reis 2026-11-02 1 min 62 words LOW VISION

Summary: Rapid urbanization has intensified health and social inequities through the uneven distribution of infrastructure, services, and public space. In St. Louis, Missouri (USA), these disparities remain geographically concentrated and rooted in long-standing patterns of racial segregation and disinvestme...

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Association Between Quality of Life and Visual Function in Inherited Retinal Disease: A Cross-sectional Study in a Retinal Genetics Centre

Robert E MacLaren 2026-11-02 1 min 27 words LOW VISION

Summary: CONCLUSIONS: Microperimetry is a patient-relevant measure in inherited retinal diseases. This study demonstrates to stakeholders, such as regulatory bodies, how visual function measures translate to patients' quality-of-life.

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Social Determinants of Kidney and Cardiovascular Outcomes: Prioritizing Action to Effectively Manage Cardiovascular-Kidney-Metabolic Syndrome

Tiffany M Powell-Wiley 2026-11-02 1 min 70 words LOW VISION

Summary: In 2005, the World Health Organization (WHO) created the inaugural Social Determinants of Health (SDOH) Commission to recognize how social and physical environments impact health for all populations. In doing so, they created an impetus for policy-making and governing bodies to improve the underlyin...

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Variability in cross-modal proprioceptive judgements impairs upper-limb function across the lifespan

Martin E Héroux 2026-07-18 1 min 69 words TACTILE ACUITY

Summary: Proprioceptive deficits are thought to contribute to age-related decline in upper-limb function. We aimed to estimate the effect of proprioceptive ability on upper-limb function under an explicitly specified causal framework and to determine whether this effect varied with age. A causal graph was de...

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A single session of high-definition transcranial direct current stimulation does not modulate effects of knee two-point discrimination training or sensorimotor function in healthy adults: Double-blind randomised controlled trial

👤 Paul W Hodges
📅 2026-07-30 📖 1 min 📝 53 words

TACTILE ACUITY

Summary: CONCLUSION: A single 20-minute session of 1 mA HD-tDCS, with or without brief TPD training, does not acutely modify knee somatosensory, sensorimotor, or functional performance in healthy adults. Existing evidence at the hand may not translate to the lower limb. Future work should investigate higher-...

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Variation in glabrous skin physiology across anatomical locations on the human hand

Martina C Meinke 2026-08-13 1 min 76 words TACTILE ACUITY

Summary: Skin physiology varies greatly between anatomical locations, with impact on tactile sensitivity. Research on the variations in morphology and physiology between different anatomical locations of glabrous skin is limited so far. This study investigates the morphology and physiology of glabrous skin o...

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Beyond Hamstring Shortness: Continuous Flexibility Is Associated With Balance and Stability but Not Sensory Measures-A Cross-Sectional Study

Senanur Akıncı 2026-08-14 1 min 66 words TACTILE ACUITY

Summary: CONCLUSIONS: Greater hamstring flexibility was associated with better dynamic balance performance and showed a weaker association with postural stability, whereas no robust associations were observed with sensory measures of postural control. These findings suggest that hamstring flexibility may be ...

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Resting-state EEG microstate dynamics reflect individual differences in tactile angle discriminability

Liu Yang 2026-08-19 1 min 63 words TACTILE ACUITY

Summary: Tactile spatial acuity, the ability to discriminate fine spatial details, is a fundamental sensory capacity modulated by both peripheral and central mechanisms. Here, we investigated whether resting-state EEG microstate dynamics are associated with individual differences in tactile spatial acuity. F...

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A grating orientation task with controlled dome elevation speed and contact force: stimulus conditions and day-to-day reliability

Hideaki Onishi 2026-09-25 1 min 66 words TACTILE ACUITY

Summary: The grating orientation discrimination task (GOT) is typically used to assess tactile spatial acuity. However, performance is influenced by stimulus speed, contact force between the stimulator and the skin, and the method of discrimination. This study aimed to explore suitable stimulus conditions fo...

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Epidermal microstructure and tactile sensitivity in the glabrous skin of hands and feet

Hannes P Saal 2026-09-25 1 min 69 words TACTILE ACUITY

Summary: Glabrous skin on the human palms and soles is specialized for mechanical interaction and is characterized by papillary ridges, an absence of hair follicles, and a thick stratum corneum. Although gross morphological differences, such as the increased thickness of the plantar stratum corneum, are well...

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Neurotrauma During Military Training: Results of the INVestigating traIning assoCiated blasT pAthology (INVICTA) Study

Walter S Carr 2026-10-01 1 min 67 words TACTILE ACUITY

Summary: Concern is growing that subconcussive blast exposure during military exercises such as heavy weapons training (HWT) may adversely affect brain function. HWT involves firing shoulder-mounted weapons that can expose the gunner, assistant gunner, and instructors (range safety officers [RSOs]) to blast ...

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Altered body representation in chronic non-specific low back pain: a testable framework linking body-schema-related disturbance, corticomotor alterations, and altered pain processing

👤 Zhifei Xie
📅 2026-10-24 📖 1 min 📝 66 words

TACTILE ACUITY

Summary: Low back pain is a leading contributor to disability worldwide; in chronic non-specific low back pain (CNSLBP), the processes sustaining pain and disability in the absence of identifiable structural pathology remain incompletely understood. Three partially overlapping research lines have documented ...

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Hyperexcitable primary somatosensory cortex in the MitoPark mouse model of Parkinson's disease

👤 Alan J Emanuel
📅 2026-10-25 📖 1 min 📝 67 words

TACTILE ACUITY

Summary: The majority of Parkinson's disease patients suffer from decreased tactile acuity or neuropathic pain, both of which contribute to low quality of life and likely exacerbate motor dysfunction. How Parkinson's disease pathology relates to altered somatosensation remains unclear. One potential contribu...

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Impact of Braille Booklet, Audio-Tactile Performance and Caregiver-supervised Training on Oral Hygiene Status, Knowledge, Attitude and Practices among Visually Challenged Children- A Randomized Clinical Trial

👤 Priyanga Chandrasekaran
📅 2026-09-22 📖 1 min 📝 52 words

BRAILLE

Summary: CONCLUSION: Both caregiver-supervised training and audio tactile performance using specialized model method of oral health education proved to be more effective than the braille booklet group on plaque control and improvement in the oral hygiene status, oral health knowledge, attitude and practice i...

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Design and functional verification of an automated Perkins Brailler accessory with audio feedback

Ali Alhaj 2026-09-23 1 min 67 words BRAILLE

Summary: CONCLUSIONS: The study did not include learners with visual impairment, Braille teachers, rehabilitation specialists, or other end users, and it did not collect quantitative measures of accuracy across the full character set, writing speed, embossing quality, latency, durability, safety, usability, ...

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Uncertainty-Aware structured decoding for Chinese Braille document transcription

Haoran Sun 2026-09-25 1 min 64 words BRAILLE

Summary: Automatic transcription of braille documents remains challenging in document digitization settings, particularly under real imaging degradations such as uneven embossing, illumination variation, paper texture, skew, blur, and ghosting, which make braille dot presence uncertain and often cause error ...

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Clinical Characteristics and Epidemiology of Isolated Dystonia: Results from a US Claims Analysis

👤 James E Thornton
📅 2026-10-08 📖 1 min 📝 54 words BRAILLE

Summary: CONCLUSIONS: This study defines an isolated dystonia cohort in the US based on real-world evidence and systematic exclusion of secondary causes of dystonia. Estimated prevalence is within range of prior epidemiological studies, and the cohort showed increased rates of comorbid psychiatric disorders....

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